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1. 产能共享：Capacity Sharing and Cost Allocation among Independent Firms with Congestion (POM, 2015)

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推荐理由：本文分析独立企业共同建设生产或服务设施的收益。设施被表示为服务率有限的排队系统，企业在延误成本、产能投资及服务水平约束之间选择最优容量，并决定共享设施的成本分摊。作者用合作博弈刻画这一问题，分别研究先到先服务和最优优先级规则下，共享有利且核心内成本分配存在的条件。研究指出，当企业的工作内容与服务变异性存在差异时，共享不一定使所有企业受益，并进一步给出部分企业组成联盟仍可获益的条件。

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Capacity Sharing and Cost Allocation among Independent Firms with Congestion

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We analyze the benefit of production/service capacity sharing for a set of independent firms. Firms have the choice of either operating their own production/service facilities or investing in a facility that is shared. Facilities are modeled as queueing systems with finite service rates. Firms decide on capacity levels (the service rate) to minimize delay costs and capacity investment costs possibly subject to service-level constraints on delay. If firms decide to operate a shared facility they must also decide on a scheme for sharing the capacity cost. We formulate the problem as a cooperative game and identify settings under which capacity sharing is beneficial and there is a cost allocation that is in the core under either the first-come, first-served policy or an optimal priority policy. We show that capacity sharing may not be beneficial in settings where firms have heterogeneous work contents and service variabilities. In such cases, we specify conditions under which capacity sharing may still be beneficial for a subset of the firms.

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1. Introduction

Capacity sharing refers to the fulfillment of demand that arises from multiple sources from a single facility instead of facilities dedicated to each demand source. In a system without capacity sharing, each dedicated facility fulfills its own demand relying solely on its capacity. It has long been known that capacity sharing can be beneficial when demand is random. This benefit can be in the form of improved service quality with the same amount of capacity or in the form of less capacity needed to provide the same quality of service. Capacity sharing can also be beneficial when there are economies of scale associated with acquiring capacity or fulfilling demand. These benefits have been shown to be true for various forms of capacity, including manufacturing, service, and inventory.

Capacity sharing has been studied mostly in situations where a single firm, or a sub-division within a firm, owns all the capacity in the system, and has responsibility for serving all the demand. This firm makes the decision about whether or not to share capacity and how much capacity to acquire. In this study, we consider a system with n independent firms, or sub-divisions within a firm, each facing its own demand and each having the option of either operating its own independent facility or joining some or all the other firms in a shared facility. The firms may vary in their demand levels and in their tolerance for capacity shortage. If some or all of the firms decide to share capacity, they must also decide on how to allocate the cost of the shared facility. They must do so in a manner that benefits everyone and prevents any of the firms from defecting and perhaps sharing a facility with a subset of the firms or staying on their own. Hence, firms that contribute more to the cost of the shared facility (because of their higher usage of capacity or lower tolerance for capacity shortage) are expected to pay a greater share of total cost.

In this study, we consider applications where facilities can be modeled as queueing systems. Demand for each firm consists of an independent stream of customers (or orders) that arrive continuously over time with random inter-arrival times. Each firm can install and operate its own facility where its customers are processed. Firms make decisions about how much service capacity to acquire in order to minimize two types of costs, delay cost due

to customers spending time at the facility prior to completing service and capacity investment cost, subject to a constraint on the amount of delay or waiting time that customers experience. Alternatively, firms may choose to collectively operate a shared facility. In that case, in addition to determining the optimal amount of capacity (taking into account the delay costs and service levels of all the firms), the firms must also determine how the corresponding costs must be allocated. In both cases, of either shared or individual facilities, the facilities are modeled as single server queues, with capacity determined by the associated service rate.

Capacity sharing among independent firms in the presence of congestion, and with heterogeneous delay costs and service level requirements, arises in a variety of settings. For example, firms (or subdivisions within a firm) can decide to share support services, such as repair and maintenance facilities, instead of investing in facilities of their own.[1] In the case of repairs, repair requests for each firm arise randomly over time (because of the randomness in breakdowns) with repair times that can be stochastic. Given the limited repair capacity, this can lead to congestion. Each firm may have its own delay cost (e.g., costs corresponding to the opportunity cost of equipment down time) or may have a specified service level it guarantees to its own customers. Capacity in this case is determined by the speed with which repairs can be undertaken (e.g., the processing rate of the main bottleneck process). Other examples include common services, such as printing, testing, and rapid prototyping, which sub-divisions within the same firm may choose to share. In this case too, the variability in the arrival of service requests and processing times can lead to congestion. Similarly, medical units within the same hospital often share radiology services, testing and diagnostic facilities, surgery suites, and appointment and billing systems. Capacity, in terms of the speed with which service requests can be processed, must again take into account the requirements of the different subdivisions, including their individual sensitivity to delay.

The main contributions of our study are summarized below.

- We provide a framework for modeling capacity sharing in queueing systems with independent firms. To our knowledge, our study is among the first to model the issue of cooperation and capacity sharing in a queueing context.
- Under the M/M/1 setting with the FCFS policy, we formulate capacity sharing as a cooperative game, in which the participating firms optimize shared capacity taking into account the characteristics of individual firms, their delay costs, and their service level requirements. We characterize settings where the *core* of the game is nonempty. That is, there exists a capacity cost allocation rule for which all the firms are better off than under any other alternative sharing arrangement, including being on their own.
- When the core exists, we identify a general class of cost allocation rules with desirable properties that is in the core. Under this class of allocation rules, each firm is charged its own delay cost and a fraction of total capacity cost. We provide an example of an allocation rule within this class where the fraction of capacity cost allocated to each firm consists of two parts, a share of the cost of base capacity, proportional to the firm's demand, and a share of the cost which it is directly responsible, its own delay cost, and a fraction of buffer capacity, consistent with the firm's contribution to this cost.
- We show that when the service level involves waiting time instead of delay, capacity sharing may not be beneficial and the core may not exist. This is because buffer capacity is not invariant to total demand as it is in the case when the service level involves delay.
- We consider systems that operate under an optimal priority policy. We show that there exists a capacity cost sharing scheme that is in core, regardless of whether the service involves delay or waiting time. This is because under the optimal priority scheme, it is possible to provide differentiated service levels for different firms.
- Under the FCFS policy, we also characterize other settings where the core may not exist because capacity sharing may not be beneficial. These settings include systems where service levels are specified in terms of waiting time instead of total delay and cases where the firms are heterogeneous in their characteristics, including their work contents, service time variability, and delay costs. For these cases, we characterize conditions, under which capacity sharing may still be beneficial for a subset of the firms. These conditions provide insights into the characteristics of firms that would benefit from forming sub-coalitions. In general, these results indicate that capacity sharing should be considered with caution when differentiated service levels are not possible.

- We extend our results to a variety of queueing systems, including M/G/1 queues, and GI/G/1 queues. In doing so, we extend known results regarding the benefit of capacity pooling in each system by endogenizing capacity instead of assuming it remains constant with and without capacity sharing.

We should note that there is a rich literature that models manufacturing and service systems as queueing systems (see sections 2 and 3 for further discussion). Surprisingly very little of this literature addresses the issue of cooperation and capacity sharing when there are independent firms. Therefore, we view our study as a step toward a more comprehensive examination of the issue of cooperation in queueing systems, whether it arises in manufacturing, services, or elsewhere. We also view it as a contribution, in the form of a potentially rich application domain, to the literature on cooperative games.

The rest of the study is organized as follows. In section 2, we provide a brief review of related literature. In section 3, we treat the case with no capacity sharing. In section 4, we analyze the case with capacity sharing. In section 5, we consider capacity sharing when there are service priorities. In section 6, we extend our analysis to systems with heterogeneous work contents and systems with general arrival and service processes. In section 7, we offer concluding comments.

2. Related Literature

There is a rich literature on capacity *pooling* in queueing systems, with applications ranging from manufacturing and service operations to telecommunications systems to computer networks. This literature can be classified broadly as relating to either the *pooling of service rates* or the *pooling of servers*. Server rate pooling refers to the consolidation of multiple servers into a single one with a faster rate (e.g., N servers, each with service rate μ and demand rate λ, are replaced by a single server with demand rate $N\lambda$). Server pooling on the other hand refers to placing multiple servers in a single facility from which all demand streams are served (e.g., N single server queues are replaced by a single multi-server queue with N servers and a demand rate $N\lambda$).

Kleinrock (1976) discusses various examples of both types of pooling. Stidham (1970) considers a design problem where the decision variables are the number of parallel servers and the service rate of each server. Smith and Whitt (1981) and Benjaafar (1995) show that server pooling, when the number of servers is exogenously determined, is beneficial as long as all customers have identical service time distributions. Buzacott (1996) considers the pooling of N servers in series, with each server dedicated to one task, into N parallel servers, with each server carrying out all the tasks. Mandelbaum and Reiman (1998) consider the pooling of general Jackson networks into single server queues with phase-type service time distributions.

Tekin et al. (2009) use approximations to evaluate the benefit of partitioning servers in multiple pools instead of a single large one. Sheikhzadeh et al. (1998), Gurumurthi and Benjaafar (2004) and Jordan et al. (2004) study the *chaining* of servers, where each server can process customers from two customer streams and each customer can be routed to two servers. They show that in systems with homogeneous demand rates and service time requirements, chaining can achieve most of the benefits of total server pooling; see also Hopp et al. (2004), Iravani et al. (2005), Bassambo et al. (2012), Aksin et al. (2008), Wallace and Whitt (2005) and the references therein. These papers belong to the growing literature on queueing systems with server flexibility (or cross-training); see Jouini et al. (2008), Aksin et al. (2007) and Koole and Pot (2005) for recent reviews.

The treatment in this study is different from the above literature in three important aspects. First, we do not assume that there is a single decision-maker that determines whether or not to pool. Instead, we consider multiple firms that decide independently on either operating their own facilities or sharing one with other firms (pooling here does not imply a merger however). Second, we do not assume that service capacity is exogenously given. We allow for this to be an outcome of an optimization carried out by the firms either individually or jointly. Third, we are concerned with identifying cost allocation schemes under which all firms prefer a single shared facility to any other capacity sharing arrangement, including remaining on their own. In this study, we focus solely on the case of server rate pooling. However, the results we obtain might still offer useful insights for systems with multiple servers. For example, in regimes where the demand rate is high, it is known that decisions regarding the optimal number of servers in multi-server systems asymptotically mimic those regarding service rate in single server systems; see for example Borst et al. (2004).

The literature dealing with capacity sharing in the context of independent firms is limited. Gonzalez and Herrero (2004), and also Garcia-Sanz et al. (2008), consider a special case of the M/M/1 model we consider. However in both cases, they do not optimize capacity (before or after pooling) and do

not consider the delay cost. In our case, the presence of delay costs significantly complicates the process of cost allocation since we seek allocations that could allow for each firm to absorb its own cost of delay. Anily and Haviv (2010) treat a related M/M/1 model where the issue is how to allocate delay cost to ensure that the allocation is in the core. They show that a Shapley allocation based on the service levels is in the core. In this literature, the common approach is to use concavity as a basis for proving that the core exists and that a Shapley allocation is in the core. In contrast to the above literature, a Shapley allocation may not be in the core in our case. This is in part due to our treatment of capacity as endogenous and to the requirement that each firm absorbs its own delay cost and only capacity costs are allocated among the firms. We also treat queueing systems other than the M/M/1 queue, including queues with service priorities, M/G/1 queues, and GI/G/1 queues.

Our work is of course related to the vast literature on cooperative game theory and, more broadly, the economics of coalition formation and joint ventures; see Moulin (1995) for a general introduction to the topic. Some of this literature has focused on cooperation involving sequencing and scheduling; see for example Moulin and Stong (2002), Maniquet (2003), and Katta and Sethuraman (2006). This literature sometimes refers to these problems as queueing problems. However, they typically involve a finite population of customers who simultaneously arrive to the system, and therefore are not concerned with steady state behavior and congestion in the way that we are in this study. In Operations Management, there is growing literature that applies cooperative game theory to joint ordering problems, particularly in the context of economic order quantity models (see Anily and Haviv 2007, Dror and Hartman 2007 and the many references therein), economic lot sizing models (see e.g., Chen and Zhang 2009 and van den Heuvel et al. 2007, among others), and news-vendor models (see Chen and Zhang 2007, Hanany and Gerchak 2008, Kemahlioglu-Ziya 2004, Muller et al. 2002, Nagarajan and Sošić 2008, Ben-Zvi and Gerchak 2012, and the references therein).

Finally, we should note that there is a rich literature on outsourcing where multiple firms may be served by the same supplier, including for settings where the outsourcing supplier is modeled as a queueing system; see for example, Cachon and Harker (2002), Allon and Federgruen (2007), Gans and Zhou (2007), and Benjaafar et al. (2007). In general, the focus of this literature is different as it does not deal with cost allocation or coalition formation.

3. Systems without Capacity Sharing

Consider a system consisting of a set $\mathcal{N} = \{1, \ldots, n\}$ of n firms. Firm i, $i \in \mathcal{N}$, faces an independent demand stream with customers arriving according to a Poisson process with rate λ_i (we treat more general arrival processes in section 6). When firms operate independently, each firm invests in a separate service facility and chooses a certain level of capacity in the form of a service rate. We refer to this scenario as the scenario without capacity sharing. Once the facilities are built, each firm serves its customers from its own facility one at a time on a first-come, first-served (FCFS) basis. We assume service times are independent and identically distributed random variables denoted by X_i where X_i is of the form Y/μ_i and Y is a random variable that is exponentially distributed with a mean equal to 1. Hence, service time is also exponentially distributed with mean $E[X_i] = 1/\mu_i$. The parameter μ_i, $(\mu_i > 0)$ is a scaling parameter that corresponds to the service rate or capacity.

The random variable Y can be viewed as the work content associated with each customer. We assume that the work content is homogeneous across firms. This assumption is justified if firms provide service to similar customers (e.g., repairing similar equipments in the case of a maintenance facility). Given the exponential nature of both customer inter-arrival times and service times, each firm behaves like an M/M/1 queue. There is a significant literature on the economics of queues with competitions that primarily focuses on the M/M/1 queue (and where the service rate is the decision variable); see Hassin and Haviv (2003) for a review of that literature and see Cachon and Harker (2002), Cachon and Zhang (2007), Benjaafar et al. (2007), and Allon and Federgruen (2007), among many others, for example, applications. Our treatment of the M/M/1 queue is consistent with assumptions made in that literature and can be viewed as complementing it for cooperative settings.

We assume that service rate can be varied continuously and that firms incur a capacity cost c per unit of service rate per unit time. This is justified in settings where capacity can be continuously scaled over a sufficiently large interval. It is consistent with treatments elsewhere in the literature (see e.g., Allon and Federgruen (2007, 2008), Cachon and Zhang (2007), Ha (2001), Kalai et al. (1992), Mendelson and Whang (1990), and the vast literature reviewed therein). This assumption can also be found in the significant literature on capacity planning, as noted recently by Bassambo et al. (2012). This is an important case that has been widely studied in the literature (see Allon , and Federgruen 2007, 2008, Bassambo et al. 2012, Cachon

and Harker 2002, Dewan and Mendelson 1990, and Stidham 1992 among others), leads to tractable analysis, and provides a useful benchmark for other cost structures.

The objective of each firm is to minimize its capacity investment while limiting the amount of delay, as measured by either total time in system, or waiting time in the queue, its customers experience. Limiting customer delay can be achieved by enforcing a service level constraint or by associating a cost with the amount of delay customers experience. A service level constraint may take several forms, including a constraint on the probability of customer delay not exceeding a specified threshold, or a constraint on expected delay not exceeding a certain maximum. service-level constraints are managerial decisions that typically reflect either a position in the marketplace that a firm would like to take or contractual obligations that a firm has negotiated with its customers. We assume that all firms choose the same type of service-level constraints since they are in the same industry.

Delay costs can reflect either direct or indirect costs. Direct costs are penalties incurred by the firm due to delays experienced by its customers (e.g., payments to customers to compensate for the total time that they spend in the system) or indirect costs due to loss of customer goodwill. Hence, delay costs are not unlike backorder costs, common in inventory settings (Zipkin 2000). The use of delay costs and service levels are both common in the literature; see for example Dewan and Mendelson (1990), Mendelson and Whang (1990), Ha (1998, 2001), Allon and Federgruen (2007, 2008), and the references therein.

In this study, we consider the case where a unit delay cost h_i is incurred for each unit of time a customer spends in the system (time either in the queue or in service in steady state) and the objective is to minimize the long run expected delay cost. Moreover, we consider the case where service level is expressed in terms of a probability that delay in the system for each customer, which we define as the sum of waiting time in the queue and time in service, does not exceed a specified threshold. This measure is consistent with service levels used elsewhere in the literature; see for example Allon and Federgruen (2007, 2008), among others. We also consider service levels expressed in terms of waiting time in the queue alone. Service level measured in terms of total time in system is appropriate in applications such as computing, telecommunication, and manufacturing where customers are concerned about the total fulfillment of their orders/service requests. Service level measured in terms of waiting time is appropriate in settings, such as call centers and other service systems, where customers are particularly sensitive to time spent in the queue.

Let $z_i(\mu_i)$ denote the expected total cost incurred by firm i given a service rate μ_i (for stability, we assume that $\lambda_i/\mu_i < 1$). Let W_i, a random variable, denote the delay (waiting time in the queue + service time) that a customer of firm i experiences and $P(W_i \le w_0)$ the probability that customer delay does not exceed w_0 where $w_0 \ge 0$ (we will consider later the case where service level is expressed in terms of waiting time). The problem faced by firm i can then be stated as follows

$$\text{Minimize } z_i(\mu_i) = c\mu_i + \frac{h_i\lambda_i}{\mu_i - \lambda_i} \tag{1}$$

$$\text{subject to } P(W_i \le w_0) = 1 - e^{-(\mu_i-\lambda_i)w_0} \ge \alpha_i, \quad \lambda_i/\mu_i \le 1. \tag{2}$$

The objective function in the above optimization problem consists of two terms: a capacity cost term and a delay cost term, where the decision variable is the capacity level of firm i as determined by the service rate μ_i. The formulation captures two important special cases: (i) the case where $\alpha_i = 0$ for all $i \in \mathcal{N}$ and (ii) the case where $h_i = 0$ for all $i \in \mathcal{N}$. The first corresponds to a pure cost-based formulation with no constraints on service levels, while the second corresponds to a service level-based formulation with no delay costs. In the absence of service-level constraints, the optimal capacity level μ_i^* can be obtained from the first order condition of optimality, since z_i is convex in μ_i, as

$$\mu_i^* = \lambda_i + \sqrt{\frac{h_i\lambda_i}{c}}. \tag{3}$$

In systems with service-level constraints but no delay costs, the optimal capacity level is given by the smallest μ_i that satisfies inequality (Equation 2). This leads to the following optimal capacity level

$$\mu_i^* = \lambda_i + \frac{\ln(\frac{1}{1-\alpha_i})}{w_0}. \tag{4}$$

Surprisingly, the buffer capacity $\frac{\ln(\frac{1}{1-\alpha_i})}{w_0}$ is independent of the demand rate, a result of the fact that the delay distribution depends on λ_i and μ_i through $\mu_i - \lambda_i$ only. This feature is also present if service levels are specified in terms of expected delay. If we let α_i now denote the threshold on the maximum expected delay, then the service level constraint is given by $\frac{1}{\mu_i - \lambda_i} \le \alpha_i$. This leads to an

optimal capacity given by $\mu_i^* = \lambda_i + \frac{1}{\alpha_i}$. Note that the buffer capacity is again independent of the demand rate.

In both cases, the optimal capacity is the sum of two components. The first corresponds to the demand rate, λ_i (since all demand must be satisfied) while the second corresponds to *buffer* capacity that increases in either the ratio $\frac{h_i\lambda_i}{c}$ or the service level α_i. The expressions in Equations (3) and (4) are not new. Similar expressions have been derived elsewhere; see for example Kleinrock (1976), Allon and Federgruen (2008), and Hassin and Haviv (2003).

In the general case, with both delay costs and service-level constraints, the optimal capacity level is given by

$$\mu_i^* = \lambda_i + \eta_i, \tag{5}$$

where

$$\eta_i = \max\left\{\frac{\ln(\frac{1}{1-\alpha_i})}{w_0}, \sqrt{\frac{h_i\lambda_i}{c}}\right\}. \tag{6}$$

Substituting μ_i^* in Equation (1), we obtain the optimal expected cost for firm i as

$$z_i^* = c(\lambda_i + \eta_i) + \frac{h_i\lambda_i}{\eta_i}.$$

This leads to a total system cost of $z^*_{1,\ldots,n} = \sum_{i\in\mathcal{N}} z_i^*$. In systems where $\sqrt{\frac{h_i\lambda_i}{c}} \geq \frac{\ln(\frac{1}{1-\alpha_i})}{w_0}$ for all $i \in \mathcal{N}$, the optimal cost simplifies to $z_i^* = c\lambda_i + 2\sqrt{h_i\lambda_i c}$. This leads to a total system cost, $z^*_{1,\ldots,n}$, given by $z^*_{1,\ldots,n} = c\sum_{i\in\mathcal{N}}\lambda_i + 2\sum_{i\in\mathcal{N}}\sqrt{h_i\lambda_i c}$. In the case of identical firms, with $\lambda_i = \lambda$ and $h_i = h$ for all $i \in \mathcal{N}$, the optimal total cost reduces to $z^*_{1,\ldots,n} = cn\lambda + 2n\sqrt{h\lambda c}$, and the total capacity in the system to $\sum_{i\in\mathcal{N}}\mu_i^* = n(\lambda + \sqrt{\frac{h\lambda}{c}})$. As we can see, both the optimal cost and the optimal buffer capacity in the system increase linearly in the number of firms n. Similar observations can be made for systems in which $\sqrt{\frac{h\lambda}{c}} \leq \frac{\ln(\frac{1}{1-\alpha})}{w_0}$. That is, in this case too, both the optimal cost and the optimal buffer capacity in the system increase linearly in n when the firms have identical cost, service level, and demand parameters.

Next, we consider the case where the service level is specified in terms of waiting time in the queue. Let Q_i, a random variable, denote the time a customer of firm i spends waiting in the queue before service starts and let $P(Q_i \leq q_0)$ be the probability that customer waiting time does not exceed q_0 where $q_0 \geq 0$. The problem faced by firm i can then be restated as

$$\text{Minimize } z_i(\mu_i) = c\mu_i + \frac{h_i\lambda_i}{\mu_i - \lambda_i} \tag{7}$$

$$\text{subject to } P(Q_i \leq q_0) = 1 - \frac{\lambda_i}{\mu_i}e^{-(\mu_i-\lambda_i)q_0} \geq \alpha_i, \quad \lambda_i/\mu_i \leq 1. \tag{8}$$

In the absence of service-level constraints, the optimal capacity level is given by Equation (3). In systems with service-level constraints but no delay costs, the optimal capacity level is given by the smallest μ_i that satisfies inequality (Equation 8) or, equivalently, the optimal capacity is the solution to the following equation

$$\ln(\lambda_i q_0) + \lambda_i q_0 - \ln(1-\alpha_i) = \ln(\mu_i q_0) + \mu_i q_0. \tag{9}$$

Unfortunately, there is no explicit solution for the above equation. However, we are able to show the following important result.

Lemma 1. *Let $\bar{\mu}_i(\alpha_i, \lambda_i)$ be the solution to Equation (9) and $\bar{\eta}_i(\alpha_i, \lambda_i) = \bar{\mu}_i(\alpha_i, \lambda_i) - \lambda_i$, the amount of buffer capacity ($\bar{\mu}_i(\alpha_i, \lambda_i) > \lambda_i$). Then, given α_i, $\bar{\eta}_i(\alpha_i, \lambda_i)$ is nondecreasing in λ_i, with $\bar{\eta}_i(\alpha_i, \lambda_i)/\lambda_i$ being nonincreasing in λ_i.*

(The proof of this and of all subsequent results can be found in the Appendix A).

This lemma indicates that, in contrast to the case where service level is delay-based, here buffer capacity is increasing in demand, although the rate of increase is less than one. This result is due to the fact that the delay is exponentially distributed with the rate of the buffer capacity, while the waiting is not exponentially distributed. As we will see in the next section, this significantly affects the benefit derived from capacity sharing.

4. Systems with Capacity Sharing

In this section, we consider the scenario where the firms decide to form a coalition and invest in a single shared facility (a joint venture) from which the demand of all the firms will then be satisfied. We assume that the rules governing the joint venture (as negotiated by members of the coalition) require that the choice of capacity, in the form of a service rate, for the shared facility takes into account the demand levels of each member of the coalition, their delay costs, and their service level requirements. In particular, we assume that the service rate is chosen by the managers of the joint venture so that it minimizes the total cost

for the coalition (the sum of expected delay costs experienced by customers of all the firms and the cost of capacity) and satisfies all service-level constraints. We assume that all members of the coalition are truthful in their reporting of their demand rates, delay costs, and service levels. We assume throughout that, although independent, the firms are not competitors so that their demands are exogenously determined and are not affected by decisions made by any of the firms.

The assumption of full information applies to settings where the information is public and can be independently verified by all the firms. For example, delay penalties and service level guarantees could be publicly advertised by the firms themselves as part of their marketing strategy. In some cases, delay penalties and service levels may also adhere to well-known industry standards. In settings where delay costs are directly incurred by the shared facility (e.g., the shared facility is responsible for handling delay penalty payments to the customers), firms would also need to provide the shared facility with the correct delay costs. Similarly, service levels must be known to the shared facility if contractual agreements with the customers regarding service levels are handled directly by the shared facility. Demand rates are in most cases verifiable since demand would eventually be satisfied from the shared facility. Firms can be induced to disclose their true demand rates by imposing high penalties if the originally reported rates are higher than the realized rates (measured over a sufficiently long period of time) once the facility is in operation. The assumption of full information is of course applicable to the case where the firms are all subdivisions of a single large firm.

We refer to the service rate in the shared facility from which the demand of all firms is satisfied as $\mu_{\mathcal{N}}$ (from heretofore, we shall index parameters associated with a set of firms with the name of that set while parameters associated with individual firms with the name of the firm). Because the superposition of independent Poisson processes is also a Poisson process, the demand process at the shared facility is Poisson with rate $\sum_{i\in\mathcal{N}}\lambda_i$. Similarly, because the work content for each customer regardless of its firm is exponentially distributed, the processing time at the shared facility is a random variable $X_{\mathcal{N}} = Y/\mu_{\mathcal{N}}$ with the exponential distribution and mean $1/\mu_{\mathcal{N}}$. We assume that customers regardless of their firm affiliation are served in a FCFS fashion. Hence, the system with the shared facility behaves again as an M/M/1 queue.

4.1. Capacity Optimization

First, we consider the case where the service level is specified in terms of delay (see section 4.3 for a discussion of waiting time-based service levels). We assume that the terms of the joint venture between the participating firms in the coalition require that the shared facility invests in capacity so as to minimize the total cost to the coalition, while satisfying the service level constraint of each firm. The total cost to the coalition consists of the sum of capacity cost and expected delay cost (experienced by customers of all the firms over the long run). Satisfying the service-level constraints of all the firms requires satisfying the highest of these service-level constraints. If we let $z_{\mathcal{N}}(\mu_{\mathcal{N}})$ denote total system cost and let $W_{\mathcal{N}}$, a random variable, refer to customer delay, then the capacity optimization problem can be stated as follows:

$$\text{Minimize } z_{\mathcal{N}}(\mu_{\mathcal{N}}) = c\mu_{\mathcal{N}} + \frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{\mu_{\mathcal{N}} - \sum_{i\in\mathcal{N}}\lambda_i} \tag{10}$$

$$\text{subject to } P(W_{\mathcal{N}} \le w_0) = 1 - e^{-(\mu_{\mathcal{N}} - \sum_{i\in\mathcal{N}}\lambda_i)w_0} \ge \alpha_{\mathcal{N}},$$
$$\sum_{i\in\mathcal{N}} \lambda_i/\mu_{\mathcal{N}} \le 1,$$

where $\alpha_{\mathcal{N}} = \max(\alpha_1, \ldots, \alpha_n)$. Then, the optimal capacity is given by

$$\mu^*_{\mathcal{N}} = \sum_{i\in\mathcal{N}}\lambda_i + \eta_{\mathcal{N}},$$

where

$$\eta_{\mathcal{N}} = \max\left\{\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0}, \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}\right\}.$$

Similar to the system without capacity sharing, the optimal capacity consists of two components. The first corresponds to the total demand rate, while the second to buffer capacity which, in this case, increases in either the sum of the ratios $\frac{h_i\lambda_i}{c}$ or the maximum service level $\alpha_{\mathcal{N}}$. The results are similar if the service level is specified in terms of a threshold on expected delay. In that case the optimal buffer capacity is given by $\eta_{\mathcal{N}} = \max\{\frac{1}{\alpha_{\mathcal{N}}}, \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}\}$.

The following theorem shows that by investing in a shared facility, the firms are able to reduce total cost in the system while investing in less capacity.

THEOREM 1. *$z^*_{\mathcal{N}} \le z^*_{1,\ldots,n}$ and $\mu^*_{\mathcal{N}} \le \sum_{i=1}^n \mu^*_i$, where $z^*_{\mathcal{N}}$ is the optimal cost in the shared facility.*

The potential magnitude of the savings from capacity sharing can be more easily seen in a system with

identical firms where $\alpha_i = \alpha$, $h_i = h$, and $\lambda_i = \lambda$ for all $i \in \mathcal{N}$. Consider the case where $\sqrt{\frac{h\lambda}{c}} \geq \frac{\ln(\frac{1}{1-z})}{w_0}$. This leads to $\mu^*_{\mathcal{N}} = n\lambda + \sqrt{\frac{nh\lambda}{c}}$, $z^*_{\mathcal{N}} = cn\lambda + 2\sqrt{cnh\lambda}$, and $E(W^*_{\mathcal{N}}) = \sqrt{\frac{c}{nh\lambda}}$ from which we can observe that both buffer capacity and expected delay, and consequently delay cost, are reduced by a factor of a square root of n (relative to those observed in the case of no capacity sharing). In the case where $\sqrt{\frac{nh\lambda}{c}} \leq \frac{\ln(\frac{1}{1-z})}{w_0}$, we have $\mu^*_{\mathcal{N}} = n\lambda + \frac{\ln(\frac{1}{1-z})}{w_0}$, $z^*_{\mathcal{N}} = c(n\lambda + \frac{\ln(\frac{1}{1-z})}{w_0}) + \frac{nh\lambda w_0}{\ln(\frac{1}{1-z})}$, and $E(W^*_{\mathcal{N}}) = \frac{w_0}{\ln(\frac{1}{1-z})}$. Here, the magnitude of savings on capacity is even larger with buffer capacity reduced by a factor of n, but expected delay remains unchanged from the case without capacity sharing.

4.2. Cost Sharing

We have so far showed that capacity sharing is system-optimal. However, whether or not it is also optimal for individual firms depends on how the cost for the shared facility would be allocated among the firms. A firm would prefer the shared facility if the sum of its share of capacity cost and its long run expected delay cost is lower than the cost it would incur without capacity sharing. Moreover, in many settings, the choice is not just between a single facility shared among all firms or facilities operated individually by each firm. There may instead be a range of facility sharing options. For example, a firm may find it more advantageous to share capacity with only a subset of the firms. This could lead firms to form groupings around multiple smaller shared facilities. A single shared facility would be preferred by all firms only if there exists a cost allocation under which the firms are better off than under any other capacity sharing arrangement, including operating individual facilities. Hence, it is desirable that the cost allocation for the shared facility would be designed so that it deters firms from breaking away and engaging in other facility sharing arrangements.

The problem of determining whether or not there exists a cost allocation scheme under which firms prefer to share a single facility to any other facility sharing configuration can be formulated as a *cooperative game* among the independent firms in the set $\mathcal{N}$. Consistent with standard terminology from cooperative game theory, let us refer to the subset of firms $\mathcal{J} \subseteq \mathcal{N}$ as *coalition* $\mathcal{J}$ and to the set $\mathcal{N}$, the largest coalition, as the *grand coalition*. A cooperative game is then defined by a characteristic function which specifies the value associated with each coalition $\mathcal{J}$. In our context, this corresponds to the total expected cost associated with a subset of firms $\mathcal{J}$ sharing a single facility. We refer to this cost as $z^*_{\mathcal{J}}$, where $z^*_{\mathcal{J}} \equiv z_{\mathcal{J}}(\mu^*_{\mathcal{J}})$. A vector $\phi = (\phi_1, \ldots, \phi_n)$ is called an allocation rule if ϕ_i corresponds to the portion of total expected cost in the grand coalition that is incurred by firm i. If $\sum_{i=1}^{n} \phi_i = z^*_{\mathcal{N}}$, then the allocation rule is said to be efficient. An allocation rule is said to be individually rational if $\phi_i \leq z^*_i$ and to be stable for a coalition J if $\sum_{i\in\mathcal{J}} \phi_i \leq z^*_{\mathcal{J}}$. An allocation is said to be a member of the core if it satisfies the following inequalities:

$$\sum_{i\in\mathcal{J}} \phi_i \leq z^*_{\mathcal{J}}, \quad \forall \mathcal{J} \subseteq \mathcal{N}, \tag{11}$$

$$\sum_{i\in\mathcal{N}} \phi_i = z^*_{\mathcal{N}}. \tag{12}$$

When an allocation rule is in the core, no subset of players would want to secede from the grand coalition and form smaller coalitions, including being on their own. Hence, the existence of an allocation rule that is in the core (the core is nonempty) is sufficient in our context to show that it is optimal for all the firms to share a single facility. This single facility is a superior arrangement to any other arrangement that may involve a set of partially pooled facilities shared among multiple subsets of the firms.

In addition to the requirement of being in the core, it is desirable for an allocation rule to be perceived as *fair*. In general, a fair allocation is one that assigns a higher portion of total cost to firms whose membership in the coalition contribute more to total cost. In particular, everything else being equal, firms with higher demand rates, higher delay costs, or higher service levels should pay a greater portion of total cost. In what follows, we first describe a general class of allocation rules that have the properties of being in the core and satisfy the above intuitive notions about fairness (for a more extensive discussion of fairness in cost allocation rules see Moulin 1995). We then provide an example of a simple allocation rule that belongs to this general class of rules.

Consider first systems without service-level constraints. Then, $z^*_{\mathcal{J}} = c\sum_{j\in\mathcal{J}} \lambda_j + 2\sqrt{c\sum_{j\in\mathcal{J}} h_j\lambda_j}$ for all $\mathcal{J} \subseteq \mathcal{N}$, from which we can easily verify that the cooperative game is a *submodular* game. That is, we have $z^*_{\mathcal{J}\cup\mathcal{T}} + z^*_{\mathcal{J}\cap\mathcal{T}} \leq z^*_{\mathcal{J}} + z^*_{\mathcal{T}}$ for all $\mathcal{J}, \mathcal{T} \subseteq \mathcal{N}$. This is important because it is well known that a submodular cooperative game admits a core cost allocation rule. In particular, the Shapley value is one such allocation (Shapley 1971); see also section 5. Let $\hat{\phi} = (\hat{\phi}_1, \ldots, \hat{\phi}_n)$ denote a cost allocation rule that is in the core. Then, it is clear that under $\hat{\phi}$ the cost

allocated to firm i must be greater than its expected delay cost, that is, $\hat{\phi}_i > h_i \frac{\lambda_i}{\sqrt{\frac{\sum_{j\in\mathcal{N}} h_j\lambda_j}{c}}}$ (otherwise, it would be optimal to exclude firm i from the grand coalition since it only adds cost).

Given any such allocation rule $\hat{\phi}$, consider now the following allocation rule for systems with service-level constraints:

$$\phi_i = \frac{h_i\lambda_i}{\eta_{\mathcal{N}}} + c\lambda_i + \gamma_i, \tag{13}$$

where

$$\gamma_i = \hat{\phi}_i - \left(\frac{h_i\lambda_i}{\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}} + c\lambda_i\right) \quad \text{if } \frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0} \le \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}} \tag{14}$$

and, otherwise $\left(\text{if } \frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0} > \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}\right)$,

$$\gamma_i = \begin{cases} \frac{1}{|\mathcal{I}_{max}|}\left(c\eta_{\mathcal{N}} - c\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}\right) + \hat{\phi}_i - \left(\frac{h_i\lambda_i}{\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}} + c\lambda_i\right) & \text{if } i \in \mathcal{I}_{max}, \\ \hat{\phi}_i - \left(\frac{h_i\lambda_i}{\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}} + c\lambda_i\right) & \text{if } i \notin \mathcal{I}_{max}, \end{cases} \tag{15}$$

where $\mathcal{I}_{max} \equiv \{i : \alpha_i = \max(\alpha_1, \ldots, \alpha_n)\}$ and $|A|$ is the cardinality of set A. Note that this cost allocation rule is modified from the core cost allocation rule for systems without service-level constraints by penalizing the firms with the highest service levels.

THEOREM 2. *The capacity cost sharing rule* $\phi = (\phi_1, \ldots, \phi_n)$ *defined in Equations* (13)–(15) *is in the core. That is, under cost allocation* ϕ*, no subset of the firms in* $\mathcal{N}$ *has an incentive to secede from the grand coalition.*

Note that under this cost allocation rule, $\gamma_i + c\lambda_i \ge 0$ for all $i \in \mathcal{N}$, although it is not guaranteed that $\gamma_i \ge 0$, as this depends on the specifics of $\hat{\phi}_i$. Below we provide an example of a simple allocation that does guarantee that $\gamma_i \ge 0$ (a non-negative γ_i implies that each firm pays for its own delay cost, the cost of capacity or which it is directly responsible, $c\lambda_i$, and a fraction of buffer capacity γ_i).

LEMMA 2. *Let* $\hat{\phi}_i = \frac{h_i\lambda_i}{\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}} + c\lambda_i + c\frac{h_i\lambda_i}{\sum_{i\in\mathcal{N}} h_i\lambda_i}\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$ *for all* $i \in \mathcal{N}$. *Then the allocation* $\hat{\phi} = (\hat{\phi}_1, \ldots, \hat{\phi}_n)$ *is in the core for systems without service-level constraints.*

Consequently, and by virtue of Theorem 2, the cost allocation rule for systems with service-level constraints specified below is in the core. Let

$$\phi_i = \frac{h_i\lambda_i}{\eta_{\mathcal{N}}} + c\lambda_i + \gamma_i, \tag{16}$$

where

$$\gamma_i = \left(\hat{\phi}_i - \left(\frac{h_i\lambda_i}{\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}} + c\lambda_i\right)\right) = c\frac{h_i\lambda_i}{\sum_{i\in\mathcal{N}} h_i\lambda_i}\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}} \quad \text{if } \frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0} \le \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}} \tag{17}$$

and, otherwise $\left(\text{if } \frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0} > \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}\right)$,

$$\gamma_i = \begin{cases} \frac{1}{|\mathcal{I}_{max}|}\left(c\eta_{\mathcal{N}} - c\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}\right) + c\frac{h_i\lambda_i}{\sum_{i\in\mathcal{N}} h_i\lambda_i}\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}} & \text{if } i \in \mathcal{I}_{max}, \text{ and} \\ c\frac{h_i\lambda_i}{\sum_{i\in\mathcal{N}} h_i\lambda_i}\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}} & \text{if } i \notin \mathcal{I}_{max}. \end{cases} \tag{18}$$

Under the above allocation rule, each firm (i) incurs its own delay cost, $\frac{h_i\lambda_i}{\eta_{\mathcal{N}}}$ and (ii) a portion of total capacity cost, $c\lambda_i + \gamma_i$. The portion of total capacity cost has itself two parts: (a) an amount proportional to the firm's demand rate that can be directly attributed to each firm (this amount corresponds to the minimum cost needed to satisfy demand from this firm) and (b) a portion of the cost of buffer capacity. This portion is nondecreasing in the demand rate, delay cost, and service level of each firm. If $\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0} \le \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$, this fraction is proportional to the firms' demand-weighted delay costs. If $\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0} > \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$ (the case where the service level constraint is more restrictive), firms with the highest service level determine the service level requirement for the entire system. Therefore, these firms are treated differently to ensure that each of them is allocated a portion of the cost that is sufficiently high so that other firms do not break away from the coalition and to satisfy fairness requirements. This allocation appears to be consistent with those

observed in practice, where combinations of volume-based and capacity/service level-based fees are common; see for example Gans and Zhou (2003, 2007) and Aksin et al. (2008).

REMARK 1. In general, a simple proportional cost allocation policy may not be in the core. For example, consider the case with pure service-level constraints and with one of the firms requiring a much higher service level than the rest (the extreme case being only one of the firms requiring a service level). Then, clearly, all but one of the firms prefer not to join the grand coalition.

4.3. The Case of Waiting Time-Based Service Levels

In this section, we consider the case where the service level is specified in terms of waiting time. Let $Q_{\mathcal{N}}$, a random variable, refer to customer waiting time in the shared facility. Then, the capacity optimization problem can be stated as

$$\begin{aligned}&\text{Minimize } z_{\mathcal{N}}(\mu_{\mathcal{N}}) = c\mu_{\mathcal{N}} + \frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{\mu_{\mathcal{N}} - \sum_{i\in\mathcal{N}} \lambda_i}\\ &\text{subject to } P(Q_{\mathcal{N}} \le q_0) = 1 - \frac{\sum_{i\in\mathcal{N}} \lambda_i}{\mu_{\mathcal{N}}} e^{-(\mu_{\mathcal{N}} - \sum_{i\in\mathcal{N}} \lambda_i)q_0} \ge \alpha_{\mathcal{N}},\\ &\qquad \sum_{i\in\mathcal{N}} \lambda_i/\mu_{\mathcal{N}} \le 1, \end{aligned} \tag{19}$$

where $\alpha_{\mathcal{N}} = \max(\alpha_1, \ldots, \alpha_n)$.

Similar to the case without capacity sharing, in the absence of service-level constraints, the optimal capacity level is given by $\sum_{i\in\mathcal{N}} \lambda_i + \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$. In systems with service-level constraints but no delay costs, the optimal capacity level is given by the smallest $\mu_{\mathcal{N}}$ that satisfies the service level constraint or, equivalently, the optimal capacity is the solution to the following equation

$$\ln(\lambda_{\mathcal{N}} q_0) + \lambda_{\mathcal{N}} q_0 - \ln(1-\alpha_{\mathcal{N}}) = \ln(\mu_{\mathcal{N}} q_0) + \mu_{\mathcal{N}} q_0, \tag{20}$$

where $\lambda_{\mathcal{N}} = \sum_{i=1}^{n} \lambda_i$. Let $\bar{\mu}_{\mathcal{N}}(\alpha_{\mathcal{N}}, \lambda_{\mathcal{N}})$ be the solution to the above equation and $\bar{\eta}_{\mathcal{N}}(\alpha_{\mathcal{N}}, \lambda_{\mathcal{N}}) = \bar{\mu}_{\mathcal{N}}(\alpha_{\mathcal{N}}, \lambda_{\mathcal{N}}) - \lambda_{\mathcal{N}}$, the associated amount of buffer capacity (note that $\bar{\mu}_{\mathcal{N}} > \lambda_{\mathcal{N}}$). In contrast to the case where service level is specified in terms of delay, the amount of buffer capacity is not invariant to $\lambda_{\mathcal{N}}$ and is indeed increasing in $\lambda_{\mathcal{N}}$. This leads to the following result.

THEOREM 3. *Capacity sharing may not be beneficial and it is possible for* $z^*_{\mathcal{N}} > z^*_{1,\ldots,n}$ *and* $\mu^*_{\mathcal{N}} > \sum_{i=1}^{n} \mu^*_i$.

The above result can be proven using a counter-example. Consider the case where $h_i = 0$ for all i and $\alpha_1 > 0$ but $\alpha_i = 0$ for all $i \neq 1$. For the case without capacity sharing, we have $\bar{\eta}_i(0, \lambda_i) = 0$ for $i \neq 1$. However, $\eta_1(\alpha_1, \lambda_1) > 0$ and is given by the solution to

$$-\ln(1-\alpha_1) = \ln(1 + \frac{z}{\lambda_1}) + zq_0.$$

For the system with a single shared facility, we have $\bar{\eta}_{\mathcal{N}}(\alpha_1, \lambda_{\mathcal{N}})$, where $\bar{\eta}_{\mathcal{N}}(\alpha_1, \lambda_{\mathcal{N}})$ is the solution to

$$-\ln(1-\alpha_1) = \ln(1 + \frac{z}{\lambda_{\mathcal{N}}}) + zq_0.$$

Then, it is clear that $\bar{\eta}_{\mathcal{N}}(\alpha_1, \lambda_{\mathcal{N}}) > \bar{\eta}_1(\alpha_1, \lambda_1)$ and, consequently, $z_{\mathcal{N}} = c\bar{\eta}_{\mathcal{N}}(\alpha_1, \lambda_{\mathcal{N}}) + c\lambda_{\mathcal{N}} > \sum_{i\in\mathcal{N}} \bar{z}_i = c\bar{\eta}_1(\alpha_1, \lambda_1) + c\lambda_{\mathcal{N}}$. This is due to fact that buffer capacity is increasing in the demand rate, with the demand rates of different firms having marginal effects on the buffer capacity.

Hence, surprisingly, capacity sharing may not be even beneficial when service-level constraints are imposed on waiting time in the queue even under the M/M/1 setting. This means that in contrast to the common sense that risk pooling is beneficial, capacity sharing should be taken with caution when the service-level constraints are imposed on waiting time in the queue.

Notice that without service-level constraints, the cooperative game can be shown as a submodular game and the core exists. As we can see that the existence of the core for systems when the service-level constraints are imposed on delay is due to the buffer capacity for the service-level constraints is independent of the total demand rate, that is, when the service-level constraints are active, the optimal buffer capacity level is determined by the firm with the highest service level only. In this case, capacity sharing always lowers the buffer capacity. However, when the service-level constraints are imposed on waiting time, in general the core may not exist. This is due to that capacity sharing may lead to higher buffer capacity as we have shown in the above example since the optimal buffer capacity level may depend on both the service-level constraints and the total demand rate. In particular, capacity sharing for a firm with high service level requirement but low demand rate and a firm with low service level requirement but high demand rate could be detrimental.

Although capacity sharing is not always beneficial in general, it is in the case where the firms have identical service level constraint and $\alpha_i = \alpha$ for all i.

THEOREM 4. *If* $\alpha_i = \alpha$, *then* $z^*_{\mathcal{N}} \le \bar{z}^*_{1,\ldots,n}$ *and* $\bar{\mu}^*_{\mathcal{N}} \le \sum_{i=1}^{n} \bar{\mu}^*_i$.

Consider the following cost allocation scheme.

$$\phi_i = \frac{h_i\lambda_i}{\bar{\eta}_{\mathcal{N}}} + c\lambda_i + \gamma_i, \tag{21}$$

where

$$\gamma_i = \frac{h_i\lambda_i}{\sum_{i\in\mathcal{N}} h_i\lambda_i} c\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}} \text{ if } \bar{\eta}_{\mathcal{N}}(\alpha,\lambda_{\mathcal{N}}) \le \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}} \tag{22}$$

and, otherwise $\left(\text{if } \eta_{\mathcal{N}}(\alpha,\lambda_{\mathcal{N}}) > \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}\right)$,

$$\gamma_i = \frac{\lambda_i}{\lambda_{\mathcal{N}}} c\bar{\eta}_{\mathcal{N}}(\alpha,\lambda_{\mathcal{N}}). \tag{23}$$

As stated in the following theorem, above allocation scheme is in the core.

THEOREM 5. *If* $\alpha_i = \alpha$ *for all* i, *then the cost allocation rule* $\phi = \pm(\phi_1, \ldots, \phi_n)$ *as specified in Equations* (21)–(23) *is in the core. That is, under this cost allocation, no subset of the firms in* $\mathcal{N}$ *has an incentive to secede from the grand coalition.*

In summary, if the service-level constraints are imposed on total delay, then capacity sharing is always beneficial and we can identify a cost allocation scheme that is in the core. However, if the service-level constraints are imposed on waiting times, then capacity sharing may not be beneficial because buffer capacity is no longer invariant to total demand. To our knowledge, this is a new result in the literature on capacity pooling in queueing systems. The result also suggests that capacity sharing should be used with caution and there might be settings where capacity sharing involving only a subset of the firms may be preferable to the grand coalition.

5. Systems with Service Priorities

We have so far assumed that customers in a shared facility, regardless of their firm affiliation, are served on a FCFS basis. This policy is simple to implement and evaluate and has the appearance of fairness. However, it is not system-optimal when there are multiple customer classes with different delay costs or different service level requirements. For example, for a system without service level requirements but with different delay costs for different customers classes, the so-called $c\mu$ rule is known to be optimal; see, for instance, Jaiswal (1968) and Klimov (1974). Under the $c\mu$ rule, customers are assigned priorities based on the product of their delay costs and their service rates (in our setting, this means that a higher service priority would be given to customers with higher delay costs). In the presence of service level requirements, the optimal priority policy is more complicated and must account for the interaction between delay costs and service levels, as well as other parameters.

In this section, we extend our analysis to settings where a priority policy is used and investigate whether or not, under an optimal priority policy, there is a cost allocation that is in the core. The analysis of systems with priorities is notoriously difficult and, to our knowledge, there are no results in the literature regarding the nature of the optimal priority policy (in the presence of both delay costs and service level requirements) and no known closed form expressions for performance evaluation for systems that operate under such a policy. Also, to our knowledge, there are no results regarding the existence of the core for queueing systems that operate under a priority policy, optimal, or otherwise. Other than assuming that a priority policy is used, the assumptions of the model we consider are the same as those of our original model described in sections 3 and 4.

In order to analyze the associated cooperative game, we resort to an indirect approach, the so-called achievable region method. Without loss of generality, we assume that $h_1 \ge h_2 \ge \cdots \ge h_n > 0$. We assume that the shared system operates under an optimal mixed preemptive priority policy (the case without preemption can be similarly analyzed). Note that for two strict preemptive priority policies $\mathcal{P}_1$ and $\mathcal{P}_2$, if at the beginning of each busy period, we use policy $\mathcal{P}_1$ with probability $1-\beta$ and policy $\mathcal{P}_2$ with probability β, the resulting policy is a mixed preemptive priority policy. Because a strict priority policy is a special case of a mixed priority policy, an optimal mixed priority policy would always perform at least as well as an optimal strict priority policy.

We assume that each firm is subject to a service level constraint on its expected delay (Unfortunately, other forms of service levels are substantially more difficult to analyze). Based on Coffman and Mitrani (1980), for coalition $\mathcal{J}$ given capacity μ (for $\mu > \lambda_{\mathcal{J}}, \lambda_{\mathcal{J}} = \sum_{i\in\mathcal{J}} \lambda_i$), the feasible region for the vector of the expected delay in $(E[W_i], i \in \mathcal{J})$ under any mixed preemptive priority

policy can be described by the following polyhedron:

$$\begin{aligned} &\sum_{i\in\mathcal{V}} \lambda_i E[W_i] \geq \frac{\lambda_{\mathcal{V}}}{\mu-\lambda_{\mathcal{V}}}, \text{ for all } \mathcal{V}\subseteq\mathcal{J}, \\ &E[W_i]\geq 0, \text{ for all } i\in\mathcal{J}. \end{aligned} \tag{24}$$

Suppose that the service level constraint for each firm i is given by $E[W_i] \leq w_i$, that is, firm i requires its expected delay not to be more than w_i. We can readily show that the problem of jointly determining the optimal capacity and the expected delays under an optimal priority policy for coalition $\mathcal{J}$ corresponds to the following optimization problem.

$$z^*_{\mathcal{J}} = \min_{E[W_i],i\in\mathcal{J};\mu} \sum_{i\in\mathcal{J}} h_i\lambda_i E[W_i] + c\mu \tag{25}$$

$$\begin{aligned} \text{subject to } &\sum_{i\in\mathcal{V}} \lambda_i E[W_i] \geq \frac{\lambda_{\mathcal{V}}}{\mu-\lambda_{\mathcal{V}}}, \text{ for all } \mathcal{V}\subseteq\mathcal{J}, \\ &0\leq E[W_i]\leq w_i, i\in\mathcal{J}, \\ &\mu>\lambda_{\mathcal{J}}. \end{aligned}$$

Let $E[W^P_{i,\mathcal{J}}], i\in\mathcal{J}$ and $\mu^P_{\mathcal{J}}$ be the optimal solution to Equation (25).

Noting that the above problem is a convex optimization problem, it can be reformulated into an equivalent problem using the Lagrangian method (see Bertsekas 1999). Let θ_i be the Lagrange multiplier for the service level constraint $E[W_i] \leq w_i$, for all $i\in\mathcal{J}$. Then, the optimal solution to the following problem is also optimal for the original problem in Equation (25):

$$\begin{aligned} \hat{z}^*_{\mathcal{J}} = &\max_{\theta_i\geq 0, i\in\mathcal{J}} \min_{E[W_i],i\in\mathcal{J},\mu} \sum_{i\in\mathcal{J}} (h_i+\theta_i)\lambda_i E[W_i] + c\mu \\ &- \sum_{i\in\mathcal{J}} \theta_i w_i \end{aligned} \tag{26}$$

$$\begin{aligned} \text{subject to } &\sum_{i\in\mathcal{V}} \lambda_i E[W_i] \geq \frac{\lambda_{\mathcal{V}}}{\mu-\lambda_{\mathcal{V}}}, \text{ for all } \mathcal{V}\subseteq\mathcal{J}, \\ &E[W_i]\geq 0, \text{ for all } i\in\mathcal{J}, \\ &\mu>\lambda_{\mathcal{J}}. \end{aligned}$$

Based on proposition 5.2.1 of Bertsekas (1999), we can show that $z^*_{\mathcal{J}} = \hat{z}^*_{\mathcal{J}}$, that is, there is no duality gap. Note that the Lagrange multiplier θ_i can be viewed as the delay cost induced by the corresponding service level constraint for firm i. Hence, the presence of service-level constraints might affect the optimal priority policy.

In the following proposition, we provide conditions under which the optimal policy is a strict priority policy. Let $\theta^*_{i,\mathcal{J}}$ be the optimal Lagrangian multiplier for firm i in coalition $\mathcal{J}$. Let $|\mathcal{J}|$ denote the cardinality of $\mathcal{J}$. The Lagrangian problem implies the following result.

Proposition 1. *For any permutation $\xi = (\xi_1, \ldots, \xi_{|\mathcal{J}|})$ of $\mathcal{J}$ such that $h_{\xi_1} + \theta^*_{\xi_1,\mathcal{J}} > h_{\xi_2} + \theta^*_{\xi_2,\mathcal{J}} > \cdots > h_{\xi_{|\mathcal{J}|}} + \theta^*_{\xi_{|\mathcal{J}|},\mathcal{J}}$, then a strict priority policy is optimal and firm ξ_i has a higher priority over firm ξ_j for any $i<j$.*

However, if there exists a subset A such that $h_i + \theta^*_{i,\mathcal{J}} = h_j + \theta^*_{j,\mathcal{J}}$ for all $i,j\in A$ and $A\in\mathcal{J}$, then a strict priority policy may not be optimal and the optimal policy may be a mixed priority policy. When Proposition 1 holds, it is interesting to note the $c\mu$ rule may not be optimal and the optimal priority order is determined by the values $h_i + \theta^*_{i,\mathcal{J}}, i\in\mathcal{J}$ (although we do not have an explicit expression for the parameters $\theta^*_{i,\mathcal{J}}$, the $\theta^*_{i,\mathcal{J}}$s can be easily computed since the dual problem is a concave optimization problem (see proposition 5.1.2 of Bertsekas 1999).

Next, we show that the optimal cost in a shared system is submodular in the set of firms involved under an optimal mixed priority policy.

Theorem 6. *The cooperative game is a submodular game, that is, $z^*_{\mathcal{J}\cup\mathcal{T}} + z^*_{\mathcal{J}\cap\mathcal{T}} \leq z^*_{\mathcal{J}} + z^*_{\mathcal{T}}$ for all $\mathcal{J},\mathcal{T}\subseteq\mathcal{N}$.*

As we mentioned in section 4.2, a submodular cooperative game admits a core cost allocation rule. The Shapley value is one such allocation. Under the Shapley value, firm i is allocated a fraction of total cost specified by

$$\phi_i = \sum_{\mathcal{J}\subseteq\mathcal{N}\setminus\{i\}} \frac{|\mathcal{J}|!(n-|\mathcal{J}|-1)!}{n!}[z^*_{\mathcal{J}\cup\{i\}} - z^*_{\mathcal{J}}], \quad i=1,\ldots,n.$$

In order to show that the Shapley value allocates each firm its own delay cost and a fraction of capacity cost, we need to show ϕ_i can be expressed as

$$\phi_i = h_i\lambda_i E[W^P_{i,\mathcal{N}}] + \beta_i, \tag{27}$$

where the first term is the expected delay cost and β_i is the capacity cost for firm i, with $\beta_i\geq 0$.

Theorem 7. *The capacity sharing cost allocation scheme defined in Equation (27) is in the core, with each firm incurring its own delay cost and a positive proportion of the total capacity cost, that is, $\beta_i\geq 0, i\in\mathcal{N}$.*

The proof follows from first noting that we can always express the Shapley value allocation ϕ_i as the sum of two terms: $h_i\lambda_i E[W^P_{i,\mathcal{N}}]$ (which is the optimal solution to Equation (25) for the grand coalition $\mathcal{N}$) and β_i, where β_i can be either positive or negative. To do so, note that $\sum_{i=1}^n \beta_i = c\mu^P_{\mathcal{N}}$. This is because $z^*_{\mathcal{N}} = \sum_{i=1}^n \phi_i = \sum_{i=1}^n [h_i\lambda_i E[W^P_{i,\mathcal{N}}] + \beta_i] = \sum_{i=1}^n h_i\lambda_i E[W^P_{i,\mathcal{N}}] + c\mu^P_{\mathcal{N}}$. Also, note that, by virtue of the submodularity of the cooperative game we have $\sum_{i\in\mathcal{J}} \phi_i \le z^*_{\mathcal{J}}$ for any sub-coalition $\mathcal{J}$. In other words, any subset of the firms are better off receiving the cost allocation under the grand coalition than receiving the cost allocation in a sub-coalition on their own. This also means that a firm that joins a subset of firms must make a positive payment toward capacity cost, in addition to incurring its own delay cost. Otherwise, the other firms would be better off without it (a firm that joins a sub-coalition increases the workload of the sub-coalition; therefore, it either increases delay for everyone or increases capacity cost). Consequently, each member of any coalition must be responsible for a non-negative fraction of capacity cost or, equivalently, we must have $\beta_i > 0$, with $\beta_i = \phi_i - h_i\lambda_i E[W^P_{i,\mathcal{N}}]$.

In contrast to systems under the FCFS policy, the cooperative game continues to be submodular for systems where the service level constraint involves the expected waiting time in the queue, and there exists a capacity cost sharing rule that is in the core; see the end of the proof of Theorem 7 in Appendix A. This is due to the fact that under a priority policy the quality of service received by different firms can be differentiated while this is not possible under the FCFS policy.

We conclude this section by noting that our analysis can be extended to systems that operate under a nonpreemptive priority policy. See for example Shanthikumar and Yao (1992) for a treatment of nonpreemptive priority policies using the achievable region method; see also the end of the proof of Theorem 7 in Appendix A for more details.

6. Systems with Heterogeneous Work Contents and General Arrival and Service Times

In this section, we examine the impact of relaxing some of the assumptions we have made so far and consider (i) systems with heterogeneous work contents and (ii) systems where the customer interarrival times and processing times are not necessarily exponentially distributed. Our objective here is not to provide a comprehensive analysis, which is outside the scope of this study, but rather to offer some insights and evaluate the extent to which results we obtained under these assumptions would continue to hold. More importantly, our objective is to highlight settings where capacity sharing may not be beneficial and/or settings where identifying a cost allocation that is in the core is not possible for systems under the FCFS policy. For such settings, we provide conditions for the stability of smaller sub-coalitions. Exact analysis for the general systems is difficult. Therefore, to obtain these insights, we rely throughout on approximations that have been extensively used in the literature. Throughout this section, we restrict our analyses to the case of pure delay costs, although some of the results can be extended to systems with service-level constraints, for example, for systems with service priorities.

6.1. Capacity Sharing among Firms with Heterogeneous Work Contents

In the models described in sections 3, 4, and 5, we assumed that work contents are homogeneous across customers from different firms, so that processing times are identically distributed regardless of firm affiliation. In this section, we consider a system where the work contents associated with customers vary from firm to firm.

Customer processing time is still a random variable of the form Y_i/μ, for firm i, but Y_i is now exponentially distributed with mean l_i (recall that, in the original model, $l_i = 1$ for all $i \in \mathcal{N}$). Customers are still served on a FCFS basis. In the shared system, with n firms sharing a single facility, service times are no longer exponentially and identically distributed. Instead the distribution of service time X is a mixture of exponential distributions (i.e., the distribution is hyper-exponential) with mean $E(X) = \sum_{i\in\mathcal{N}} p_i l_i/\mu_{\mathcal{N}}$ and second moment $E(X^2) = 2\sum_{i\in\mathcal{N}} p_i l_i^2/\mu^2_{\mathcal{N}}$, where $p_i = \lambda_i/\sum_{i\in\mathcal{N}}\lambda_i$. The resulting squared coefficient of variation $C_s^2 = \frac{E(X^2)-E(X)^2}{E(X)^2}$ is strictly greater than 1 ($C_s^2 = 1$ is the case where firms have identical work contents) and increases with the differences in the amount of work content of different firms.

Without capacity sharing, the problem faced by firm i can be stated as

$$\text{minimize } z_i(\mu_i) = c\mu_i + \frac{h_i\lambda_i}{\mu_i/l_i - \lambda_i}$$
$$\text{subject to } \lambda_i l_i/\mu_i \le 1,$$

whereas with capacity sharing under the FCFS policy, the problem can be stated as

$$\text{minimize } z_{\mathcal{N}}(\mu_{\mathcal{N}}) = \sum_{i\in\mathcal{N}} h_i\lambda_i\left[\frac{(1+C_s^2)E(X)}{2(1-\rho)} + \frac{l_i}{\mu_{\mathcal{N}}}\right] + c\mu_{\mathcal{N}}$$
$$\text{subject to } \rho = \frac{\sum_{i\in\mathcal{N}}\lambda_i l_i}{\mu_{\mathcal{N}}} \le 1.$$

The optimization problems in both cases are convex and therefore can be easily solved numerically. However, a closed form analytical solution is difficult to obtain in the case of the shared system because the shared facility no longer behaves like an M/M/1 queue and, instead, behaves like an M/G/1 queue. Note that expected delay now depends on both the mean and the variance of service time via the coefficient of variation C_s.

In Appendix B, we provide a numerical example showing that capacity sharing may not be optimal in this case. Moreover, we also show that sub-coalitions could be optimal in this case. In the following proposition, we provide a sufficient condition for capacity sharing to be beneficial.

THEOREM 8. *Capacity sharing is beneficial if*

$$\left(\frac{h_i}{l_i} - \frac{h_j}{l_j}\right)(l_i - l_j) \ge 0, \quad \forall i \ne j. \tag{28}$$

In particular, if $h_i/l_i = h_j/l_j$ *for all i,j, then the capacity cost allocation scheme* $c\lambda_i + \frac{h_i\lambda_i l_i}{\sum_{k=1}^{n} h_k\lambda_k l_k}\sqrt{\frac{\sum_{k=1}^{n} h_k\lambda_k l_k}{c}}$ *for firm i is in the core.*

To our knowledge, the above result is new to the literature. The result states that for capacity sharing to be beneficial a firm with a larger (smaller) work content should also have higher (lower) value of $\frac{h_i}{l_i}$, that is, the delay cost rate h_i should increase at least proportionally to l_i. We note that Whitt (1999) also shows that pooling demand streams with different characteristics could be harmful. He shows this for settings without capacity optimization. van Dijk and Sluis (2008) also show that pooling two M/M/1 systems into an M/M/2 system may not be beneficial if the mean service times for two M/M/1 systems are different. They also provide a sufficient condition under which pooling is beneficial.

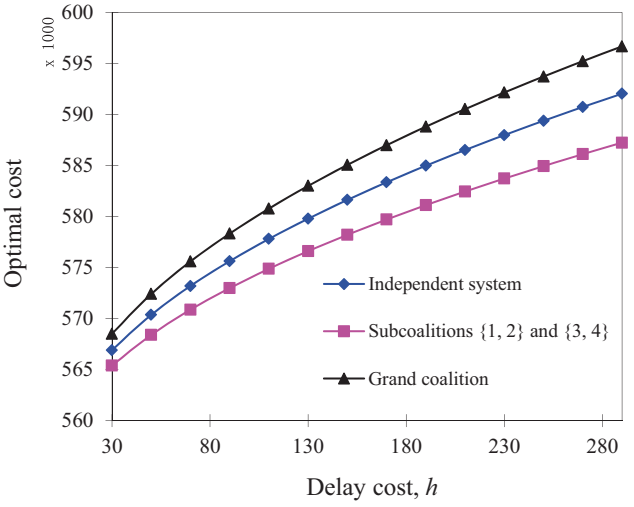
The case such that $h_i/l_i = h_j/l_j$ for all i, j corresponds to settings where delay cost rates are proportional to work content, so that firms that bring more work to the system per customer are also those that are more sensitive to delay. This prevents situations where a firm with a small work content but a high delay cost rate leads the shared facility to invest in a capacity level under which the facility utilization is low to mitigate the delay cost of that firm. In that case, the other firms may do better by excluding the high delay cost rate/low work content firm from the coalition. Note that the condition in the theorem is independent of the demand rates. This points to the dominant role that differences in work content and delay cost rates play in determining whether or not capacity sharing is beneficial. When the core does not exist, then the above result suggests we shall pool firms with similar $\frac{h_i}{l_i}$ together, which is also observed in Figure 1.

6.2. Capacity Sharing among Firms with General Demand and Processing Times

We consider systems where customer inter-arrival times for each firm $i \in \mathcal{N}$ are independent and identically distributed (i.e., arrivals form a renewal process) with mean $1/\lambda_i$ and coefficient of variation C_{a_i}. Customer processing times are independent, identically distributed, and described by a random variable of the form Y/μ, where Y has a mean equal to one and coefficient of variation C_s. The parameter μ is again a scaling factor that corresponds to the service rate. Exact analysis is difficult for the system under a FCFS policy. Therefore, to obtain insights we rely on approximations that have been extensively used in the literature.

In systems without capacity sharing, each independent facility can thus be modeled as a $GI/G/1$ queue. To obtain an explicit expression for the expected delay cost, we rely on an approximation that is

Figure 1 The Benefit of Partial Pooling ($c = 50$, $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 20$, $l_1 = l_2 = 2.5$, $l_3 = 100$, $l_4 = 450$)



asymptotically correct when the demand rates are high (i.e., when $\lambda_i \to \infty$). In particular, we approximate the expected number of customers at firm i, given capacity level μ_i, as follows

$$E[L_i(\mu_i)] \approx \sigma_i^2 \frac{\lambda_i}{\mu_i - \lambda_i}, \tag{29}$$

where $\sigma_i = \sqrt{\frac{C_{a_i}^2 + C_s^2}{2}}$. Motivation and supporting arguments for the approximation can be found in Harrison (1985) and more recently in Bassambo et al. (2012) and the references therein. The problem faced by each firm can then be restated as

$$\text{Minimize } z_i(\mu_i) \approx \frac{h_i \lambda_i}{\mu_i - \lambda_i} \sigma_i^2 + c\mu_i,$$

subject to $\lambda_i/\mu_i \leq 1$. This leads to an optimal capacity given by $\mu_i^* = \lambda_i + \sigma_i \sqrt{\frac{h_i \lambda_i}{c}}$, and corresponding optimal cost

$$z_i^* = c\lambda_i + 2\sigma_i \sqrt{c h_i \lambda_i}. \tag{30}$$

Bassambo et al. (2012) show that this capacity is asymptotically optimal when the demand rate is high (i.e., $\lambda_i \to \infty$). It also reduces to the optimal capacity for the $M/M/1$ case (in that case, $\sigma_i = 1$). Note that the above expressions capture now explicitly the effect of both demand and processing time variability.

For systems with capacity sharing, the analysis is more complicated since the superposition of renewal processes is not necessarily a renewal process. To handle this difficulty, we approximate superposed renewal processes by a renewal process whose coefficient of variation is obtained via a two-moment approximation; see Albin (1984) and Whitt (1982). In particular, we approximate the arrival process to a facility shared by the n firms by a renewal process with rate $\sum_{i \in \mathcal{N}} \lambda_i$ and coefficient of variation $C_{a_\mathcal{N}}^2 = \sum_{i \in \mathcal{N}} \frac{\lambda_i {C_{a_i}}^2}{\sum_{i=1}^n \lambda_i}$. The capacity optimization problem can be stated as

$$\text{Minimize } z_\mathcal{N}(\mu_\mathcal{N}) \approx c\mu_\mathcal{N} + \sigma_\mathcal{N}^2 \frac{\sum_{i \in \mathcal{N}} h_i \lambda_i}{\mu_\mathcal{N} - \sum_{i \in \mathcal{N}} \lambda_i},$$

subject to $\sum_{i \in \mathcal{N}} \lambda_i/\mu_\mathcal{N} \leq 1$, where $\sigma_\mathcal{N} = \sqrt{\frac{C_{a_\mathcal{N}}^2 + C_s^2}{2}}$. Hence, the optimal capacity is given by $\mu_\mathcal{N}^* = \sum_{i \in \mathcal{N}} \lambda_i + \sigma_\mathcal{N} \sqrt{\frac{\sum_{i \in \mathcal{N}} h_i \lambda_i}{c}}$ and the optimal cost is given by

$$z_\mathcal{N}^* = c \sum_{i \in \mathcal{N}} \lambda_i + 2\sigma_\mathcal{N} \sqrt{c \sum_{i \in \mathcal{N}} h_i \lambda_i}. \tag{31}$$

Observation 1. *Capacity sharing can lead to higher total cost in the system. That is, it is possible to have* $z_\mathcal{N}^* > \sum_{i \in \mathcal{N}} z_i^*$.

To see why the observation is true, consider an example with two firms, where firm 1 has negligible unit delay cost ($h_1 \approx 0$) but high variability ($\sigma_1 \gg 0$) while firm 2 has high unit delay cost ($h_2 \gg 0$) but negligible variability ($\sigma \approx 0$). Then, without capacity sharing, each firm would invest in negligible buffer capacity as either the unit delay cost is negligible or variability is negligible. When the firms share the same facility, the overall variability is positive. Therefore, there is congestion which, in turn, leads to delay costs being incurred by customers of firm 2, requiring an investment in buffer capacity.

We note that Whitt (1999) also shows that pooling demand streams with different characteristics could be harmful. He shows this for settings without capacity optimization. Our results show that this effect is also present when capacity is optimized. We also show it for the specific setting of section 6.1 where variability is driven by differences in work contents.

Although capacity sharing is not always beneficial, it is still possible to identify plausible ranges of parameter values for which capacity sharing is beneficial. A necessary and sufficient condition for capacity sharing to be preferable is given by

$$\sigma_\mathcal{N} \sqrt{\sum_{i=1}^n h_i \lambda_i} \leq \sum_{i \in \mathcal{N}} \sigma_i \sqrt{h_i \lambda_i}, \tag{32}$$

which follows from a direct comparison of the total buffer capacity cost in systems with and without capacity sharing. A sufficient condition for the inequality in Equation (32) to hold is for

$$\frac{\sigma_\mathcal{N}}{\sigma_i} \leq \frac{\sum_{i \in \mathcal{N}} \sqrt{h_i \lambda_i}}{\sqrt{\sum_{i=1}^n h_i \lambda_i}} \quad \text{for all } i \in \mathcal{N}.$$

In other words, capacity sharing is beneficial if the variability associated with capacity sharing among all the firms does not exceed a certain threshold relative to the variability associated with firms investing in their own independent facilities. The following theorem provides a more explicit sufficient condition under which capacity sharing is beneficial.

THEOREM 9. *Capacity sharing is beneficial if, for each pair of firms i and j, $(h_i - h_j)(\sigma_i - \sigma_j) \geq 0$. In other words, capacity sharing is beneficial if firms with higher delay costs also have higher demand variability. In particular, the cost allocation defined by*

$$\phi_i = \frac{h_i\lambda_i}{\sqrt{\frac{\sum_{k=1}^{n} h_k\lambda_k}{c}}} + c\lambda_i + \frac{h_i\lambda_i}{\sum_{k=1}^{n} h_k\lambda_k}\sigma_{\mathcal{N}}\sqrt{\frac{\sum_{k=1}^{n} h_k\lambda_k}{c}} \qquad (33)$$

is in the core if $\sigma_{\mathcal{N}}^2/\sigma_{\mathcal{J}}^2 \leq \gamma_{\mathcal{J}}$ where $\gamma_{\mathcal{J}} = \frac{\sum_{k\in\mathcal{N}} h_k\lambda_k}{\sum_{k\in\mathcal{J}} h_k\lambda_k}$ for all $\mathcal{J} \subseteq \mathcal{N}$.

To our knowledge, the condition in Theorem 9 is also new to the literature. Note that the variability parameter here is affected by differences in inter-arrival time variability. Hence, this result complements the result of section 6.1 where the variability parameter was determined by differences in work contents. The condition is independent of the firms' demand rates. In particular, if $\sigma_i = \sigma_j$ for all i and j, then capacity sharing is always beneficial. The case of exponential inter-arrival times and processing times of course satisfies this condition, but it is also satisfied by a broader class of problems.

We conclude this section by noting that for systems that operate under the FCFS policy, in addition to highlighting conditions under which the core is empty, the results presented here provide conditions under which capacity sharing continues to be beneficial. These conditions can be used to identify settings where stable sub-coalitions might arise. They can also be used to guide decisions for whether or not individual firms should invest in shared facilities with specific firms.

7. Summary and Concluding Comments

In this study, we studied the benefit of capacity sharing among independent firms when capacity is optimized to take into account the costs and service level requirements of the different firms. We characterized settings under which capacity sharing is beneficial and proposed cost allocation schemes that guarantee that the grand coalition is stable for systems operate under either a FCFS policy or an optimal priority policy. However, for some systems that operate under a FCFS policy, we found that capacity sharing may not be beneficial. This is the case when service levels are specified in terms of waiting time instead of total delay and when the firms are heterogeneous in their characteristics, including work contents, service time variability, and delay costs. For these cases, we characterize conditions, under which capacity sharing may still be beneficial for a subset of the firms. These conditions provide insights into the characteristics of firms that would benefit from sharing capacity. In Appendix B, we also numerically explored the benefits of capacity sharing under different system parameters and obtained some insights correspondingly.

There are various avenues for future research. It would be useful to generalize the analysis to systems where capacity sharing involves the sharing of a network of facilities whose capacities can be individually optimized. It would also be useful to treat more general cost functions, including those that may exhibit economies or diseconomies of scale. Finally, it would be useful to consider systems where capacities can be varied only in discrete amounts. We expect the treatment of each of these cases to be considerable more difficult and to require perhaps the development of alternative approaches or approximations.

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Appendix A. Proofs

PROOF OF LEMMA 1. Noting that $\bar{\eta}_i(\alpha_i, \lambda_i)$ is the solution to

$$-\ln(1-\alpha_i) = \ln((\lambda_i + z)q_0) + (\lambda_i + z)q_0 - \ln(\lambda_i q_0) - \lambda_i q_0,$$

or equivalently

$$-\ln(1-\alpha_i) = \ln(1 + \frac{z}{\lambda_i}) + zq_0,$$

it is to verify that $\bar{\eta}_i(\alpha_i, \lambda_i)$ is nondecreasing in λ_i for all $\lambda_i > 0$.

We show that $\bar{\eta}_i(\alpha_i, \lambda_i)/\lambda_i$ is nonincreasing in λ_i by contradiction. Suppose that $\bar{\eta}_i(\alpha_i, x)/x$ is strictly increasing in x in the neighborhood of λ_i for some

$\lambda_i > 0$, that is, $\bar{\eta}_i(\alpha_i, \lambda_i + \delta)/(\lambda_i + \delta) > \bar{\eta}_i(\alpha_i, \lambda_i)/\lambda_i$ for any small enough $\delta > 0$. Then we must have

$$\ln(1 + \frac{\bar{\eta}_i(\alpha_i, \lambda_i + \delta)}{\lambda_i + \delta}) + (\lambda_i + \delta)\frac{\bar{\eta}_i(\alpha_i, \lambda_i + \delta)}{\lambda_i + \delta} q_0 > \ln(1 + \frac{\bar{\eta}_i(\alpha_i, \lambda_i)}{\lambda_i}) + \lambda_i \frac{\bar{\eta}_i(\alpha_i, \lambda_i)}{\lambda_i} q_0 = -\ln(1 - \alpha_i)$$

for all $\delta > 0$ since $\ln(1 + x)$ is an increasing function, which contradicts with the fact that

$$\ln(1 + \frac{\bar{\eta}_i(\alpha_i, \lambda_i + \delta)}{\lambda_i + \delta}) + (\lambda_i + \delta)\frac{\bar{\eta}_i(\alpha_i, \lambda_i + \delta)}{\lambda_i + \delta} q_0 = -\ln(1 - \alpha_i).$$

Hence, $\bar{\eta}_i(\alpha_i, \lambda_i)/\lambda_i$ is nonincreasing in λ_i. □

Proof of Theorem 1. To prove that $\sum_{i\in\mathcal{N}} \mu_i^* \geq \mu_{\mathcal{N}}^*$, note that

$$\begin{aligned}\sum_{i\in\mathcal{N}} \mu_i^* &= \sum_{i\in\mathcal{N}} \max\left\{\frac{\ln(\frac{1}{1-\alpha_i})}{w_0}, \sqrt{\frac{h_i\lambda_i}{c}}\right\} \\ &\geq \max\left\{\frac{\ln(\frac{1}{1-\alpha_{i_{max}}})}{w_0}, \sqrt{\frac{h_{i_{max}}\lambda_{i_{max}}}{c}}\right\} + \sum_{i\in\mathcal{N}, i\neq i_{max}} \sqrt{\frac{h_i\lambda_i}{c}} \geq \max\left\{\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0}, \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}\right\} = \mu_{\mathcal{N}}^*,\end{aligned}$$

where $i_{max} \in \{i : \alpha_i = \max(\alpha_1, \ldots, \alpha_n)\}$. The last inequality is due to that $\sqrt{\frac{h_{i_{max}}\lambda_{i_{max}}}{c}} + \sum_{i\in\mathcal{N}, i\neq i_{max}} \sqrt{\frac{h_i\lambda_i}{c}} \geq \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$. In order to prove that $z_{\mathcal{N}}^* \leq z_{1,\ldots,n}^*$, we distinguish two cases.

(1) $\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0} \leq \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$: In this case, we have

$$\begin{aligned}z_{\mathcal{N}}^* &= \frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}} + c\sum_{i\in\mathcal{N}} \lambda_i + c\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}} \\ &\leq \sum_{i\in\mathcal{N}} \left(\frac{h_i\lambda_i}{\sqrt{\frac{h_i\lambda_i}{c}}} + c\lambda_i + c\sqrt{\frac{h_i\lambda_i}{c}}\right) \\ &\leq \sum_{i\in\mathcal{N}} \left(c(\lambda_i + \eta_i) + \frac{h_i\lambda_i}{\eta_i}\right) \\ &\leq z_{1,\ldots,n}^*.\end{aligned}$$

The first inequality is due to that $\sqrt{\sum_{i\in\mathcal{N}} h_i\lambda_i c} \leq \sum_{i\in\mathcal{N}} \sqrt{h_i\lambda_i c}$. The second inequality is due to that the presence of the service-level constraints would not decrease the optimal costs for firms.

(2) $\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0} > \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$: In this case, we have

$$\begin{aligned}z_{\mathcal{N}}^* &= \frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0}} + c\sum_{i\in\mathcal{N}} \lambda_i + c\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0} \\ &\leq \frac{h_{i_{max}}\lambda_{i_{max}}}{\frac{\ln(\frac{1}{1-\alpha_{i_{max}}})}{w_0}} + c\lambda_{i_{max}} + c\frac{\ln(\frac{1}{1-\alpha_{i_{max}}})}{w_0} + \sum_{i\in\mathcal{N}, i\neq i_{max}} \left(\frac{h_i\lambda_i}{\sqrt{\frac{h_i\lambda_i}{c}}} + c\lambda_i + c\sqrt{\frac{h_i\lambda_i}{c}}\right) \\ &\leq \sum_{i\in\mathcal{N}} \left(c(\lambda_i + \eta_i) + \frac{h_i\lambda_i}{\eta_i}\right) \leq z_{1,\ldots,n}^*.\end{aligned}$$

The first inequality is due to that $\alpha_{i_{max}} = \alpha_{\mathcal{N}}$. The second inequality is due to that $\eta_{i_{max}} = \frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0}$ and the presence of the service-level constraints would not decrease the optimal costs for firms. □

Proof of Theorem 2. We distinguish two cases here.

(1) $\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0} \leq \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$. In this case, $z_{\mathcal{J}}^* \geq \sum_{i\in\mathcal{J}} \hat{\phi}_i$ since $\hat{\phi}$ is a core cost allocation rule for systems without service-level constraints (noting that the service-level constraints are inactive).

(2) $\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0} > \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$. For coalition $\mathcal{J} \subseteq \mathcal{N}\backslash\mathcal{I}_{max}$, we have

$$\begin{aligned}z_{\mathcal{J}}^* &\geq \sum_{j\in\mathcal{J}} \hat{\phi}_j = \sum_{j\in\mathcal{J}} \frac{h_j\lambda_j}{\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}} + c\sum_{j\in\mathcal{J}} \lambda_j + \sum_{j\in\mathcal{J}} \left(\hat{\phi}_j - \left(\frac{h_j\lambda_j}{\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}} + c\lambda_j\right)\right) \\ &\geq \frac{\sum_{j\in\mathcal{J}} h_j\lambda_j}{\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0}} + c\sum_{j\in\mathcal{J}} \lambda_j + \sum_{j\in\mathcal{J}} \left(\hat{\phi}_j - \left(\frac{h_j\lambda_j}{\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}} + c\lambda_j\right)\right) = \sum_{j\in\mathcal{J}} \phi_j^G.\end{aligned}$$

The first inequality is due to the fact that the optimal cost for a system with service-level constraints should be no less than that of the same system without service-level constraints and $\hat{\phi}$ is a core cost allocation rule for systems without service-level constraints.

If $\mathcal{J} \subseteq \mathcal{N}$ and $\mathcal{I}_{max} \cap \mathcal{J} \neq \emptyset$, then

$$z^*_{\mathcal{J}} = \frac{\sum_{j\in\mathcal{J}} h_j\lambda_j}{\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0}} + c\sum_{j\in\mathcal{J}}\lambda_j + c\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0}$$
$$\geq \frac{\sum_{j\in\mathcal{J}} h_j\lambda_j}{\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0}} + c\sum_{j\in\mathcal{J}}\lambda_j + \sum_{j\in\mathcal{J}}(\hat{\phi}_j - (\frac{h_j\lambda_j}{\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}} + c\lambda_j))$$
$$+ \frac{|\mathcal{J}\cap\mathcal{I}_{max}|}{|\mathcal{I}_{max}|}(c\frac{\ln(\frac{1}{1-\alpha_{\mathcal{N}}})}{w_0} - c\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}) = \sum_{j\in\mathcal{J}}\phi^G_j.$$

(Note that $\sum_{j\in\mathcal{J}}(\hat{\phi}_j - (\frac{h_j\lambda_j}{\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}} + c\lambda_j)) \leq \sqrt{c\sum_{i\in\mathcal{N}} h_i\lambda_i}$, which is due to the fact that $\sum_{j\in\mathcal{J}}\hat{\phi}_j \leq c\sum_{j\in\mathcal{J}}\lambda_j + 2\sqrt{c\sum_{j\in\mathcal{J}} h_j\lambda_j} \leq c\sum_{j\in\mathcal{J}}\lambda_j + \sqrt{c\sum_{i\in\mathcal{N}} h_i\lambda_i} + \frac{\sum_{j\in\mathcal{J}} h_j\lambda_j}{\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}}$). Consequently, the allocation rule is in the core.

Hence, ϕ^G must be in the core. In addition, we must have $\phi^G_i - \frac{h_i\lambda_i}{\eta_{\mathcal{N}}} > 0$ for all $i \in \mathcal{N}$. Otherwise it is optimal to exclude firm i from the grand coalition and as a result it contradicts with the fact that ϕ^G is in the core. □

Proof of Lemma 2. Note that

$$\sum_{i\in\mathcal{J}}\hat{\phi}_i - \left[c\sum_{i\in\mathcal{J}}\lambda_i + 2\sqrt{c\sum_{i\in\mathcal{J}} h_i\lambda_i}\right]$$
$$= 2\left[\sum_{i\in\mathcal{J}} h_i\lambda_i\sqrt{\frac{c}{\sum_{i\in\mathcal{N}} h_i\lambda_i}} - \sqrt{c\sum_{i\in\mathcal{J}} h_i\lambda_i}\right] \leq 0,$$
$$\forall \mathcal{J} \subseteq \mathcal{N}.$$

Hence, $\hat{\phi}$ in this case is a core cost allocation for the case without service-level constraints. □

Proof of Theorem 4. Noting that $\bar{\eta}_i(\alpha,\lambda_i)$ is the solution to

$$-\ln(1-\alpha) = \ln(1+\frac{z}{\lambda_i}) + zq_0,$$

and $\bar{\eta}_{\mathcal{N}}(\alpha,\lambda_{\mathcal{N}})$ is the solution to

$$-\ln(1-\alpha) = \ln(1+\frac{z}{\lambda_{\mathcal{N}}}) + zq_0,$$

it is easy to verify that

$$\frac{\sum_{i\in\mathcal{N}}\bar{\eta}_i(\alpha,\lambda_i)}{\sum_{i\in\mathcal{N}}\lambda_i} \geq \min_{i\in\mathcal{N}}\frac{\bar{\eta}_i(\alpha,\lambda_i)}{\lambda_i}.$$

Hence, since $\bar{\eta}_i(\alpha,\lambda)/\lambda$ is nonincreasing in λ for all i, we must have

$$\sum_{i\in\mathcal{N}}\bar{\eta}_i(\alpha,\lambda_i) > \bar{\eta}_{\mathcal{N}}(\alpha,\lambda_{\mathcal{N}}).$$

To prove that $\sum_{i\in\mathcal{N}}\bar{\mu}^*_i \geq \bar{\mu}^*_{\mathcal{N}}$, note that

$$\sum_{i\in\mathcal{N}}\bar{\mu}^*_i - \lambda_{\mathcal{N}} = \sum_{i\in\mathcal{N}}\max\{\bar{\eta}_i(\alpha,\lambda_i), \sqrt{\frac{h_i\lambda_i}{c}}\}$$
$$\geq \max\{\sum_{i\in\mathcal{N}}\bar{\eta}_i(\alpha,\lambda_i), \sum_{i\in\mathcal{N}}\sqrt{\frac{h_i\lambda_i}{c}}\}$$
$$\geq \max\{\bar{\eta}_{\mathcal{N}}(\alpha,\lambda_{\mathcal{N}}), \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}\} = \bar{\mu}^*_{\mathcal{N}} - \lambda_{\mathcal{N}}.$$

In order to prove that $\bar{z}^*_{\mathcal{N}} \leq \bar{z}^*_{1,\ldots,n}$, we distinguish two cases.

(1) $\bar{\eta}_{\mathcal{N}}(\alpha,\lambda_{\mathcal{N}}) \leq \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$. In this case, we have

$$\bar{z}^*_{\mathcal{N}} = \frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}} + c\sum_{i\in\mathcal{N}}\lambda_i + c\sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$$
$$\leq \sum_{i\in\mathcal{N}}(\frac{h_i\lambda_i}{\sqrt{\frac{h_i\lambda_i}{c}}} + c\lambda_i + c\sqrt{\frac{h_i\lambda_i}{c}}) \leq \sum_{i\in\mathcal{N}}(c(\lambda_i+\bar{\eta}_i) + \frac{h_i\lambda_i}{\bar{\eta}_i}) = \bar{z}^*_{1,\ldots,n}.$$

(2) $\bar{\eta}_{\mathcal{N}}(\alpha,\lambda_{\mathcal{N}}) > \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$. In this case, we have

$$\bar{z}^*_{\mathcal{N}} = \frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{\bar{\eta}_{\mathcal{N}}(\alpha,\lambda_{\mathcal{N}})} + c\sum_{i\in\mathcal{N}}\lambda_i + c\bar{\eta}_{\mathcal{N}}(\alpha,\lambda_{\mathcal{N}})$$
$$\leq \frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{\sum_{i\in\mathcal{N}}\bar{\eta}_i} + c\sum_{i\in\mathcal{N}}\lambda_i + c\sum_{i\in\mathcal{N}}\bar{\eta}_i$$
$$\leq \sum_{i\in\mathcal{N}}\left(c(\lambda_i+\bar{\eta}_i) + \frac{h_i\lambda_i}{\bar{\eta}_i}\right) = \bar{z}^*_{1,\ldots,n}.$$

where the first inequality is due to

$$\left\{\bar{\eta}_i = \max\{\eta_i(\alpha,\lambda_i), \sqrt{\frac{h_i\lambda_i}{c}}\right\}$$

and $\sum_{i\in\mathcal{N}}\bar{\eta}_i \geq \eta_{\mathcal{N}}(\alpha,\lambda_{\mathcal{N}}) > \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$. □

PROOF OF THEOREM 5. We distinguish two cases.

(1) $\bar{\eta}_{\mathcal{N}}(\alpha, \lambda_{\mathcal{N}}) \le \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$. First note that

$$z^*_{\mathcal{J}} \ge c\sum_{i\in\mathcal{J}} \lambda_i + 2\sqrt{c\sum_{i\in\mathcal{J}} h_i\lambda_i}.$$

Since $\sum_{i\in\mathcal{J}} \phi_i - \left[c\sum_{i\in\mathcal{J}} \lambda_i + 2\sqrt{c\sum_{i\in\mathcal{J}} h_i\lambda_i}\right] = 2\left[\sum_{i\in\mathcal{J}} h_i\lambda_i\sqrt{\frac{c}{\sum_{i\in\mathcal{N}} h_i\lambda_i}} - \sqrt{c\sum_{i\in\mathcal{J}} h_i\lambda_i}\right] \le 0$, we have $z^*_{\mathcal{J}} \ge \sum_{i\in\mathcal{J}} \phi_i, \forall \mathcal{J} \subseteq \mathcal{N}$. It follows that the allocation rule is in the core.

(2) $\bar{\eta}_{\mathcal{N}}(\alpha, \lambda_{\mathcal{N}}) > \sqrt{\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{c}}$. Note that

$$\frac{\sum_{i\in\mathcal{J}} h_i\lambda_i}{\bar{\eta}_{\mathcal{N}}(\alpha, \lambda_{\mathcal{N}})} + c\frac{\sum_{i\in\mathcal{J}} \lambda_i}{\lambda_{\mathcal{N}}}\bar{\eta}_{\mathcal{N}}(\alpha, \lambda_{\mathcal{N}}) \le \frac{\sum_{i\in\mathcal{J}} h_i\lambda_i}{\bar{\eta}_{\mathcal{J}}} + c\bar{\eta}_{\mathcal{J}}.$$

The inequality is due to $\bar{\eta}_{\mathcal{N}}(\alpha, \lambda_{\mathcal{N}}) \ge \bar{\eta}_{\mathcal{J}}$ and $\frac{\bar{\eta}_{\mathcal{N}}(\alpha,\lambda_{\mathcal{N}})}{\lambda_{\mathcal{N}}} \le \frac{\bar{\eta}_{\mathcal{J}}(\alpha,\lambda_{\mathcal{J}})}{\lambda_{\mathcal{J}}}$ by Lemma 1. Consequently, the allocation is in the core.

PROOF OF LEMMA 5.1. In order to utilize the achievable region method to analyze our problem, we first introduce the notion of *polymatroid*. Let $\mathcal{N} = \{1, \ldots, n\}$. A function $g: 2^{\mathcal{N}} \to \mathbb{R}$ is defined as a rank function if it has the following properties: (i) $g(\phi) = 0$; (ii) g is nondecreasing, that is, $g(\mathcal{S}) \le g(\mathcal{T})$ for any $\mathcal{S} \subseteq \mathcal{T} \subseteq \mathcal{N}$; (iii) g is supermodular, that is, $g(\mathcal{J}\cup\mathcal{T}) + g(\mathcal{J}\cap\mathcal{T}) \ge g(\mathcal{J}) + g(\mathcal{T})$. Given a subset $T \subseteq \mathcal{N}$ and a function $g: 2^{\mathcal{N}} \to \mathbb{R}$, the following polyhedron

$$\{(x_i, i \in \mathcal{T}) : \sum_{i\in\mathcal{V}} x_i \ge g(\mathcal{V}), \forall \mathcal{V} \subseteq \mathcal{T}; x_i \ge 0, i \in \mathcal{T}\}$$

is called a *polymatroid* if g is a rank function (see Edmonds 1970).

Define $\pi_i = \{1, \ldots, i\}$. Consider the following linear programming problem.

$$\min \sum_{i=1}^{n} c_i x_i$$
$$\text{subject to } \sum_{i\in\mathcal{V}} x_i \ge g(\mathcal{V}), \forall \mathcal{V} \subseteq \mathcal{N}, x_i \ge 0, i \in \mathcal{N},$$

where g is a rank function. □

LEMMA 7.1. *(Edmonds 1970). Suppose that* $c_1 \ge c_2 \ge \cdots \ge c_n$. *Then* $x^*_1 = g(\pi_1), x^*_i = g(\pi_i) - g(\pi_{i-1})$, $i = 2, \ldots, n$ *is an optimal solution to Equation (A1). Moreover, if* $c_1 > c_2 > \cdots > c_n$, *then this is the unique optimal solution.*

Hence, if we optimize a linear objective function over a polymatriod, then a greedy algorithm is optimal.

Note that the feasible region of Equation (25) is a polymatroid. Suppose that $(\xi_1, \ldots, \xi_{|\mathcal{J}|})$ is a permutation of the set of indices $\{i : i \in \mathcal{J}\}$ such that $h_{\xi_1} + \theta^*_{\xi_1,\mathcal{J}} > \cdots > h_{\xi_{|\mathcal{J}|}} + \theta^*_{\xi_{|\mathcal{J}|},\mathcal{J}}$. Let $\hat{\pi}_i = \{\xi_1, \ldots, \xi_i\}$. Based on Lemma 7.1, $\lambda_{\xi_i} E[W^P_{\xi_i,\mathcal{J}}] = \frac{\lambda_{\hat{\pi}_i}}{\mu-\lambda_{\hat{\pi}_i}} - \frac{\lambda_{\hat{\pi}_{i-1}}}{\mu-\lambda_{\hat{\pi}_{i-1}}}, i = 2, \ldots, |\mathcal{J}|$ and $\lambda_{\xi_1} W^P_{\xi_1,\mathcal{J}} = \frac{\lambda_{\hat{\pi}_1}}{\mu-\lambda_{\hat{\pi}_1}}$. The optimal priority policy for coalition $\mathcal{J}$ is a strict preemptive priority policy and the priority order is decreasing in $h_i + \theta^*_{i,\mathcal{J}}$, with the firm with the highest value of $h_i + \theta^*_{i,\mathcal{J}}$ having the highest priority, which is due to the fact that $\frac{\lambda_{\mathcal{V}}}{\mu-\lambda_{\mathcal{V}}}$ is nondecreasing and strictly supermodular in $\mathcal{V}$, and Lemma 7.1. □

PROOF OF THEOREM 6. Without loss of generality, in the following we assume that $\lambda_{\mathcal{N}} \le \frac{1}{4}\frac{h_n}{c}$. This can be accomplished by rescaling the time (i.e., using a small enough unit of time) since $\lambda_{\mathcal{N}}$ is the expected total arrival rate per unit of time and $\frac{h_n}{c}$ is unitless. Note that h_i is the delay cost incurred by firm *i for each unit of time* a customer spends in the system while c is the capacity cost per unit of service rate *per unit time*. This implies that cost is measured in money per unit time and is consistent with treatment elsewhere in the literature (see e.g., Bassambo et al. 2012, Hassin and Haviv 2003).

First, we consider the case without service-level constraints. In this case, the objective function is given by $h_1\frac{\lambda_{\pi_1\cap\mathcal{J}}}{\mu-\lambda_{\pi_1\cap\mathcal{J}}} + \sum_{i=2}^{n} h_i[\frac{\lambda_{\pi_i\cap\mathcal{J}}}{\mu-\lambda_{\pi_i\cap\mathcal{J}}} - \frac{\lambda_{\pi_{i-1}\cap\mathcal{J}}}{\mu-\lambda_{\pi_{i-1}\cap\mathcal{J}}}] + c\mu = \sum_{i=1}^{n}\gamma_i(\frac{\lambda_{\pi_i\cap\mathcal{J}}}{\mu-\lambda_{\pi_i\cap\mathcal{J}}}) + c\mu$, where $\gamma_i = h_i - h_{i+1} \ge 0$, $i = 1, \ldots, n-1$ and $\gamma_n = h_n$. As a result, the optimal capacity decision for coalition $\mathcal{J}$ is given by $\min_{\mu>\lambda_{\mathcal{J}}}[\sum_{i=1}^{n}\gamma_i(\frac{\lambda_{\pi_i\cap\mathcal{J}}}{\mu-\lambda_{\pi_i\cap\mathcal{J}}}) + c\mu]$.

Define

$$z^*(\lambda_1, \ldots, \lambda_n) = \min_{\mu>\lambda_{\mathcal{N}}}[\sum_{i=1}^{n}\gamma_i(\frac{\lambda_{\pi_i}}{\mu-\lambda_{\pi_i}}) + c\mu],$$

as the optimal cost for the grand coalition under demand rates $\lambda_1, \ldots, \lambda_n$. Next, we will show that z^* is submodular in $(\lambda_1, \ldots, \lambda_n)$. But the objective function inside the minimization operator is not submodular in $(\mu, \lambda_1, \ldots, \lambda_n)$. Instead we show it by using the envelope theorem. Let μ^* be the optimal capacity. First, it is clear that μ^* is nondecreasing in $h_1, \ldots, h_n$ since the objective function is submodular in $(h_1, \ldots, h_n, \mu)$. It

follows that $\sqrt{\frac{h_n\lambda_{\mathcal{N}}}{c}}+\lambda_{\mathcal{N}} \le \mu^* \le \sqrt{\frac{h_1\lambda_{\mathcal{N}}}{c}}+\lambda_{\mathcal{N}}$ (noting that if the holding cost rates for all firms are h_n, then $\mu^* = \sqrt{\frac{h_n\lambda_{\mathcal{N}}}{c}}+\lambda_{\mathcal{N}}$. But as $h_1, \ldots, h_n$ increase, the optimal capacity also increases and hence the upper bound of μ^* is $\sqrt{\frac{h_1\lambda_{\mathcal{N}}}{c}}+\lambda_{\mathcal{N}}$). Without loss of generality, we assume that $m,k \in \mathcal{N}$ and $m<k$. By the envelope theorem (see Milgrom and Segal 2002 for more details), we have

$$\frac{\partial z^*(\lambda_1, \ldots, \lambda_n)}{\partial \lambda_k} = \sum_{i=k}^{n} \gamma_i\left(\frac{1}{\mu^* - \lambda_{\pi_i}}\right) + \sum_{i=k}^{n} \gamma_i\left(\frac{\lambda_{\pi_i}}{(\mu^* - \lambda_{\pi_i})^2}\right).$$

Then we have the following equation:

$$\begin{aligned}\frac{\partial^2 z^*(\lambda_1, \ldots, \lambda_n)}{\partial \lambda_k \partial \lambda_m} &= -\sum_{i=k}^{n} \gamma_i \frac{1}{(\mu^* - \lambda_{\pi_i})^2}\left(\frac{\partial \mu^*}{\partial \lambda_m} - 1\right) \\ &\quad - \sum_{i=k}^{n} 2\gamma_i \frac{\lambda_{\pi_i}}{(\mu^* - \lambda_{\pi_i})^3}\left(\frac{\partial \mu^*}{\partial \lambda_m} - 1\right) \\ &\quad + \sum_{i=k}^{n} \gamma_i\left(\frac{1}{(\mu^* - \lambda_{\pi_i})^2}\right),\end{aligned}$$

since $m<k$. But $\frac{\partial \mu^*}{\partial \lambda_m} \ge \frac{\partial \sqrt{\frac{h_n\lambda_{\mathcal{N}}}{c}}}{\partial \lambda_m} + 1 = \frac{1}{2}\sqrt{\frac{h_n}{c\lambda_{\mathcal{N}}}} + 1$ based on the aforementioned properties of the optimal capacity (noting that $\sqrt{\frac{h_n\lambda_{\mathcal{N}}}{c}}+\lambda_{\mathcal{N}} \le \mu^*$ and μ^* is nondecreasing in $h_1, \ldots, h_n$). Hence, if $\frac{1}{2}\sqrt{\frac{h_n}{c\lambda_{\mathcal{N}}}} \ge 1$ or equivalently $\lambda_{\mathcal{N}} \le \frac{1}{4}\frac{h_n}{c}$, then $\frac{\partial^2 z^*(\lambda_1, \ldots, \lambda_n)}{\partial \lambda_k \partial \lambda_m} \le 0$ for all $k \ne m,k$, $m \in \mathcal{N}$. Therefore, the cooperative game must be a submodular game.

Next, we show that the optimal cost is still submodular even with service-level constraints. Suppose that $\mathcal{S}, \mathcal{T} \in \mathcal{N}$. Let $(\theta^*_{\mathcal{J}}, E[W^P_{i,\mathcal{J}}], i \in \mathcal{J}, \mu^P_{\mathcal{J}})$, $(\theta^*_{\mathcal{T}}, E[W^P_{i,\mathcal{T}}], i \in \mathcal{T}, \mu^P_{\mathcal{T}})$ be the optimal solutions for the subsets $\mathcal{J}$ and $\mathcal{T}$ for the Lagrangian problem in Equation (26), respectively. Let $\hat{\theta}$ be the vector such that under $\hat{\theta}$, the optimal solution $E[\hat{W}^P_{i,\mathcal{J}\cup\mathcal{T}}], i \in \mathcal{J}\cup\mathcal{T}$ for

$$\begin{aligned} z(\hat{\theta}, \mathcal{J}\cup\mathcal{T}) = &\min_{E[W_i], i\in\mathcal{J}\cup\mathcal{T}, \mu} \sum_{i\in\mathcal{J}\cup\mathcal{T}} (h_i + \hat{\theta}_i)\lambda_i E[W_i] \\ &+ c\mu - \sum_{i\in\mathcal{J}\cup\mathcal{T}} \hat{\theta}_i w_i \\ \text{subject to } &\sum_{i\in\mathcal{V}} \lambda_i E[W_i] \ge \frac{\lambda_{\mathcal{V}}}{\mu - \lambda_{\mathcal{V}}}, \mathcal{V} \subseteq \mathcal{J}\cup\mathcal{T}, \\ &E[W_i] \ge 0, i \in \mathcal{J}\cup\mathcal{T}, \mu > \lambda_{\mathcal{J}\cup\mathcal{T}}. \end{aligned}$$

satisfies $E[\hat{W}^P_{i,\mathcal{J}\cup\mathcal{T}}] \le w_i, i \in \mathcal{J}\cup\mathcal{T}$ and the optimal solution $E[\hat{W}^P_{i,\mathcal{J}\cap\mathcal{T}}], i \in \mathcal{S}\cap\mathcal{T}$ for

$$\begin{aligned} z(\hat{\theta}, \mathcal{J}\cap\mathcal{T}) = &\min_{E[W_i], i\in\mathcal{J}\cap\mathcal{T}, \mu} \sum_{i\in\mathcal{J}\cap\mathcal{T}} (h_i + \hat{\theta}_i)\lambda_i E[W_i] \\ &+ c\mu - \sum_{i\in\mathcal{J}\cap\mathcal{T}} \hat{\theta}_i w_i \\ \text{subject to } &\sum_{i\in\mathcal{V}} \lambda_i E[W_i] \ge \frac{\lambda_{\mathcal{V}}}{\mu - \lambda_{\mathcal{V}}}, \mathcal{V} \subseteq \mathcal{J}\cap\mathcal{T}, \\ &E[W_i] \ge 0, i \in \mathcal{J}\cap\mathcal{T}, \mu > \lambda_{\mathcal{J}\cap\mathcal{T}}. \end{aligned}$$

satisfies $E[\hat{W}^P_{i,\mathcal{J}\cap\mathcal{T}}] \le w_i, i \in \mathcal{J}\cap\mathcal{T}$ (this is possible by setting $\hat{\theta}_i$ large enough and under which the optimal capacities are large enough since as $\min_i(h_i + \hat{\theta}_i) \to \infty$, the corresponding optimal capacity also approaches ∞).

Then it is clear that

$$\begin{aligned} z^*_{\mathcal{J}} + z^*_{\mathcal{T}} &= \sum_{i\in\mathcal{J}} (h_i + \theta^*_{i,\mathcal{J}})\lambda_i E[W^P_{i,\mathcal{J}}] + c\mu^P_{\mathcal{J}} - \sum_{i\in\mathcal{J}} \theta^*_{i,\mathcal{J}} w_i \\ &\quad + \sum_{i\in\mathcal{T}} (h_i + \theta^*_{i,\mathcal{T}})\lambda_i E[W^P_{i,\mathcal{T}}] + c\mu^P_{\mathcal{T}} - \sum_{i\in\mathcal{T}} \theta^*_{i,\mathcal{T}} w_i \\ &\ge \sum_{i\in\mathcal{J}} (h_i + \hat{\theta}_i)\lambda_i E[W^P_{i,\mathcal{J}}] + c\mu^P_{\mathcal{J}} - \sum_{i\in\mathcal{J}} \hat{\theta}_i w_i \\ &\quad + \sum_{i\in\mathcal{T}} (h_i + \hat{\theta}_i)\lambda_i E[W^P_{i,\mathcal{T}}] + c\mu^P_{\mathcal{T}} - \sum_{i\in\mathcal{T}} \hat{\theta}_i w_i \\ &\ge z(\hat{\theta}, \mathcal{J}) + z(\hat{\theta}, \mathcal{T}) \ge z(\hat{\theta}, \mathcal{J}\cup\mathcal{T}) + z(\hat{\theta}, \mathcal{J}\cap\mathcal{T}) \\ &\ge z^*_{\mathcal{J}\cup\mathcal{T}} + z^*_{\mathcal{J}\cap\mathcal{T}}. \end{aligned}$$

The first inequality is due to the fact that $\hat{\theta}$ is a feasible solution to either $\max_{\theta\ge 0}[\sum_{i\in\mathcal{J}}(h_i+\theta_i)\lambda_i E[W^P_{i,\mathcal{J}}] + c\mu^P_{\mathcal{J}} - \sum_{i\in\mathcal{J}}\theta_i w_i]$ or $\max_{\theta\ge 0}[\sum_{i\in\mathcal{T}}(h_i+\theta_i)\lambda_i E[W^P_{i,\mathcal{T}}] + c\mu^P_{\mathcal{T}} - \sum_{i\in\mathcal{T}}\theta_i w_i]$; the second inequality is due to the fact that $(E[W^P_{i,\mathcal{J}}], i \in \mathcal{J}, \mu^P_{\mathcal{J}})$ is a feasible solution to

$$\begin{aligned} z(\hat{\theta}, \mathcal{J}) = &\min_{E[W_i], i\in\mathcal{J}; \mu} \sum_{i\in\mathcal{J}} (h_i + \hat{\theta}_i)\lambda_i E[W_i] + c\mu - \sum_{i\in\mathcal{J}} \hat{\theta}_i w_i \\ \text{subject to } &\sum_{i\in\mathcal{V}} \lambda_i E[W_i] \ge \frac{\lambda_{\mathcal{V}}}{\mu - \lambda_{\mathcal{V}}}, \mathcal{V} \subseteq \mathcal{J}, \\ &E[W_i] \ge 0, i \in \mathcal{J}, \end{aligned}$$

and $(E[W^P_{i,\mathcal{T}}], i \in \mathcal{T}, \mu^P_{\mathcal{T}})$ is a feasible solution to

$$\begin{aligned} z(\hat{\theta}, \mathcal{T}) = &\min_{E[W_i], i\in\mathcal{T}, \mu} \sum_{i\in\mathcal{T}} (h_i + \hat{\theta}_i)\lambda_i E[W_i] \\ &+ c\mu - \sum_{i\in\mathcal{T}} \hat{\theta}_i w_i \\ \text{subject to } &\sum_{i\in\mathcal{V}} \lambda_i E[W_i] \ge \frac{\lambda_{\mathcal{V}}}{\mu - \lambda_{\mathcal{V}}}, \mathcal{V} \subseteq \mathcal{T}, \\ &\lambda_i E[W_i] \ge 0, i \in \mathcal{T}; \end{aligned}$$

the third inequality is due to the fact that given any delay costs the optimal cost z is submodular (note that $\sum_{i\in S} \hat{\theta}_i w_i$ is a linear function of $\mathcal{J}$ and

hence $z(\hat{\theta}, \mathcal{J})$ is submodular in $\mathcal{J}$ given $\hat{\theta} \geq 0$). Finally the fourth inequality is due to the fact that $z^*_{\mathcal{J}\cup\mathcal{T}}$ and $z^*_{\mathcal{J}\cap\mathcal{T}}$ are the optimal costs and under $\hat{\theta}$, the optimal solutions for coalitions $\mathcal{J}\cup\mathcal{T}$ and $\mathcal{J}\cap\mathcal{T}$ are feasible, and hence $z^*_{\mathcal{J}\cup\mathcal{T}} \leq z(\hat{\theta}, \mathcal{J}\cup\mathcal{T})$, $z^*_{\mathcal{J}\cap\mathcal{T}} \leq z(\hat{\theta}, \mathcal{J}\cap\mathcal{T})$.

Now we extend our results to systems with the nonpreemptive priority policy. Under the optimal nonpreemptive priority policy, the optimal cost is given by

$$\tilde{z}^*(\lambda_1, \ldots, \lambda_n) = \min_{\mu > \lambda_{\mathcal{N}}} \left[\sum_{i=1}^{n} \gamma_i \left(\frac{\lambda_{\mathcal{N}}}{\mu} \frac{\lambda_{\pi_i}}{\mu - \lambda_{\pi_i}} + \frac{\lambda_{\pi_i}}{\mu} \right) + c\mu \right]$$
$$= \min_{\mu > \lambda_{\mathcal{N}}} \left[\sum_{i=1}^{n} \gamma_i \left(\frac{\lambda_{\pi_i}}{\mu - \lambda_{\pi_i}} + \frac{\lambda_{\pi_i}\lambda_{\mathcal{N}\setminus\pi_i}}{\mu(\mu - \lambda_{\pi_i})} \right) + c\mu \right].$$

Let $\tilde{\mu}^*$ be the optimal capacity. Without loss of generality, we assume that $m, k \in \mathcal{N}$ and $m < k$. By the envelope theorem, we have

$$\frac{\partial \tilde{z}^*(\lambda_1, \ldots, \lambda_n)}{\partial \lambda_k} = \sum_{i=k}^{n} \gamma_i \left(\frac{1}{\tilde{\mu}^* - \lambda_{\pi_i}} + \frac{\lambda_{\pi_i}}{(\tilde{\mu}^* - \lambda_{\pi_i})^2} \right)$$
$$+ \sum_{i=k}^{n} \gamma_i \left(\frac{\lambda_{\mathcal{N}\setminus\pi_i}}{\tilde{\mu}^*(\tilde{\mu}^* - \lambda_{\pi_i})} + \frac{\lambda_{\pi_i}\lambda_{\mathcal{N}\setminus\pi_i}}{\tilde{\mu}^*(\tilde{\mu}^* - \lambda_{\pi_i})^2} \right).$$

Based on the foregoing result,

$$\frac{\partial \tilde{z}^*(\lambda_1, \ldots, \lambda_n)}{\partial \lambda_k \partial \lambda_m} = \sum_{i=k}^{n} \gamma_i \left(-\frac{1}{(\tilde{\mu}^* - \lambda_{\pi_i})^2} \left(\frac{\partial \tilde{\mu}^*}{\partial \lambda_m} - 1 \right) \right.$$
$$+ \frac{1}{(\tilde{\mu}^* - \lambda_{\pi_i})^2} - 2\frac{\lambda_{\pi_i}}{(\tilde{\mu}^* - \lambda_{\pi_i})^3} \left(\frac{\partial \tilde{\mu}^*}{\partial \lambda_m} - 1 \right) \Bigg)$$
$$+ \sum_{i=k}^{n} \gamma_i \left(-\frac{\lambda_{\mathcal{N}\setminus\pi_i}}{(\tilde{\mu}^*)^2(\tilde{\mu}^* - \lambda_{\pi_i})} \frac{\partial \tilde{\mu}^*}{\partial \lambda_m} \right.$$
$$- \frac{\lambda_{\mathcal{N}\setminus\pi_i}}{(\tilde{\mu}^*)(\tilde{\mu}^* - \lambda_{\pi_i})^2} \left(\frac{\partial \tilde{\mu}^*}{\partial \lambda_m} - 1 \right) + \frac{\lambda_{\mathcal{N}\setminus\pi_i}}{\tilde{\mu}^*(\tilde{\mu}^* - \lambda_{\pi_i})^2}$$
$$- \frac{\lambda_{\pi_i}\lambda_{\mathcal{N}\setminus\pi_i}}{(\tilde{\mu}^*)^2(\tilde{\mu}^* - \lambda_{\pi_i})^2} \frac{\partial \tilde{\mu}^*}{\partial \lambda_m}$$
$$\left. - 2\frac{\lambda_{\pi_i}\lambda_{\mathcal{N}\setminus\pi_i}}{(\tilde{\mu}^*)(\tilde{\mu}^* - \lambda_{\pi_i})^3} \left(\frac{\partial \tilde{\mu}^*}{\partial \lambda_m} - 1 \right) \right),$$

since $m < k$. It is clear that $\sqrt{\frac{h_n \lambda_{\mathcal{N}}}{c}} + \lambda_{\mathcal{N}} \leq \tilde{\mu}^* \leq \sqrt{\frac{h_1 \lambda_{\mathcal{N}}}{c}} + \lambda_{\mathcal{N}}$ and $\frac{\partial \tilde{\mu}^*}{\partial \lambda_m} \geq \frac{\partial \sqrt{\frac{h_n \lambda_{\mathcal{N}}}{c}}}{\partial \lambda_m} + 1 = \frac{1}{2}\sqrt{\frac{h_n}{c\lambda_{\mathcal{N}}}} + 1$ based on a similar argument on the proof for systems under preemptive priority policies. Hence if $\frac{1}{2}\sqrt{\frac{h_n}{c\lambda_{\mathcal{N}}}} \geq 1$ or equivalently $\lambda_{\mathcal{N}} \leq \frac{1}{4}\frac{h_n}{c}$, then $\frac{\partial^2 \tilde{z}^*(\lambda_1, \ldots, \lambda_n)}{\partial \lambda_k \partial \lambda_m} \leq 0$ for all $k \neq m, k, m \in \mathcal{N}$. Hence, the corresponding cooperative game must be submodular. Based on Shanthikumar and Yao (1992), that is, the performance measures for an M/M/1 system under the nonpreemptive priority policy can still have an achievable region. Then the analysis for the case with service-level constraints on the expected delay can be accomplished by a similar argument as in the case for the systems with the preemptive priority policy.

Finally, suppose that the service level constraint for each firm i is given by $E[Q_i] \leq \tilde{w}_i$, that is, firm i requires its expected waiting time in the queue not to be more than $\tilde{w}_i$. We know that $E[W_i] = E[Q_i] + \frac{1}{\mu}$. We can readily show that the problem of jointly deciding on the optimal capacity and the optimal mixed priority policy for coalition $\mathcal{J}$ is specified by the solution to the following optimization problem.

$$z^{*'}_{\mathcal{J}} = \min_{E[W_i], i\in\mathcal{J}; \mu} \sum_{i\in\mathcal{J}} h_i \lambda_i E[W_i] + c\mu$$
$$\text{subject to } \sum_{i\in\mathcal{V}} \lambda_i E[W_i] \geq \frac{\lambda_{\mathcal{V}}}{\mu - \lambda_{\mathcal{V}}}, \text{ for all } \mathcal{V} \subseteq \mathcal{J},$$
$$0 \leq E[W_i] \leq \tilde{w}_i + \frac{1}{\mu}, i \in \mathcal{J}, \ \mu > \lambda_{\mathcal{J}}. \quad \text{(A1)}$$

Let $\tilde{\theta}_i$ be the Lagrange multiplier for the service level constraint $E[W_i] \leq \tilde{w}_i + \frac{1}{\mu}$, which is equivalent to $E[Q_i] \leq \tilde{w}_i$, for all $i \in \mathcal{J}$. Then, the optimal solution to the following problem is also optimal for the original problem in Equation (A1):

$$\tilde{z}^*_{\mathcal{J}} = \max_{\tilde{\theta}_i \geq 0, i\in\mathcal{J}} \min_{E[W_i], i\in\mathcal{J}, \mu} \sum_{i\in\mathcal{J}} (h_i + \theta_i)\lambda_i E[W_i] + c\mu$$
$$- \sum_{i\in\mathcal{J}} \tilde{\theta}_i (\tilde{w}_i + \frac{1}{\mu})$$
$$\text{subject to} \sum_{i\in\mathcal{V}} \lambda_i E[W_i] \geq \frac{\lambda_{\mathcal{V}}}{\mu - \lambda_{\mathcal{V}}}, \text{ for all } \mathcal{V} \subseteq \mathcal{J},$$
$$E[W_i] \geq 0, \text{ for all } i \in \mathcal{J}, \mu > \lambda_{\mathcal{J}}. \quad \text{(A2)}$$

By the envelope theorem, for $\lambda_{\mathcal{N}} \leq \frac{1}{4}\frac{h_n}{c}$ we can show that

$$\hat{z}^*(\lambda_1, \ldots, \lambda_n) = \min_{\mu > \lambda_{\mathcal{N}}} [\sum_{i=1}^{n} \gamma_i (\frac{\lambda_{\pi_i}}{\mu - \lambda_{\pi_i}}) + c\mu - \theta\frac{1}{\mu}],$$

is still submodular in $(\lambda_1, \ldots, \lambda_n)$ for any constant θ (the proof is the same as that for the previous case when $\theta = 0$). Then it is easy to show that $z^{*'}_{\mathcal{J}}$ is submodular by a similar argument as in the case when the service level is imposed on the expected delay. □

Proof of Theorem 8. First, we can show that $(\frac{h_i}{l_i} - \frac{h_j}{l_j})(l_i - l_j) \geq 0, \forall i \neq j$, which is equivalent to $(h_i l_j - h_j l_i)(l_i - l_j) \leq 0, \forall i \neq j$, imply that

$$\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i}{\sum_{i\in\mathcal{N}} h_i\lambda_i l_i} \le \frac{\sum_{i\in\mathcal{N}} \lambda_i l_i}{\sum_{i\in\mathcal{N}} \lambda_i l_i^2}. \tag{A3}$$

Then it is sufficient to show that the total expected delay cost in the independent system, given capacities $\mu_i > \lambda_i$ for $i \in \mathcal{N}$, is greater than or equal to the expected delay cost in a shared system with capacity $\mu = \sum_{i\in\mathcal{N}} \mu_i$, that is,

$$\sum_{i\in\mathcal{N}} \frac{h_i\lambda_i l_i}{\mu_i - \lambda_i l_i} \ge \frac{(\sum_{i\in\mathcal{N}} h_i\lambda_i)(\sum_{i\in\mathcal{N}} \lambda_i l_i^2)}{\mu(\mu - \sum_{i\in\mathcal{N}} \lambda_i l_i)} + \frac{\sum_{i\in\mathcal{N}} h_i\lambda_i l_i}{\mu}.$$

Then note that with Equation (A3) we have

$$\begin{aligned}
&\frac{(\sum_{i\in\mathcal{N}} h_i\lambda_i)(\sum_{i\in\mathcal{N}} \lambda_i l_i^2)}{\mu(\mu - \sum_{i\in\mathcal{N}} \lambda_i l_i)} + \frac{\sum_{i\in\mathcal{N}} h_i\lambda_i l_i}{\mu} \\
&= \frac{\mu\sum_{i\in\mathcal{N}} h_i\lambda_i l_i + \sum_{i\in\mathcal{N}} h_i\lambda_i \sum_{i\in\mathcal{N}} \lambda_i l_i^2 - \sum_{i\in\mathcal{N}} h_i\lambda_i l_i \sum_{i\in\mathcal{N}} \lambda_i l_i}{\mu(\mu - \sum_{i\in\mathcal{N}} \lambda_i l_i)} \\
&\le \frac{\sum_{i\in\mathcal{N}} h_i\lambda_i l_i}{\mu - \sum_{i\in\mathcal{N}} \lambda_i l_i}.
\end{aligned}$$

However, we have

$$\frac{\sum_{i\in\mathcal{N}} h_i\lambda_i l_i}{\sum_{i\in\mathcal{N}} \mu_i - \sum_{i\in\mathcal{N}} \lambda_i l_i} \le \max_i\{\frac{h_i\lambda_i l_i}{\mu_i - \lambda_i l_i}\} \le \sum_{i\in\mathcal{N}} \frac{h_i\lambda_i l_i}{\mu_i - \lambda_i l_i},$$

which completes the proof.

If $h_i/h_j = l_i/l_j$ for all i,j, then $\frac{(\sum_{i\in\mathcal{N}} h_i\lambda_i)(\sum_{i\in\mathcal{N}} \lambda_i l_i^2)}{\mu(\mu - \sum_{i\in\mathcal{N}} \lambda_i l_i)} + \frac{\sum_{i\in\mathcal{N}} h_i\lambda_i l_i}{\mu} = \frac{\sum_{i=1}^n h_i\lambda_i l_i}{\mu - \sum_{i\in\mathcal{N}} \lambda_i l_i}$. Arguments similar to those made in the proof of the M/M/1 case complete the proof.

Proof to Theorem 9. A sufficient condition for Equation (32) to hold is for

$$\sqrt{\frac{\sum_{i=1}^n \lambda_i\sigma_i^2}{\sum_{i=1}^n \lambda_i}} \sqrt{\sum_{i=1}^n h_i\lambda_i} \le \sum_{i=1}^n \sqrt{h_i\lambda_i\sigma_i^2},$$

or equivalently, by taking the square of both sides of the inequality, for

$$\frac{\sum_{i=1}^n \lambda_i\sigma_i^2 \sum_{i=1}^n h_i\lambda_i}{\sum_{i=1}^n \lambda_i} \le \sum_{i=1}^n h_i\lambda_i\sigma_i^2 + 2\sum_{j\ne i} \sqrt{h_i\lambda_i\sigma_i^2 h_j\lambda_j\sigma_j^2}.$$

To prove that $\sum_{i=1}^n \lambda_i(\sum_{i=1}^n h_i\lambda_i\sigma_i^2 + 2\sum_{j\ne i} \sqrt{h_i\lambda_i\sigma_i^2 h_j\lambda_j\sigma_j^2}) \ge \sum_{i=1}^n \lambda_i\sigma_i^2 \sum_{i=1}^n h_i\lambda_i$, it is sufficient to show that $\sum_{i=1}^n \lambda_i \sum_{i=1}^n h_i\lambda_i\sigma_i^2 \ge \sum_{i=1}^n \lambda_i\sigma_i^2 \sum_{i=1}^n h_i\lambda_i$. Note that

$$\sum_{i=1}^n \lambda_i \sum_{i=1}^n h_i\lambda_i\sigma_i^2 = \sum_{i=1}^n h_i\lambda_i^2\sigma_i^2 + 2\sum_{j\ne i} h_i\lambda_i\lambda_j\sigma_i^2$$

and

$$\sum_{i=1}^n \lambda_i\sigma_i^2 \sum_{i=1}^n h_i\lambda_i = \sum_{i=1}^n h_i\lambda_i^2\sigma_i^2 + 2\sum_{j\ne i} h_i\lambda_i\lambda_j\sigma_j^2.$$

Now let $a(i,j) = h_i\lambda_i\sigma_i^2\lambda_j, a(j,i) = h_j\lambda_j\sigma_j^2\lambda_i$ and $b(i,j) = h_i\lambda_i\lambda_j\sigma_j^2, b(j,i) = h_j\lambda_j\lambda_i\sigma_i^2$. Then,

$$a(i,j) + a(j,i) - b(i,j) - b(j,i) = (h_i - h_j)\lambda_i\lambda_j(\sigma_i^2 - \sigma_j^2).$$

Hence, if we have $(h_i - h_j)(\sigma_i^2 - \sigma_j^2) \ge 0$, we must also have

$$\sum_{j\ne i}(h_i - h_j)(\sigma_i^2 - \sigma_j^2) = \sum_{j\ne i} h_i\lambda_i\lambda_j\sigma_i^2 \ge \sum_{j\ne i} h_i\lambda_i\lambda_j\sigma_j^2 \ge 0.$$

Consequently, we also have $\sum_{i=1}^n \lambda_i \sum_{i=1}^n h_i\lambda_i\sigma_i^2 \ge \sum_{i=1}^n \lambda_i\sigma_i^2 \sum_{i=1}^n h_i\lambda_i$, which completes the proof.

Note that

$$\begin{aligned}
&\frac{\sum_{i\in\mathcal{J}} h_i\lambda_i}{\sum_{i=1}^n h_i\lambda_i}\sigma_{\mathcal{N}}\sqrt{\frac{\sum_{i=1}^n h_i\lambda_i}{c}} \\
&= \frac{\sum_{i\in\mathcal{J}} h_i\lambda_i}{\sum_{i=1}^n h_i\lambda_i}\sqrt{\frac{\sum_{i=1}^n \lambda_i\sigma_i^2}{\sum_{i=1}^n \lambda_i}}\sqrt{\frac{\sum_{i=1}^n h_i\lambda_i}{c}} \\
&\le \sqrt{\frac{\sum_{i\in S} \lambda_i\sigma_i^2}{\sum_{i\in\mathcal{J}} \lambda_i}}\sqrt{\frac{\sum_{i\in\mathcal{J}} h_i\lambda_i}{c}},
\end{aligned}$$

where the inequality is due to the assumption in the theorem. Hence,

$$\begin{aligned}
\sum_{i\in\mathcal{J}} \phi_i &= \sum_{i\in\mathcal{J}} \lambda_i + \sigma_{\mathcal{N}} \frac{\sum_{i\in\mathcal{J}} h_i\lambda_i}{\sqrt{\frac{\sum_{i=1}^n h_i\lambda_i}{c}}} + c\frac{\sum_{i\in\mathcal{J}} h_i\lambda_i}{\sum_{i=1}^n h_i\lambda_i}\sigma_{\mathcal{N}}\sqrt{\frac{\sum_{i=1}^n h_i\lambda_i}{c}} \\
&= \sum_{i\in\mathcal{J}} \lambda_i + 2c\frac{\sum_{i\in\mathcal{J}} h_i\lambda_i}{\sum_{i=1}^n h_i\lambda_i}\sigma_{\mathcal{N}}\sqrt{\frac{\sum_{i=1}^n h_i\lambda_i}{c}} \\
&\le \sum_{i\in\mathcal{J}} \lambda_i + 2c\sqrt{\frac{\sum_{i\in\mathcal{J}} \lambda_i\sigma_i^2}{\sum_{i\in\mathcal{J}} \lambda_i}}\sqrt{\frac{\sum_{i\in\mathcal{J}} h_i\lambda_i}{c}} \\
&= z_{\mathcal{J}}^*.
\end{aligned}$$

Appendix B. Numerical Studies

B.1. The Benefits of Capacity Sharing

In this section, we numerically investigate the benefits of capacity sharing under both the FCFS policy and the optimal preemptive priority policy.

First, we consider the case with three firms. There are three systems: the independent system; the shared system under the FCFS policy, and the shared system under the optimal preemptive priority policy. Specifically, the arrival rates are $\lambda_1 = 10\gamma$, $\lambda_2 = 20\gamma$, $\lambda_3 = 30\gamma$, and the delay costs are given by β,2β,3β, respectively. All firms subject to minimum expected delays 0.2θ, 0.2θ, 0.3θ, respectively. In the following three figures, the vertical axis denotes the percentage of cost saving of a shared system with respect to that of the independent system.

In Figure B1, we can see that as γ increases (hence all demand rates increase), then benefits of capacity sharing decrease for both the FCFS system and the priority system. This implies that capacity sharing is more effective when demand rates are small. Moreover, the benefits in are more significant when the minimum delay requirements are smaller. Hence, firms are more willing to participate capacity sharing when they have very stringent service-level constraints, which is reasonable since capacity sharing can take advantage of the risk pooling.

In Figure B2, as β increases (hence all delay costs increase), then the benefits of capacity sharing mostly have an upward trend, that is, as the delay costs increase, the benefits of capacity sharing also increase. Moreover, the benefits of capacity sharing are more significant when γ is smaller, which is consistent with Figure B1. But in priority system 2, as we can see that the benefit of capacity sharing is not monotone with respect to β. Hence, how the delay costs affect the benefit of capacity sharing should be taken with caution.

In Figure B3, as the capacity cost c increases, the benefits of capacity sharing decrease and when c is large enough these benefits converge to a constant level. This is due to that as capacity cost is large enough, then the shared system cannot take full advantage of the risk pooling effect. In addition, mostly the benefits of capacity sharing are more significant when β is larger.

Finally, in each of all these three figures, the benefit of capacity sharing for the priority system is more significant than that of the FCFS system. But the gap is not as big as the benefit of capacity sharing for the FCFS system. In summary, in general firms have more incentive to participate in capacity sharing when the delay costs are large, demand rates and capacity costs are not so large and service-level constraints are very stringent.

B.2. Numerical Examples for Capacity Sharing with Heterogeneous Work Contents

It is difficult to characterize analytically the optimal capacity in a shared facility. Therefore, it is also difficult to characterize analytically conditions under which capacity sharing is beneficial (and the core may or may not be empty). In order to get some insights, and isolate the effect of work content variability, let us consider the case where $h_i = h$ and $\lambda_i = \lambda$ for all $i \in \mathcal{N}$. The expression for C_s^2 then simplifies to $C_s^2 = 2C_l^2 + 1$ where $C_l = \sqrt{\frac{\sum_{i\in\mathcal{N}} nl_i^2 - (\sum_{i\in\mathcal{N}} l_i)^2}{(\sum_{i\in\mathcal{N}} l_i)^2}}$ is the coefficient of variation of the means of the individual work contents. The total cost in the shared system can now be expressed as

Figure B1 The Benefits of Capacity Sharing ($\beta = 1$, $c = 50$; System 1: $\theta = 1$; System 2: $\theta = 10$)

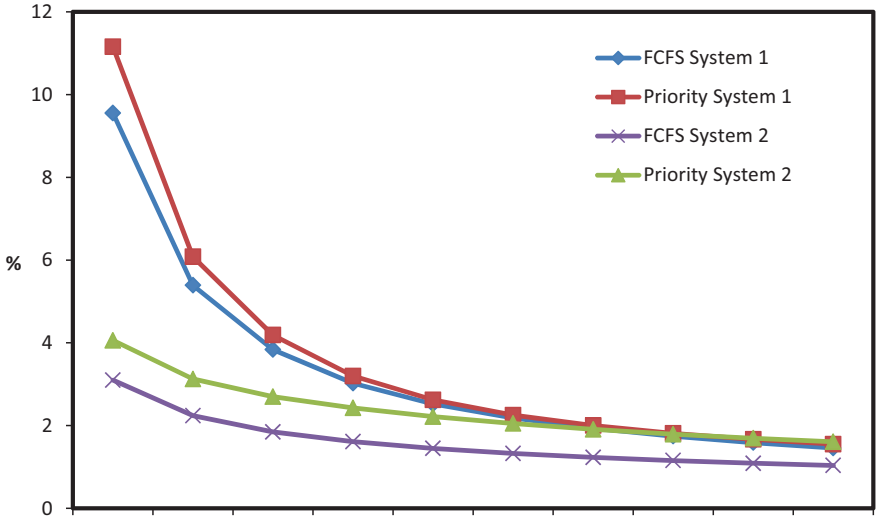


Figure B2 The Benefits of Capacity Sharing ($\theta = 1$, $c = 50$; System 1: $\gamma = 1$; System 2: $\gamma = 10$)

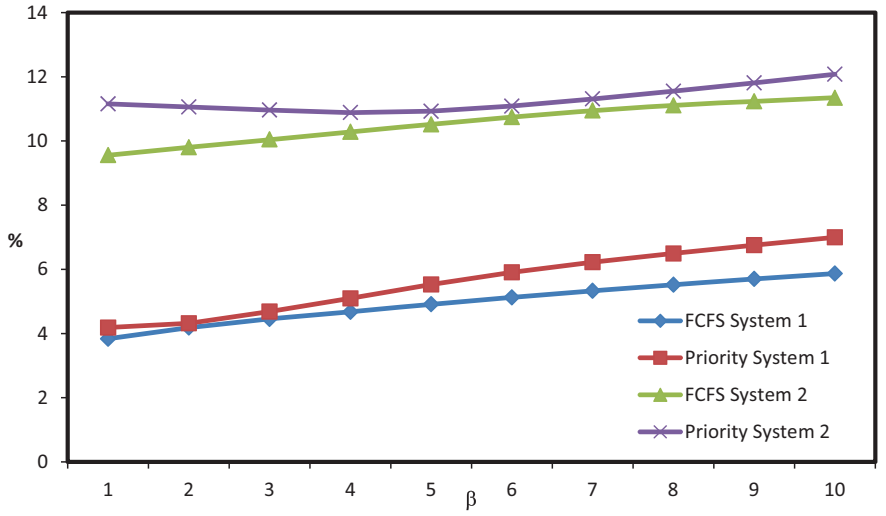
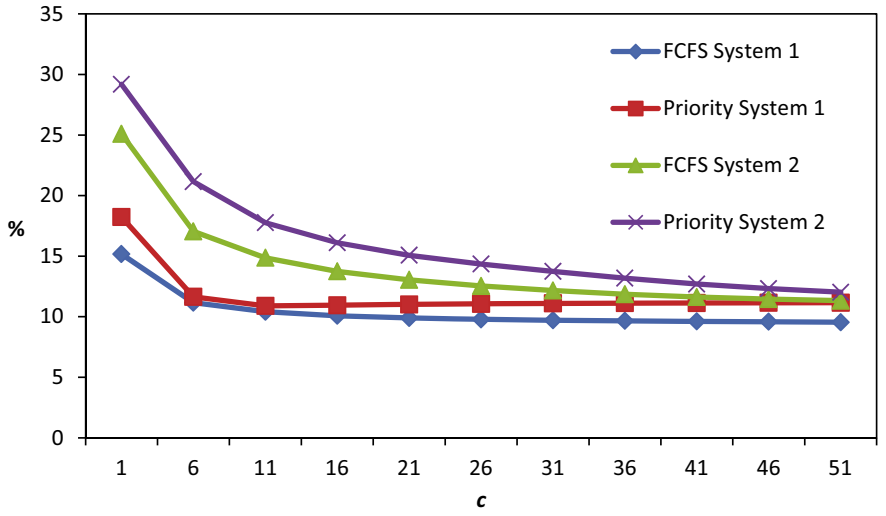


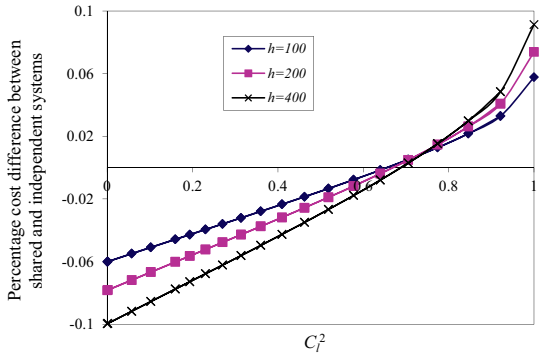
Figure B3 The Benefits of Capacity Sharing ($\gamma = 1$, $\theta = 1$; System 1: $\beta = 1$; System 2: $\beta = 10$)



$$z_{\mathcal{N}}(\mu_{\mathcal{N}}) = c\mu_{\mathcal{N}} + h\left[\frac{(1+C_l^2)\rho^2}{(1-\rho)} + \rho\right].$$

It is easy to see that, everything else remaining the same, the total cost for a given service rate $\mu_{\mathcal{N}}$ is increasing in C_l and, consequently, the optimal capacity level is also increasing in C_l. This suggests that for sufficiently high C_l, the shared system could be less desirable than a system with independent facilities. This observation is confirmed by the numerical results shown in Figure B1 where the percentage cost difference between systems with and without capacity sharing, $(z^*_{\mathcal{N}} - z^*_{1,\ldots,n})/z^*_{1,\ldots,n}$, is shown for different values of C_l for an example system with two firms (to

Figure B4 The Effect of Work Content Variability ($c = 30$, $\lambda_1 = \lambda_2 = 20$, $l_1 + l_2 = 20$)



obtain different values of C_l, we vary l_1 and l_2 while keeping $l_1 + l_2 = 20$). There is a threshold on C_l above which the system without capacity sharing becomes preferable. The effect of increasing C_l is particularly significant when delay cost is high (as shown in Figure B4) or when the demand rates or capacity cost are low (as observed in additional numerical results not shown here for brevity). these are the settings where the difference between pooled and distributed systems tends to be most significant.

Note

[1]For example, several airlines share repair and maintenance facilities; see recent announcements by Air France and Lufthansa http://www.lufthansa-technik.com/spairliners and Delta and Aero Mexico http://news.delta.com/index.php?s=43&item=1698.

References

Aksin, O. Z., F. Karaesmen, E. L. Ormeci. 2007. A review of workforce cross-training in call centers from an operations management perspective. D. Nembhard, ed. *Workforce Cross Training Handbook*, CRC Press, Boca Raton, FL, 211–240.

Aksin, O. Z., F. de Vericourt, F. Karaesmen. 2008. Call center outsourcing contract design and choice. *Manage. Sci.* **54**: 354–368.

Albin, S. L. 1984. Approximating a point process by a renewal process, II: Superposition arrival processes to queues. *Oper. Res.* **32**: 1133–1162.

Allon, G., A. Federgruen. 2007. Competition in service industries. *Oper. Res.* **55**: 37–55.

Allon, G., A. Federgruen. 2008. Service competition with general queueing facilities. *Oper. Res.* **56**: 827–849.

Anily, S., M. Haviv. 2007. Cost-allocation for the first order interaction joint replenishment model. *Oper. Res.* **55**: 292–302.

Anily, S., M. Haviv. 2010. Cooperation in service systems. *Oper. Res.* **58**: 660–673.

Bassambo, A., R. S. Randhawa, J. A. van Mieghem. 2012. A little flexibility is all you need: On the asymptotic value of flexible capacity in parallel queuing systems. *Oper. Res.* **60**: 1423–1435.

Ben-Zvi, N., Y. Gerchak, 2012. Inventory centralization in a newsvendor setting when shortage costs differ: Priorities and costs allocation. T-M. Choi, ed. *Handbook of Newsvendor Problems*. Springer, New York, Ch.11, 263–276.

Benjaafar, S. 1995. Performance bounds for the effectiveness of pooling in multi-processing systems. *Eur. J. Oper. Res.* **87**: 375–388.

Benjaafar, S. , E. Elahi, K. Donohue. 2007. Outsourcing via service quality competition. *Manage. Sci.* **53**: 241–259.

Bertsekas, D. 1999. *Nonlinear Programming*. Athena Scientific, Belmont, MA.

Borst, S., A. Mandelbaum, M. I. Reiman. 2004. Dimensioning large call centers. *Oper. Res.* **52**: 17–34.

Buzacott, J. A. 1996. Commonalities in reengineered business processes: Models and issues. *Manage. Sci.* **42**: 768–782.

Cachon, G., P. Harker. 2002. Competition and outsourcing with scale economies. *Manage. Sci.* **48**: 1314–1333.

Cachon, G., F. Zhang. 2007. Obtaining fast service in a queuing system via performance-based allocation of demand. *Manage. Sci.* **53**: 408–420.

Chen, X., J. Zhang. 2007. Duality approaches to economic lot sizing games. Working paper, New York University.

Chen, X., J. Zhang. 2009. A stochastic programming duality approach to inventory centralization games. *Oper. Res.* **57**: 840–851.

Coffman Jr., E., I. Mitrani. 1980. A characterization of waiting time performance realizable by single-server queues. *Oper. Res.* **28**: 810–821.

Dewan, S., H. Mendelson. 1990. User delay costs and internal pricing for a service facility. *Manage. Sci.* **36**: 1502–1517.

van Dijk, N. M., E. Sluis. 2008. To pool or not to pool in call centers. *Prod. Oper. Manag.* **17**(3): 296–305.

Dror, M., B. Hartman. 2007. Shipment consolidation: Who pays for it and how much? *Manage. Sci.* **53**: 78–87.

Edmonds, J. 1970. Submodular functions, matroids, and certain polyhedra. *Combinatorial structures and their applications*. Gordon and Breach, New York, 69–87.

Gans, N., Y-P. Zhou. 2003. A call-routing problem with service-level constraints. *Oper. Res.* **51**: 255–271.

Gans, N., Y-P. Zhou. 2007. Call-routing schemes for call-center outsourcing. *Manuf. Serv. Oper. Manag.* **9**: 33–50.

Garcia-Sanz, M. D., F. R. Fernandez, M. G. Fiestras-Janeiro, I. Garcia-Jurado, J. Puerto. 2008. Cooperation in markovian queueing models. *Eur. J. Oper. Res.* **188**: 485–495.

Gonzalez, P., C. Herrero. 2004. Optimal sharing of surgical costs in the presence of queues. *Math. Methods Oper. Res.* **59**: 435–446.

Gurumurthi, S., S. Benjaafar. 2004. Modeling and analysis of flexible queueing systems. *Nav. Res. Log.* **51**: 755–782.

Ha, A. Y. 1998. Incentive-compatible pricing for a service facility with joint production and congestion externalities. *Manage. Sci.* **44**: 1623–1636.

Ha, A. Y. 2001. Optimal pricing that coordinates queues with customer-chosen service requirements. *Manage. Sci.* **47**: 915–930.

Hanany, E., Y. Gerchak. 2008. Nash bargaining over inventory and pooling contracts. *Nav. Res. Log.* **55**: 541–550.

Harrison, J. M. 1985. *Brownian Motion and Stochastic Flow Systems*. John Wiley, New York.

Hassin, R., M. Haviv. 2003. *To Queue or Not to Queue*. Kluwer, Boston.

van den Heuvel, W., P. E. M. Borm, H. Hamers. 2007. Economic lot-sizing games. *Eur. J. Oper. Res.* **176**: 1117–1130.

Hopp, W. J., E. Tekin, M. P. Van Oyen. 2004. Benefits of skill chaining in production lines with cross-trained workers. *Manage. Sci.* **50**: 83–98.

Iravani, S. M., M. P. Van Oyen, K. T. Sims. 2005. Structural flexibility: A new perspective on the design of manufacturing and service operations. *Manage. Sci.* **51**: 151–166.

Jaiswal, N. K. 1968. *Priority Queues*. Academic Press, New York.

Jordan, W. C., R. R. Inman, D. E. Blumenfeld. 2004. Chained cross-training of workers for robust performance. *IIE Trans.* **36**: 953–967.

Jouini, O., Y. Dallery, R. Nait-Abdallah. 2008. Analysis of the impact of team-based organizations in call center management. *Manage. Sci.* **54**: 400–414.

Kalai, E., M. I. Kamien, M. Rubinovitch. 1992. Optimal service speeds in a competitive environment. *Manage. Sci.* **38**: 1154–1163.

Katta, A., J. Sethuraman. 2006. Cooperation in queues. Working paper, Columbia University.

Kemahlioglu-Ziya, E. 2004. Formal methods of value sharing in supply chains. Ph.D Thesis, Georgia Institute of Technology.

Kleinrock, L. 1976. *Queueing Systems, Computer Applications*, Vol. 2. John Wiley & Sons, New York.

Klimov, G. P. 1974. Time sharing service systems I. *Theory Prob. Appl.* **19**: 532–551.

Koole, G., A. Pot. 2005. An overview of routing and staffing in multi-skill customer contact centers. Working paper, Vrije Universiteit.

Mandelbaum, A., M. I. Reiman. 1998. On pooling in queueing networks. *Manage. Sci.* **44**: 971–981.

Maniquet, F. 2003. A characterization of the shapley value in queueing problems. *J. Econ. Theory* **109**: 90–103.

Mendelson, H., S. Whang. 1990. Optimal incentive-compatible priority pricing for the m/m/1 queue. *Oper. Res.* **38**: 870–883.

Milgrom, P., I. Segal. 2002. Envelope theorems for arbitrary choice sets. *Econometrica* **70**: 583–601.

Moulin, H. 1995. *Cooperative Microeconomics: A Game-Theoretic Introduction*. Princeton University Press, Princeton, NJ.

Moulin, H., R. Stong. 2002. Fair queuing and other probabilistic allocation methods. *Math. Oper. Res.* **27**: 1–30.

Muller, A., M. Scarsini, M. Shaked. 2002. The newsvendor game has a nonempty core. *Games Econ. Behav.* **38**: 118–126.

Nagarajan, M., G. Sošić. 2008. Game-theoretic analysis of cooperation among supply chain agents: Review and extensions. *Eur. J. Oper. Res.* **187**: 719–745.

Shanthikumar, J. G., D. D. Yao. 1992. Multiclass queueing systems: Polymatroidal structure and optimal scheduling control. *Oper. Res.* **40**: S293–S299.

Shapley, L. S. 1971. Cores of convex games. *Int. J. Game Theory* **1** (1): 11–26.

Sheikhzadeh, M., S. Benjaafar, D. Gupta. 1998. Machine sharing in manufacturing systems: Flexibility versus chaining. *Int. J. Flex. Manuf. Sys.* **10**: 351–378.

Smith, D. R., W. Whitt. 1981. Resource sharing for efficiency in traffic systems. *Bell Syst. Tech. J.* **60**: 39–55.

Stidham, S. 1970. On the optimality of single-server queueing systems. *Oper. Res.* **18**: 708–732.

Stidham, S. 1992. Pricing and capacity decisions for a service facility: Stability and multiple local optima. *Manage. Sci.* **38**: 1121–1139.

Tekin, E., W. Hopp, M. V. Oyen. 2009. Pooling strategies for call center agent crosstraining. *IIE Trans.* **41**: 546–561.

Wallace, R. B., W. Whitt. 2005. A staffing algorithm for call centers with skill-based routing. *Manuf. Serv. Oper. Manag.* **7**: 276–294.

Whitt, W. 1982. Approximating a point process by a renewal process, I: Two basic methods. *Oper. Res.* **30**: 125–147.

Whitt, W. 1999. Partitioning customers into service groups. *Manage. Sci.* **45**(11): 1579–1592.

Zipkin, P. H. 2000. *Foundation of Inventory Management*. McGraw-Hill, Boston, MA.

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推荐人：刘许成

推荐理由：本文提出一类源于生产库存问题的新合作博弈。若干参与者需要在有限计划期内满足各自的需求，并允许欠交。每个参与者均具有各自的单位生产成本、库存持有成本和欠交成本。参与者通过共享生产流程和仓储设施开展合作：联盟内的参与者以最低生产成本进行生产，并以最低库存成本储存。本文证明，由此产生的合作博弈是完全平衡的，而且欧文集退化为单点，即欧文点。基于这一分配，本文为这类博弈构造了一个人口单调分配方案。最后，指出欧文点与核仁、夏普利值等其他著名分配规则之间的关系。

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Production-inventory games: A new class of totally balanced combinatorial optimization games

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Abstract

In this paper we introduce a new class of cooperative games that arise from production-inventory problems. Several agents have to cover their demand over a finite time horizon and shortages are allowed. Each agent has its own unit production, inventory-holding and backlogging cost. Cooperation among agents is given by sharing production processes and warehouse facilities: agents in a coalition produce with the cheapest production cost and store with the cheapest inventory cost. We prove that the resulting cooperative game is totally balanced and the Owen set reduces to a singleton: the Owen point. Based on this type of allocation we find a population monotonic allocation scheme for this class of games. Finally, we point out the relationship of the Owen point with other well-known allocation rules such as the nucleolus and the Shapley value.
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1. Introduction

One of the main objectives of management of firms is cost reduction. In order to achieve this goal, groups of firms might form coalitions to diminish operation costs making dynamic

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decisions throughout a finite planning horizon. In tactical planning of enterprises which produce indivisible goods, operation costs mainly consist of production, inventory-holding, and backlogging costs. These coalitions should induce individual and collective cost reductions; thus, stability is achieved in the process of enterprise cooperation.

In our framework a coalition allows each of its members of have access to the technologies owned by the other members of the coalition. Thus, members of a coalition can use the lowest-cost technology of the firms in the coalition. Planning is done throughout a finite time horizon; therefore, at the beginning of each period, the costs to the members of a coalition, which depend on the best technology at that point, may change.

The model that represents that situation is the dynamic, discrete, finite planning horizon production-inventory problem with backlogging. The objective of any group of firms is to satisfy the demand for indivisible goods in each period at a minimum cost. This is a well-known combinatorial optimization problem for which the algorithm by Wagner and Whitin (1958) provides optimal solutions by dynamic programming techniques. The optimal solutions of this problem lead to the best production-inventory policy for the group of firms. These policies generate an optimal operation cost for the entire group. The question is what portion of this cost is to be supported by each firm. Cooperative game theory provides the natural tools for answering this question.

The analysis of inventory situations is not new. Thus, one can find in the literature several centralization inventory models approached from this point of view. The interested reader is referred to Eppen (1979), Kohli and Park (1989), Gerchak and Gupta (1991), Robinson (1993), Hartman and Dror (2003, 2005, 1996), Hartman et al. (2000), Anupindi et al. (2001), Müller et al. (2002), Meca et al. (2003, 2004), Meca (2007), Minner (2007), Tijs et al. (2005) and Slikker et al. (2005) among others, for comprehensive literature on this subject. Other operations research games are studied as well. For a clear and detailed presentation of operations research games (including inventory games) we refer to Borm et al. (2001). We are not aware of any reference in the literature of centralization in inventory models that analyzes inventory models for which optimal operation is only defined implicitly as the optimal solution of a combinatorial problem (as it is the case in the discrete review model in this paper). In this regards, our approach makes a step forward.

The study of cooperative combinatorial optimization games, which are defined through characteristic functions given as optimal values of combinatorial optimization problems, is a fruitful topic (see, for instance, Shapley and Shubik, 1972; Dubey and Shapley, 1984; Granot, 1986; Tamir, 1992; Deng et al., 1999, 2000; Faigle and Kern, 2000). There are characterizations of the total balancedness of several classes of these games. Inventory games and combinatorial optimization games are up to date disjoint classes of games. While in the former class there is always an explicit form for the characteristic function of each game, the characteristic function of the games in the latter class it is defined implicitly as the optimal value of an optimization problem in integer variables.

In this paper we introduce a class of production-inventory games which combine the characteristics of inventory and combinatorial optimization games: this class models cooperation on production and storage of indivisible goods and its characteristic function is defined implicitly as the optimal value of a combinatorial optimization problem. It turns out to be a new class of totally balanced combinatorial optimization games.

We start by introducing definitions and notations in Section 2. In Section 3 we give a complete description of the production-inventory problem (PI-problem). A natural variant of this problem is addressed in Section 4. Several agents, each one facing a PI-problem, decide to cooperate to

reduce costs, and then a production-inventory situation (PI-situation) arises. Then, for each PI-situation, the corresponding cooperative game structure, namely production-inventory game (PI-game), is defined. The main results (total balancedness and an explicit form for the characteristic function) are stated in this section. Section 5 completes the study of PI-games by showing that the Owen set of a PI-situation (the set of allocations that are achievable through dual solutions) shrinks to a singleton: the Owen point. Its explicit form is also provided, and moreover, it is proved that the Owen point can be reached through a population monotonic allocation scheme. In addition, a necessary and sufficient condition for the core of a PI-game to be a singleton: the Owen point, is presented. We propose the Owen point as a core-allocation for a PI-game which is easy to calculate and satisfies good properties. Finally, we point out the relationship of the Owen point with some well-known point solutions in cooperative game theory.

2. Preliminaries

Production-inventory games constitute a class of cooperative cost games with transferable utility (TU games). A TU cost game is a pair (N, c), where $N = \{1, 2, \ldots, n\}$ is the finite player set and $c : \mathcal{P}(N) \to \mathbb{R}$ the characteristic function satisfying $c(\emptyset) = 0$. The subgame related to coalition S, c_S, is the restriction of the mapping c to the subcoalitions of S. We denote by lower case letter s the cardinality of set S, i.e. $\text{card}(S) = s$, for all $S \subseteq N$. A cost-sharing vector will be $x \in \mathbb{R}^n$ and, for every coalition $S \subseteq N$ we shall write $x(S) := \sum_{i \in S} x_i$ the cost-sharing to coalition S (where $x(\emptyset) = 0$). The core of the game (N, c) consists of those cost-sharing vectors which allocate the cost of the grand coalition in such a way that every other coalition pays at most its cost, given by the characteristic function: $\text{Core}(N, c) = \{x \in \mathbb{R}^n / x(N) = c(N)$ and $x(S) \leqslant c(S)$ for all $S \subset N\}$. In the following, cost-sharing vectors belonging to the core will be called core-allocations. A cost game (N, c) has a nonempty core if and only if it is balanced (see Bondareva, 1963; Shapley, 1967). It is a totally balanced game if the core of every subgame is nonempty. Totally balanced games were introduced by Shapley and Shubik in the study of market games (see Shapley and Shubik, 1969).

A population monotonic allocation scheme (see Sprumont, 1990), or pmas, for the game (N, c) is a collection of vectors $y^S \in \mathbb{R}^s$ for all $S \subseteq N$, $S \neq \emptyset$ such that $y^S(S) = c(S)$ for all $S \subseteq N$, $S \neq \emptyset$, and $y_i^S \geqslant y_i^T$ for all $S \subseteq T \subseteq N$ and $i \in S$. Note that if $(y^S)_{\emptyset \neq S \subseteq N}$ is a pmas for (N, c), then $y^S \in \text{Core}(N, c_S)$ for all $S \subseteq N$, $S \neq \emptyset$. Hence, the set of payoff vectors that can be reached through a pmas can be seen as a refinement of the core. Every cost game with pmas is totally balanced. However, it is not possible to get a pmas with any random selection of cost-sharings and there are totally balanced cost games without pmas. A core-allocation for (N, c), denoted by $x \in \text{Core}(N, c)$, is reached through a pmas if there exists $(y^S)_{\emptyset \neq S \subseteq N}$ for the game (N, c) such that $y_i^N = x_i$ for all $i \in N$.

A game is said to be subadditive when for all disjoint coalitions S and T, $c(S \cup T) \leqslant c(S) + c(T)$ holds. In a subadditive game, it will always be beneficial for two disjoint coalitions to cooperate and form a larger coalition. Balanced cost games might not be subadditive but they always satisfy subadditive inequalities involving the grand coalition. However, totally balanced cost games are subadditive. A well-known class of balanced and subadditive games is the class of concave games (see Shapley, 1971).

Finally to simplify the presentation, for a generic problem (P) we will denote in the following by $val(P)$, $F(P)$ and $O(P)$, the optimal value, the set of feasible solutions and the set of optimal solutions of problem (P), respectively.

3. Production-inventory problems

To start with, the basic form of the production-inventory model is now described (the interested reader is referred to Johnson and Montgomery, 1974 for further details). Demand for a single product occurs during each of T consecutive time periods that are numbered 1 through T. The demand that occurs during a given period can be satisfied by production during that period, during any earlier period (as inventory is carried forward) or can be backlogged to be covered by production at future periods (as backlogged demand is accumulated). Inventory at period 1 is zero, and inventory at the end of period T is required to be zero. The model includes production, inventory and backlogging costs. The objective is to schedule production so as to satisfy demand at minimum cost. Formally, a production-inventory problem (PI-problem for short) is a 5-tuple (T, d, h, b, p) where:

- T is the planning horizon.
- $d = (d_1, \ldots, d_T) \geqslant 0$, $d_t =$ demand during period t, $t = 1, \ldots, T$.
- $h = (h_1, \ldots, h_T) \geqslant 0$, $h_t =$ unit inventory carrying costs in period t, $t = 1, \ldots, T$.
- $b = (b_1, \ldots, b_T) \geqslant 0$, $b_t =$ unit backlogging carrying costs in period t, $t = 1, \ldots, T$.
- $p = (p_1, \ldots, p_T) \geqslant 0$, $p_t =$ unit production costs in period t, $t = 1, \ldots, T$.

The decision variables of the model are:

- $q_t =$ production during period t.
- $I_t =$ inventory at hand at the end of period t.
- $E_t =$ backlogged demand at the end of period t.

The reader may notice that I and E variables are instrumental and help to clarify presentation while q is the only actual set of decision variables. (Once q is known the others can be obtained solving a system of linear equations.) These decision variables are required to be in integer quantities. The resolution amounts to the following mathematical programming formulation:

$$\text{(PI)} \quad \min \sum_{t=1}^{T} (p_t q_t + h_t I_t + b_t E_t)$$

$$\text{s.t.} \quad I_0 = I_T = E_0 = E_T = 0, \tag{1}$$

$$I_t - E_t = I_{t-1} - E_{t-1} + q_t - d_t, \quad t = 1, \ldots, T, \tag{2}$$

$$q_t,\ I_t, E_t, \quad \text{nonnegative, integer,}\ t = 1 \ldots, T. \tag{3}$$

Constraint (1) ensures initial and final conditions on inventory and backlogged demand. Matter is conserved, and (2) requires that the sum of the inventory at the end of a period is the sum of the inventory at the end of previous period minus consumption. Finally, constraint (3) ensures nonnegativity and integrality of the variables in the problem.

We associate with problem (PI) its linear relaxation (LPI). This problem turns out to be

$$\text{(LPI)} \quad \min \sum_{t=1}^{T} (p_t q_t + h_t I_t + b_t E_t)$$

$$\text{s.t.} \quad I_0 = I_T = E_0 = E_T = 0, \tag{4}$$

$$I_t - E_t = I_{t-1} - E_{t-1} + q_t - d_t, \quad t = 1, \ldots, T, \tag{5}$$

$$q_t \geqslant 0,\ I_t \geqslant 0,\ E_t \geqslant 0, \quad t = 1, \ldots, T. \tag{6}$$

First of all, we note in passing that if vector d has integer components all the extreme points of the feasible region of (LPI) are integers. Indeed, the constraint matrix of problem (LPI) is a 0, ± 1 valued matrix where each column has at most one $+1$ and at most one -1, then it is totally unimodular and the result is well known (see, e.g., Garfinkel and Nemhauser, 1972).

Let (DLPI) be the dual of (LPI). This problem is given as

$$\text{(DLPI)} \quad \max \sum_{t=1}^{T} y_t d_t \tag{7}$$

$$\text{s.t.} \quad y_t \leqslant p_t, \quad t = 1, \ldots, T, \tag{8}$$

$$y_{t+1} - y_t \leqslant h_t, \quad t = 1, \ldots, T-1, \tag{9}$$

$$-y_{t+1} + y_t \leqslant b_t, \quad t = 1, \ldots, T-1. \tag{10}$$

The consequence of the above discussion is that

$$val(\text{PI}) = val(\text{LPI}) = val(\text{DLPI}).$$

Moreover, problems (PI) and (LPI) have integer optimal solutions.

Let

$$h_{kt} = \sum_{r=k}^{t-1} h_r \quad \text{for any } k < t,\ t = 2, \ldots, T,\ h_{k1} = 0,\ k < 1, \tag{11}$$

$$b_{tk} = \sum_{r=t}^{k-1} b_r, \quad \text{for any } k > t,\ t = 1, \ldots, T-1,\ b_{Tk} = 0,\ k > T. \tag{12}$$

According to this notation one can check that a feasible solution for (DLPI) is given as

$$y_t^* = \min\Big\{ p_t, \min_{k<t}\{p_k + h_{kt}\}, \min_{k>t}\{p_k + b_{tk}\}\Big\}, \quad t = 1, \ldots, T, \tag{13}$$

where

$$p_k = \begin{cases} p_1 & \text{if } k < 1, \\ p_T & \text{if } k > T. \end{cases}$$

The reader may notice that this solution induces a feasible solution in the primal problem. In this solution the demand d_t that occurs in period t is produced according to the following scheme:

$$\begin{cases} \text{in period } t & \text{if } y_t^* = p_t, \\ \text{in period } k_h & \text{if } y_t^* = p_{k_h} + h_{k_h t},\ k_h < t, \\ \text{in period } k_b & \text{if } y_t^* = p_{k_b} + b_{t k_b},\ k_b > t. \end{cases}$$

Lemma 3.1. *The vector y^* given in* (13) *is an optimal solution of problem* (DLPI).

Proof. The reader can easily check that y^* satisfies all the constraints of problem (DLPI). This solution induces a feasible production plan (q^*, I^*, E^*) in the primal problem (PI) so that $\sum_{t=1}^{T}(p_t q_t^* + h_t I_t^* + b_t E_t^*) = \sum_{t=1}^{T} d_t y_t^*$. Therefore, both solutions are optimal in their corresponding problems. □

The reader may notice that if it were assumed that all the unit costs h, b and p are nonnegative integers then the solution y^* would also be an integer. Moreover, we would like to have more

information about the structure of the optimal solution set to (DLPI). The next results show that the optimal solution y^* given by (13) is the componentwise upper bound of any optimal solution of (DLPI). Hence, it is called *maximal optimal solution* for (DLPI).

The following technical lemma will be useful to prove the theorem. In fact, it reveals that vector y^* is the componentwise maximum of all feasible solutions. Recall that by $F(\text{DLPI})$ we denote the feasible solution set for (DLPI).

Lemma 3.2. *For each* $y \in F(\text{DLPI})$, $y_t \leqslant y_t^*$ *for all* $t \in \{1, \ldots, T\}$.

Proof. Take $y \in F(\text{DLPI})$. Let $p_{k_h} + h_{k_h t} = \min_{k<t}\{p_k + h_{kt}\}$ and $p_{k_b} + b_{tk_b} = \min_{k>t}\{p_k + b_{tk}\}$, $t = 1, \ldots, T$.

For any $t \in \{1, \ldots, T\}$ we can distinguish three situations:

(S1) $p_t \leqslant p_{k_h} + h_{k_h t}$ and $p_t \leqslant p_{k_b} + h_{k_b t}$. Then, $y_t^* = p_t$, and by (8) $y_t \leqslant y_t^*$.
(S2) $p_{k_h} + h_{k_h t} < p_t$ and $p_{k_h} + h_{k_h t} \leqslant p_{k_b} + b_{tk_b}$. Then, $y_t^* = h_{k_h t} + p_{k_h}$.

Taking into account that

$$\begin{aligned}
y_{k_h} &\leqslant p_{k_h},\\
y_{k_h+1} &\leqslant h_{k_h} + y_{k_h},\\
y_{k_h+2} &\leqslant h_{k_h+1} + y_{k_h+1} \leqslant h_{k_h+1} + (h_{k_h} + p_{k_h}) = h_{k_h k_h+2} + p_{k_h},\\
&\vdots\\
y_t &\leqslant h_{t-1} + y_{t-1} \leqslant h_{t-1} + (h_{k_h t-1} + p_{k_h}) = h_{k_h t} + p_{k_h}
\end{aligned}$$

we conclude that $y_t \leqslant y_t^*$.

(S3) $p_{k_b} + b_{tk_b} < p_t$ and $p_{k_b} + b_{tk_b} < p_{k_h} + h_{k_h t}$. Then, $y_t^* = b_{tk_b} + p_{k_b}$; and

$$\begin{aligned}
y_{k_b} &\leqslant p_{k_b},\\
y_{k_b-1} &\leqslant b_{k_b-1} + y_{k_b},\\
y_{k_b-2} &\leqslant b_{k_b-2} + y_{k_b-1} \leqslant b_{k_b-2} + (b_{k_b-1} + p_{k_b}) = b_{k_b-2k_b} + p_{k_b},\\
&\vdots\\
y_t &\leqslant b_t + y_{t+1} \leqslant b_t + (b_{t+1k_b} + p_{k_b}) = b_{tk_b} + p_{k_b}.
\end{aligned}$$

Hence, $y_t \leqslant y_t^*$. □

Theorem 3.3. *The optimal solution set of problem* (*DLPI*) *is given by*

$$O(\text{DLPI}) = \left\{ y \in F(\text{DLPI}); \quad \begin{array}{ll} y_t \leqslant y_t^* & \text{if } d_t = 0 \\ y_t = y_t^* & \text{if } d_t > 0 \end{array} \right\}.$$

Proof. ($\subseteq$) Take $y \in O(\text{DLPI})$. By Lemma 3.2, $y_t \leqslant y_t^*$ for all $t \in \{1, \ldots, T\}$. Suppose there exists a period $t \in \{1, \ldots, T\}$ with $d_t > 0$ and $y_t < y_t^*$. Then by Lemma 3.1, $val(\text{DLPI}) = y_1^* d_1 + \cdots + y_T^* d_T > y_1 d_1 + \cdots + y_T d_T$, which is a contradiction.

($\supseteq$) It is obvious that for all $y \in F(\text{DLPI})$ such that $y_t \leqslant y_t^*$, if $d_t = 0$, and $y_t = y_t^*$ otherwise,

$$\sum_{t=1}^{T} d_t y_t = \sum_{\substack{t=1 \\ d_t>0}}^{T} d_t y_t^* = \sum_{t=1}^{T} d_t y_t^*.$$

Hence by duality theorem in linear programming, we can conclude that $y \in O(\text{DLPI})$. □

4. Production-inventory games

Once we have revisited the classical version of the production-inventory problem, we address a natural variant of this problem in which several agents, each one facing a PI-problem, decide to cooperate to reduce costs. Here the cooperation is driven by sharing technologies in production, inventory carrying and backlogged demand. Thus, if a group of agents agree on cooperation then at each period they will produce and pay inventory carrying and backlogged demand at the cheapest costs among the members of the coalition. Cooperation in holding and production costs is usual and has appeared already in literature. Our mode of cooperation in backlogging is also natural: once a coalition is formed, all its members pay compensation to customers for delayed delivering (backlogging cost) of their demands according to the cheapest cost among the members in the coalition. In some regards, larger coalitions are stronger and "squeeze" a bit more their clients. Formally, a production-inventory situation (PI-situation) is a 5-tuple (N, D, H, B, P) where N is the set of players $N = \{1, \ldots, n\}$ and for each player i (T, d^i, h^i, b^i, p^i) is a PI-problem.

The reader may notice that

$$D = \left[d^1, \ldots, d^n\right]', \qquad H = \left[h^1, \ldots, h^n\right]',$$
$$B = \left[b^1, \ldots, b^n\right]', \qquad P = \left[p^1, \ldots, p^n\right]';$$

and

$$d^i = \left[d_1^i, \ldots, d_T^i\right]', \qquad h^i = \left[h_1^i, \ldots, h_T^i\right]',$$
$$b^i = \left[b_1^i, \ldots, b_T^i\right]', \qquad p^i = \left[p_1^i, \ldots, p_T^i\right]'.$$

Note that we can associate with each PI-situation (N, D, H, B, P) a cost TU-game (N, c) with characteristic function c defined as follows: $c(\emptyset) = 0$ and for any $S \subseteq N$, $c(S) = \text{val}(\text{LPI}(S))$, where

$$\begin{aligned}
(\text{LPI}(S)) \quad & \min \sum_{t=1}^{T} \left(p_t^S q_t + h_t^S I_t + b_t^S E_t\right) \\
& \text{s.t.} \quad I_0 = I_T = E_0 = E_T = 0, \\
& \qquad I_t - E_t = I_{t-1} - E_{t-1} + q_t - d_t^S, \quad t = 1, \ldots, T, \\
& \qquad q_t \geqslant 0,\ I_t \geqslant 0,\ E_t \geqslant 0, \quad t = 1, \ldots, T;
\end{aligned}$$

with

$$p_t^S = \min_{i \in S}\left\{p_t^i\right\}, \qquad h_t^S = \min_{i \in S}\left\{h_t^i\right\}, \qquad b_t^S = \min_{i \in S}\left\{b_t^i\right\}, \qquad d_t^S = \sum_{i \in S} d_t^i.$$

Every cost TU-game defined in this way is what we call a production-inventory game (PI-game).

The main result in this section states that every PI-game is totally balanced, hence subadditive.

Theorem 4.1. *Let (N, D, H, B, P) be a PI-situation with D being an integer matrix. The corresponding PI-game (N, c) is totally balanced.*

Proof. For any $S \subseteq N$ consider the reduced game (S, c_S).

Let y^* be an optimal solution of (DLPI(S)). Define $u^* = (y^{*\prime} d^i)_{i \in S} \in \mathbb{R}^s$. It is clear that

$$\sum_{i \in S} u_i^* = \sum_{t=1}^{T} y_t^* d_t^S = val\big(\text{DLPI}(S)\big) = val\big(\text{LPI}(S)\big) = c_S(S).$$

Moreover, for any $R \subseteq S$, y^* is a feasible solution for (DLPI(R)) since the constraint matrix does not depend on S and the right-hand side of (DLPI(S)) is componentwise smaller than or equal to the one of (DLPI(R)). Hence,

$$c_S(R) = val\big(\text{DLPI}(R)\big) = \max_{y \in F(\text{DLPI}(R))} \sum_{t=1}^{T} d_t^R y_t \geqslant \sum_{t=1}^{T} d_t^R y_t^* = \sum_{i \in R} y^{*\prime} d^i = \sum_{i \in R} u_i^*.$$

Therefore, $\text{Core}(S, c_S) \neq \emptyset$. □

We note in passing that although PI-games are totally balanced games, in general these games are not concave (see Example 4.4).

Recently, Deng et al. (1999, 2000) have studied some families of combinatorial optimization games, namely packing and covering games, for which they prove total balancedness. Here, we have presented a different class exhibiting the same property. The core of the above mentioned classes of combinatorial optimization games coincides with the set of dual solutions, i.e. optimal solutions of the dual problem to the one that defines the characteristic function. However, the same property does not hold in our class of PI-games as will be shown in the next examples. In order to study the core set within the class of PI-games we introduce the so-called *Owen set*: the set of allocations that are achievable through dual solutions (see Gellekom et al., 2000). Formally, the Owen set of (N, D, H, B, P) is defined by

$$\text{Owen}(N, D, H, B, P) = \big\{\big(y' d^i\big)_{i \in N} \colon\ y \in O\big(\text{DLPI}(N)\big)\big\}. \tag{14}$$

As a straightforward consequence of Theorem 4.1, we obtain that

$$\text{Owen}(N, D, H, B, P) \subseteq \text{Core}(N, c).$$

Our next example shows that in general there are core-allocations not achieved with dual solutions.

Example 4.2. Consider the following PI-situation with two periods and two players, namely P1 and P2:

	Demand		Production		Inventory		Backlogging	
P1	10	10	2	2	1	2	1	2
P2	8	12	1	1	2	4	2	2

The data above gives rise to the game with characteristic function in the following table.

	d_1^S	d_2^S	p_1^S	p_2^S	h_1^S	h_2^S	b_1^S	b_2^S	c
{1}	10	10	2	2	1	2	1	2	40
{2}	8	12	1	1	2	4	2	2	20
{1, 2}	18	22	1	1	1	2	1	2	40

The unique optimal solution of (DLPI({1, 2})) is $y^{*\prime} = (1, 1)$. This gives Owen$(N, D, H, B, P) = \{(20, 20)\}$. However, Core$(N, c) = \{(x_1, x_2)\colon 0 \leqslant x_1 \leqslant 40, 0 \leqslant x_2 \leqslant 20, x_1 + x_2 = 40\}$.

In order to get the Owen set one has to solve three optimization problems (DLPI({1, 2})), (DLPI({1})) and (DLPI({2})). In general, for a n-player game $2^n - 1$ optimization problems would have had to be solved.

For the sake of simplicity, we obtain an explicit form for the characteristic function of PI-games.

Consider (DLPI(S)) the dual of (LPI(S)). Recall that the problem (DLPI(S)) is defined as (DLPI) (see (7)–(10)) where cost coefficients and right-hand side vector are replaced by p^S, h^S and b^S, and d^S, respectively.

Let us denote

$$h_{kt}^S = \sum_{r=k}^{t-1} h_r^S \quad \text{for any } k < t,\ t = 2, \ldots, T,\ h_{k1}^S = 0,\ k < 1, \tag{15}$$

$$b_{tk}^S = \sum_{r=t}^{k-1} b_r^S \quad \text{for any } k > t,\ t = 1, \ldots, T-1,\ b_{Tk}^S = 0,\ k > T. \tag{16}$$

Similar arguments to the ones given in Section 3 show that for all $S \subseteq N$, $S \neq \emptyset$, the optimal solution set of problem (DLPI(S)) is

$$O\big(\text{DLPI}(S)\big) = \left\{ y(S) \in F\big(\text{DLPI}(S)\big);\quad \begin{array}{ll} y_t(S) \leqslant y_t^*(S) & \text{if } d_t^S = 0 \\ y_t(S) = y_t^*(S) & \text{if } d_t^S > 0 \end{array} \right\},$$

where $y^*(S)$, the maximal optimal solution, is now given by

$$y_t^*(S) = \min\Big\{ p_t^S, \min_{k<t}\big\{p_k^S + h_{kt}^S\big\}, \min_{k>t}\big\{p_k^S + b_{tk}^S\big\}\Big\}, \quad t = 1, \ldots, T, \tag{17}$$

with

$$p_k^S = \begin{cases} p_1^S & \text{if } k < 1, \\ p_T^S & \text{if } k > T. \end{cases}$$

Let

$$p_{k_h}^S + h_{k_h t}^S = \min_{k<t}\big\{p_k^S + h_{kt}^S\big\}, \qquad p_{k_b}^S + b_{tk_b}^S = \min_{k>t}\big\{p_k^S + b_{tk}^S\big\}, \quad t = 1, \ldots, T,$$

and

$$H_1(S) = \big\{t \in \{1, \ldots, T\} \big/ p_t^S \leqslant p_{k_h}^S + h_{k_h t}^S \text{ and } p_t^S \leqslant p_{k_b}^S + b_{tk_b}^S\big\},$$
$$H_2(S) = \big\{t \in \{2, \ldots, T\} \big/ p_{k_h}^S + h_{k_h t}^S < p_t^S \text{ and } p_{k_h}^S + h_{k_h t}^S \leqslant p_{k_b}^S + b_{tk_b}^S\big\},$$
$$H_3(S) = \big\{t \in \{1, \ldots, T-1\} \big/ p_{k_b}^S + b_{tk_b}^S < p_t^S \text{ and } p_{k_b}^S + b_{tk_b}^S < p_{k_h}^S + h_{k_h t}^S\big\}.$$

Remark that $\{H_1(S), H_2(S), H_3(S)\}$ is a partition of the planning horizon set $\{1, \ldots, T\}$. According to this notation, it can be checked that an explicit form for the characteristic function of a PI-game is given by (18).

Proposition 4.3. *Let* (N, D, H, B, P) *be a PI-situation with* D *being an integer matrix, and* (N, c) *the corresponding PI-game. Then, for each* $S \subseteq N$, $S \neq \emptyset$

$$c(S) = \sum_{t \in H_1(S)} p_t^S d_t^S + \sum_{t \in H_2(S)} d_t^S \left(p_{k_h}^S + h_{k_h t}^S\right) + \sum_{t \in H_3(S)} d_t^S \left(p_{k_b}^S + b_{t k_b}^S\right). \tag{18}$$

Proof. Taking into account that for any $S \subseteq N$, $c(S) = val(\text{DLPI}(S))$ and $y^*(S) \in O(\text{DLPI}(S))$, we obtain $c(S) = \sum_{t=1}^{T} y_t^*(S) d_t^S$. Hence, (18) holds. □

The explicit form above turns out to be expectable. It means that if we are in a period in which production costs are less than production-inventory carrying and production-backlogging carrying costs ($t \in H_1(S)$), we should produce all the demand in that period. On the contrary, if the period is such that production and production-backlogging carrying costs are greater than production-inventory carrying ones ($t \in H_2(S)$), all the demand should have been produced in the earlier period with minimum cost (k_h) and inventory carried until that period (t). Finally, in a period where production and production-inventory carrying costs are greater than production-backlogging carrying ones ($t \in H_3(S)$), all the demand should be produced in the future period with minimum cost (k_b) and backlogging carried until that period (t).

The reader may notice that if the matrices H, B and P are in integer values then the characteristic function of these games is also integer.

The next example illustrates all the results obtained in Section 4.

Example 4.4. Consider the following PI-situation with three periods and three players.

	Demand			Production			Inventory			Backlogging		
P1	0	2	5	3	1	2	2	0	1	1	2	1
P2	0	8	6	4	2	3	2	0	1	1	3	1
P3	0	6	2	3	1	2	2	0	1	1	2	1

It can be easily checked that for all $S \subseteq N$, $H_1(S) = \{2\}$, $H_2(S) = \{3\}$, $H_3(S) = \{1\}$, and $c(S) = p_2^S d_2^S + d_3^S (h_{23}^S + p_2^S) + d_1^S (b_{21}^S + p_2^S)$. Hence, the data above gives rise to the game with characteristic function in the following table:

	d_1^S	d_2^S	d_3^S	p_1^S	p_2^S	p_3^S	h_1^S	h_2^S	h_3^S	b_1^S	b_2^S	b_3^S	c
{1}	0	2	5	3	1	2	2	0	1	1	2	1	7
{2}	0	8	6	4	2	3	2	0	1	1	3	1	28
{3}	0	6	2	3	1	2	2	0	1	1	2	1	8
{1, 2}	0	10	11	3	1	2	2	0	1	1	2	1	21
{1, 3}	0	8	7	3	1	2	2	0	1	1	2	1	15
{2, 3}	0	14	8	3	1	2	2	0	1	1	2	1	22
{1, 2, 3}	0	16	13	3	1	2	2	0	1	1	2	1	29

The optimal solution set of (DLPI($\{1, 2, 3\}$)) is $\{(y_1, 1, 1)/-1 \leqslant y_1 \leqslant 2\}$. However, the Owen set and the core reduces to a singleton: $\text{Owen}(N, D, H, B, P) = \text{Core}(N, c) = \{(7, 14, 8)\}$. Moreover, the game above is not concave since $c(\{1, 2\}) - c(\{2\}) = -7 < c(N) - c(\{2, 3\}) = 7$.

Note that even though there are infinite optimal solutions of the problem (DLPI(N)), the Owen set of the PI-situation above reduces to a singleton again. We wonder if every PI-situation exhibits this property. Next section gives an affirmative answer.

5. Owen point

It is well known that the larger the player set, the more difficult it is to determine the core of a cost TU-game (i.e. the number of inequalities involved increases rapidly). The goal of this section is to find a core-allocation for a PI-game which is easy to calculate and satisfies good properties.

The first result reveals that the Owen set of a PI-situation shrinks to a singleton and provides an explicit form to compute it.

Theorem 5.1. *Let (N, D, H, B, P) be a PI-situation with D being an integer matrix. Then,* $\text{Owen}(N, D, H, B, P) = \{(o_1, \ldots, o_n)\}$ *where, for each $i \in N$,*

$$o_i = \sum_{t \in H_1(N)} p_t^N d_t^i + \sum_{t \in H_2(N)} d_t^i \big(p_{k_h}^N + h_{k_h t}^N\big) + \sum_{t \in H_3(N)} d_t^i \big(p_{k_b}^N + b_{t k_b}^N\big). \tag{19}$$

Proof. Take $y^*(N), y(N) \in O(\text{DLPI}(N))$, $y^*(N) \neq y(N)$. Proving that $\sum_{t=1}^T y_t(N) d_t^i = \sum_{t=1}^T y_t^*(N) d_t^i$, it follows $o_i := \sum_{t=1}^T y_t^*(N) d_t^i$ which proves (19).

Taking into account that $d_t^N = 0$ if and only if $d_t^i = 0$ for all $i \in N$, and $y_t(N) = y_t^*(N)$ if $d_t^N > 0$,

$$\sum_{t=1}^T \big[y_t(N) - y_t^*(N)\big] d_t^i = \sum_{\substack{1 \leqslant t \leqslant T \\ d_t^N = 0}} \big[y_t(N) - y_t^*(N)\big] d_t^i + \sum_{\substack{1 \leqslant t \leqslant T \\ d_t^N > 0}} \big[y_t(N) - y_t^*(N)\big] d_t^i = 0.$$

□

Every cost allocation $o = (o_i)_{i \in N}$ of a PI-game defined in this way is what we call the *Owen point* for (N, c). Obviously, the Owen point is a core-allocation. The Owen point, for each player i, can be interpreted as the cost he/she has to pay when producing at the minimum operation cost. Hence, we propose the Owen point as an alternative value for PI-games. We note in passing that if the matrices P, H and B are integers then the core-allocation given by the Owen point is also in integer values.

The Owen point for the situation given in Example 4.4 is obtained by

$$o_1 = p_2^N d_2^1 + d_3^1 \big(h_{23}^N + p_2^N\big) + d_1^1 \big(b_{21}^N + p_2^N\big) = 7,$$
$$o_2 = p_2^N d_2^2 + d_3^2 \big(h_{23}^N + p_2^N\big) + d_1^2 \big(b_{21}^N + p_2^N\big) = 14,$$
$$o_3 = p_2^N d_2^3 + d_3^3 \big(h_{23}^N + p_2^N\big) + d_1^3 \big(b_{21}^N + p_2^N\big) = 8.$$

Next theorem shows that there is a pmas that realizes the Owen point.

Theorem 5.2. *Let (N, D, H, B, P) be a PI-situation with D being an integer matrix, and (N, c) the corresponding PI-game. The Owen point can be reached through a pmas.*

Proof. Define for all $i \in S$, $S \subseteq N$ and $S \neq \emptyset$,

$$y_i^S := \sum_{t=1}^{T} y_t^*(S) d_t^i.$$

Then for all $S \subseteq N$, $S \neq \emptyset$

$$y^S(S) = \sum_{i \in S} \left(\sum_{t=1}^{T} y_t^*(S) d_t^i \right) = \sum_{t=1}^{T} y_t^*(S) d_t^S = c(S),$$

and for all $S \subseteq R \subseteq N$, $S, R \neq \emptyset$ and for all $i \in S$,

$$y_i^S = \sum_{t=1}^{T} y_t^*(S) d_t^i \geqslant \sum_{t=1}^{T} y_t^*(R) d_t^i = y_i^R,$$

since $y_t^*(S) \geqslant y_t^*(R)$ (a consequence of $F(\mathrm{DLPI}(R)) \subseteq F(\mathrm{DLPI}(S))$ and Lemma 3.2 extended to the coalition $S \subseteq N$).

Finally, we see that $y_i^N = o_i$ for all $i \in N$. So, the Owen point o can be reached through the pmas $(y^S)_{\emptyset \neq S \subseteq N}$. □

Existence of pmas for the entire class of production-inventory games is important by itself but in addition has an interesting consequence. It allows us to prove that PI-games are strictly included in the class of totally balanced games since in general there are totally balanced games without pmas.

It is also possible to prove the existence of pmas in PI-games using the following analysis: consider a situation with N players and T time periods, but only 1 out of the $|N| \times |T|$ demands is positive. The player with this demand pays the costs, which obviously decreases if the coalition he belongs to increases. This is a natural pmas (this player is some sort of a veto-player). Extending this to a general setting by adding $|N| \times |T|$ of these games provides a pmas in general.

Another observation, from the proof of the above theorem, is that a pmas for any PI-game can be built just taking the Owen point of each subgame and gathering all of them as a collection of vectors.

The Owen point for the game given in Example 4.4 can be reached through the pmas

$$\big((7)^{\{1\}}, (28)^{\{2\}}, (8)^{\{3\}}, (7, 14)^{\{1,2\}}, (7, 8)^{\{1,3\}}, (14, 8)^{\{2,3\}}, (7, 14, 8)^{\{1,2,3\}}\big).$$

In that example the core is a singleton: the Owen point. This is not a general property for PI-games. Next theorem provides a necessary and sufficient condition for the core of a PI-game to be a singleton. The following concept is required.

Definition 5.3. Let (N, D, H, B, P) be a PI-situation with D being an integer matrix. We say that $i \in N$ is an *essential player* for (N, D, H, B, P) if there exists at least one period $t \in \{1, \ldots, T\}$ with $d_t^{N \setminus \{i\}} > 0$ such that $y_t^*(N) < y_t^*(N \setminus \{i\})$.

The reader may notice that an essential player is the one for which at least one period exists, in which it is needed by the rest of players in order to produce at a minimum cost a certain demand. On the other hand, an *inessential player* i is the one which is unnecessary by the grand coalition to operate at minimum cost.

We denote by $\mathcal{E}$ the set of essential players and by $N\backslash\mathcal{E}$ its complementary set. In the following we prove that the core of a PI-game shrinks to the Owen point just only when all players are inessential for the PI-situation.

Theorem 5.4. *Let (N, D, H, B, P) be a PI-situation with D being an integer matrix and (N, c) the corresponding PI-game. Then,* $\text{Core}(N, c) = \{o\}$ *if and only if* $\mathcal{E} = \emptyset$.

Proof. (*Only if*) Take (N, D, H, B, P) a PI-situation with D being an integer matrix and all players are inessential for it. Then, for all $i \in N$, $y_t^*(N \setminus \{i\}) = y_t^*(N)$ for all $t \in \{1, \ldots, T\}$ with $d_t^{N\setminus\{i\}} > 0$, and $y_t^*(N \setminus \{i\}) \geqslant y_t^*(N)$ for all $t \in \{1, \ldots, T\}$ with $d_t^{N\setminus\{i\}} = 0$. Hence, $c(N \setminus \{i\}) = o(N \setminus \{i\})$.

Take $x \in \text{Core}(N, c)$ and recall that $o \in \text{Core}(N, c)$. Then, for all $i \in N$, $c(N) - c(N \setminus \{i\}) = o_i \leqslant x_i$ and $\sum_{i=1}^n x_i = \sum_{i=1}^n o_i$. Hence, $x_i = o_i$ for all $i \in N$. So we can conclude that $\text{Core}(N, c) = \{o\}$.

(*If*) Suppose that $\mathcal{E} \neq \emptyset$. Take $j \in \mathcal{E}$, then there exists $t^* \in \{1, \ldots, T\}$ with $d_{t^*}^{N\setminus\{j\}} > 0$ such that $y_{t^*}^*(N) < y_{t^*}^*(N \setminus \{j\})$. For any period t satisfying the above condition, there exists $i \in N \setminus \{j\}$ such that $d_t^i > 0$. It can be checked that for all $R \subset N \setminus \{j\}$, $i \in R$ it holds $y_t^*(N) < y_t^*(N \setminus \{j\}) \leqslant y_t^*(R)$. Hence $o(R) < c(R)$.

Define $\Delta := \{R \subset N \setminus \{j\}\colon i \in R\}$, $0 < \alpha := \min_{R\in\Delta}\{c(R) - o(R)\}$, and for all $k \in N$

$$o_k^* = \begin{cases} o_k & k \in N \setminus \{i, j\}, \\ o_i + \alpha & k = i, \\ o_j - \alpha & k = j. \end{cases} \tag{20}$$

It can be easily checked that $o^* \in \text{Core}(N, c)$. Hence, we can conclude that $\text{Core}(N, c) \neq \{o\}$. □

The above proof sheds light onto the structure of the core. If there exists at least one essential player, the rest of players compensate him for cooperation by reducing the cost when producing at minimum production cost according to (20). On the contrary, if no player is essential the unique core-allocation is the one given by the cost generated when producing at minimum production cost.

We would like to conclude this section positioning the Owen point in comparison with other well-known allocations in cooperative game theory. Although due to space requirements we do not include examples in the paper, the interested reader can find examples supporting our claims in the working paper by Guardiola et al. (2004). First of all, in the class of PI-games the Shapley value (Shapley, 1953) is not, in general, a core allocation. Moreover, even in those cases when Shapley value belongs to the core it does not have to coincide with the Owen point. Finally, we point out that the Owen point does not coincide either with the nucleolus (Schmeidler, 1969).

6. Concluding remarks

In this paper we have presented a model of cooperation among several firms arising from dynamic production-inventory situations with discrete demand and finite planning horizon. In this model any group of firms can agree to cooperate because fair allocations of operation costs which are stable always exist. In this sense none of the firms would have an incentive to leave the group. From among all the above fair allocations, we propose the Owen point: the allocation

in which every firm has to pay the minimum cost of operation. This allocation is especially appealing since it can be calculated in polynomial time and, moreover, it can be reached through a pmas.

We note in passing that incorporating selling prices to the considered model would have not modified the structure of the problem, so that all the conclusions would have been the same.

We would like to finish these remarks mentioning some additional topics for further research on the cooperation model considered in this paper:

(1) to find out descriptions of the structure of the core,
(2) axiomatic characterizations of the Owen point,
(3) alternative allocation schemes, and
(4) other forms of cooperation and/or competition, as well as models with concave cost functions.

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References

Anupindi, R., Bassok, Y., Zemel, E., 2001. A general framework for the study of decentralized distribution systems. Manufacturing Service Operations Management 3, 349–368.

Bondareva, O.N., 1963. Some applications of linear programming methods to the theory of cooperative games. Prob. Kibernety 10, 119–139 (in Russian).

Borm, P.E.M., Hamers, H., Hendrickx, R., 2001. Operations research games: A survey. TOP 9, 139–216.

Deng, X., Ibaraki, T., Nagamochi, H., 1999. Algorithmic aspect of the core of combinatorial optimization games. Math. Oper. Res. 24, 751–766.

Deng, X., Ibaraki, T., Nagamochi, H., Zang, W., 2000. Totally balanced combinatorial optimization games. Math. Programm. 87, 441–452.

Dubey, P., Shapley, L.S., 1984. Totally balanced games arising from controlled programming problems. Math. Programm. 29, 245–267.

Eppen, G.D., 1979. Effect of centralization on expected cost in a multi-location newsboy problem. Manage. Sci. 25, 498–501.

Faigle, U., Kern, W., 2000. On the core of ordered submodular cost games. Math. Programm. 87, 483–499.

Garfinkel, R.S., Nemhauser, G.L., 1972. Integer Programming. Wiley, New York.

Gellekom, J.R.G., Potters, J.A.M., Reijnierse, J.H., Engel, M.C., Tijs, S.H., 2000. Characterization of the Owen set of linear production processes. Games Econ. Behav. 32, 139–156.

Gerchak, Y., Gupta, D., 1991. On apportioning costs to customers in centralized continuous review inventory systems. J. Oper. Manage. 10, 546–551.

Granot, D., 1986. A generalized linear production model: A unified model. Math. Programm. 34, 212–222.

Guardiola, L.A., Meca, A., Puerto, J., 2004. Production-inventory games: A new class of totally balanced combinatorial optimization games. Trabajos de I+D I-2004-8, CIO Universidad Miguel Hernández, Elche.

Hartman, B.C., Dror, M., 1996. Cost allocation in continuous review inventory models. Naval Res. Logist. 43, 549–561.

Hartman, B.C., Dror, M., 2003. Optimizing centralized inventory operations in a cooperative game theory setting. IIE Trans. 35, 243–257.

Hartman, B.C., Dror, M., 2005. Allocation of gains from inventory centralization in newsvendor environments. IIE Trans. 37, 93–107.

Hartman, B.C., Dror, M., Shaked, M., 2000. Cores of inventory centralization games. Games Econ. Behav. 31, 26–49.

Johnson, L.A., Montgomery, D.C., 1974. Operations Research in Production Planning, Scheduling, and Inventory Control. John Wiley & Sons, Chichester and New York.

Kohli, R., Park, H., 1989. A cooperative game model of quantity discounts. Manage. Sci. 35, 693–707.
Meca, A., 2007. A core-allocation family for generalized holding cost games. Math. Methods Oper. Res. In press.
Meca, A., García-Jurado, I., Borm, P.E.M., 2003. Cooperation and competition in inventory games. Math. Methods Oper. Res. 57, 481–493.
Meca, A., Timmer, J., García-Jurado, I., Borm, P.E.M., 2004. Inventory games. Europ. J. Operations Res. 156, 127–139.
Minner, S., 2007. Bargaining for cooperative economic ordering. Decis. Support Syst. 43, 569–583.
Müller, A., Scarsini, M., Shaked, M., 2002. The newsvendor game has a nonempty core. Games Econ. Behav. 38, 118–126.
Robinson, L.W., 1993. Comment on "On apportioning costs to customers in centralized continuous review inventory systems", by Gerchak and Gupta. J. Oper. Manage. 11, 99–102.
Schmeidler, D., 1969. The nucleolus of a characteristic function game. SIAM J. Appl. Math. 17, 1163–1170.
Shapley, L.S., 1953. A value for n-person games. In: Kuhn, H., Tucker, A.W. (Eds.), Contributions to the Theory of Games, vol. II. Princeton Univ. Press, Princeton, NJ, pp. 307–317.
Shapley, L.S., 1967. On balanced sets and cores. Naval Res. Logist. 14, 453–460.
Shapley, L.S., 1971. Cores of convex games. Int. J. Game Theory 1, 11–26.
Shapley, L.S., Shubik, M., 1969. On market games. J. Econ. Theory 1, 9–25.
Shapley, L.S., Shubik, M., 1972. The assignment game. Int. J. Game Theory. 1, 111–130.
Slikker, M., Fransoo, J., Wouters, M., 2005. Cooperation between multiple news-vendors with transshipments. Europ. J. Operations Res. 167, 370–380.
Sprumont, Y., 1990. Population monotonic allocation schemes for cooperative games with transferable utility. Games Econ. Behav. 2, 378–394.
Tamir, A., 1992. On the core of cost allocation games defined on location problems. Transp. Sci. 27, 81–86.
Tijs, S.H., Meca, A., López, M.A., 2005. Benefit sharing in holding situations. Europ. J. Operations Res. 162, 251–269.
Wagner, H.M., Whitin, T.M., 1958. Dynamic version of the economic lot size model. Manage. Sci. 5, 89–96.

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关键词：可转移效用合作博弈；平衡博弈；核心；凸多面体；组合多胞体

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推荐理由：本文研究具有非空核心的可转移效用合作博弈集合的几何结构。邦达列娃与夏普利将这类博弈刻画为平衡博弈。本文证明，该集合是一个非尖多面锥，并确定其极射线与面。对于大联盟价值固定的平衡博弈集合，本文也开展了同样的研究，得到一个仿射非尖多面锥。最后，考察大联盟价值固定的非负平衡博弈集合。这一集合是具有若干显著性质的凸多胞体。本文刻画其顶点与面，研究顶点的邻接结构，提出按均匀分布随机生成顶点的算法，并证明该多胞体是组合多胞体，其邻接图为哈密顿图。最后，给出核心退化为单点的博弈集合的刻画。

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Given v. what's C(v)? → What is the geometry of v. s.t. C(v)≠∅ ?

On the Set of Balanced Games

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Abstract. We study the geometric structure of the set of cooperative transferable utility games having a nonempty core, characterized by Bondareva and Shapley as balanced games. We show that this set is a nonpointed polyhedral cone, and we find the set of its extremal rays and facets. This study is also done for the set of balanced games whose value for the grand coalition is fixed, which yields an affine nonpointed polyhedral cone. Finally, the case of nonnegative balanced games with fixed value for the grand coalition is tackled. This set is a convex polytope, with remarkable properties. We characterize its vertices and facets, study the adjacency structure of vertices, develop an algorithm for generating vertices in a random uniform way, and show that this polytope is combinatorial and its adjacency graph is Hamiltonian. Last, we give a characterization of the set of games having a core reduced to a singleton.

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Keywords: cooperative TU games • balanced games • core • convex polyhedra • combinatorial polytope

1. Introduction

Given a set of players N, a game with transferable utility, abbreviated hereafter as TU game or simply game, is a mapping v assigning to each subset $S \subseteq N$ (called a coalition) a quantity $v(S)$, representing, for example, the benefit of cooperation of the players in S. The core of a game appears to be a fundamental concept, introduced by Gillies [6]: It is the set of payment vectors $x \in \mathbb{R}^N$ to players, such that every coalition S receives at least the quantity $v(S)$, under the constraint that the total payment which is distributed is equal to $v(N)$. This set, however, may be empty. Bondareva [1], and independently Shapley [19], found a sufficient and necessary condition for a game to have a nonempty core. Games satisfying this condition are called *balanced* by Shapley, because it is based on particular arrangements of players, called *(minimal) balanced collections*, which generalize the notion of partition of a set. As the number of minimal balanced collections is finite, this condition amounts to check a finite number of linear inequalities, one for each possible minimal balanced collection. This shows that the set of balanced games is a (closed convex) polyhedron. To the best of our knowledge, there is only one article studying related polyhedra (Kroupa and Studený [10]), where the facets of the polyhedra of totally balanced, balanced, and exact games are studied, but in that paper, it is assumed that $v(\emptyset)$ might attain a nonnull value, and hence the structures of the polyhedra are different.

The main aim of our paper is precisely to fill this gap. The game-theoretic motivation behind this mathematical study, apart its own interest, is to solve (in a future study) the problem of finding the closest balanced game to a nonbalanced game, as that would provide a new solution concept for games with an empty core. This problem amounts to an orthogonal projection on a polyhedron, a problem that has no analytical solution in general (Rutkowski [17]). The only hope to solve it is a deep understanding of the structure of the polyhedron under consideration, which is exactly what the present paper does.

Before elaborating on our findings, we emphasize that this topic is not limited to (cooperative) game theory. Games as defined above are merely set functions vanishing on the empty set and are encountered in many domains linked to operations research, in particular decision theory, voting theory, combinatorial optimization and reliability (see Grabisch [7] for details). In decision theory, *capacities* (Choquet [2]) are a particular class of games that is monotonic and represents uncertainty on the set of states of nature (Schmeidler [18]). In this context, the core is the set of probability measures that are "compatible with" (i.e., eventwise dominating) a given capacity. Voting theory considers only 0-1–valued monotone games, which are called *simple games* (Peters [16]). They represent the power of coalitions of players to win an election. They are in fact special Boolean functions, whereas games are nothing other than pseudo-Boolean functions, as introduced by Hammer (Foldes and

Hammer [4]). In combinatorial optimization, submodular games are considered, as they are encountered for example as rank function of a polymatroid (Edmonds [3]), and the core corresponds to the base polyhedron of a matroid (see the monograph of Fujishige devoted to this topic [5]). Simple games are also used in reliability theory where they indicate the state of a system (functioning or not), depending on the states of its components. We mention finally that our study is also linked to combinatorics and the study of polytopes and polyhedra. Indeed, the concept of minimal balanced collection generalizes the concept of partitions, and their enumeration remains an open problem. Also, they are closely related to many geometrical properties of the games and the core (see Laplace Mermoud et al. [11] for details). As a conclusion, our results are not limited to (cooperative) game theory but concern many fields of operations research, as well as combinatorics and polyhedra.

We split our study in three parts, considering three different sets of balanced games. The first one is simply the set of all possible balanced games on N, which we denote by $\mathcal{BG}(n)$, with $|N| = n$. For the second one we impose the restriction that $v(N) = \alpha$ for some fixed $\alpha \in \mathbb{R}$. We denote this set by $\mathcal{BG}_\alpha(n)$. Last, we impose in addition that v should be a nonnegative function, with $\alpha = 1$, without loss of generality (w.l.o.g.). We denote the set of such balanced game as $\mathcal{BG}_+(n)$. It follows that $\mathcal{BG}_+(n)$ is a polytope, and we give the complete characterization of its vertices, as well as an algorithm to randomly generate them in a uniform way. Interestingly, the number of vertices appears to be a known integer sequence related to Boolean functions. We also characterize the adjacency of vertices, and finally we show that this polytope is combinatorial (in the sense of Naddef and Pulleyblank [14]), that is, the adjacency graph of its vertices is Hamilton connected.

We show that $\mathcal{BG}(n)$ is a nonpointed cone and identify its lineality space, all its extremal rays, and all its facets, obtaining thus a complete description of the polyhedron. The study of $\mathcal{BG}_\alpha(n)$ is similar and shows that it is an affine nonpointed cone. Again, its structure is completely elucidated.

Last, we address the following question: For which balanced games is the core reduced to a singleton? The question is of interest because a core reduced to a singleton provides a very simple and handy solution to a game. We solve completely this question for $\mathcal{BG}(n)$ and for $\mathcal{BG}_\alpha(n)$. Some open questions remain for $\mathcal{BG}_+(n)$.

The paper is organized as follows. Section 2 introduces the necessary material on games and balanced collections. In Section 3, we study the cone $\mathcal{BG}(n)$, its lineality space, and its extremal rays and facets, whereas Section 4 is devoted to the study of the affine cone $\mathcal{BG}_\alpha(n)$. Section 5 is devoted to the study of the polytope $\mathcal{BG}_+(n)$: dimension, vertices and their enumeration, adjacency, and facets. We also provide a random procedure to generate vertices of this polytope. In Section 6, we address the problem of finding balanced games whose core is reduced to a singleton. Section 7 is devoted to point some possible applications of these results. Section 8 gives some concluding remarks.

2. Basic Concepts

We refer the reader to Peleg and Sudhölter [15] and Grabisch [7] for full details, proofs, and references. We limit here ourselves to the essential. Throughout the paper, we consider a (fixed) set N of n players, simply denoted by $N = \{1, \dots, n\}$. *Coalitions* are nonempty subsets of N, denoted by capital letters S, T, and so on. A *TU game* (N, v) (or simply a game v) is a function $v : 2^N \to \mathbb{R}$ satisfying $v(\emptyset) = 0$. The value $v(S)$ represents the maximal value (benefit) that the coalition S can guarantee, no matter what players outside S might do. We will denote by $\mathcal{G}(N)$ (or simply $\mathcal{G}(n)$, as no subgame will be considered) the set of games v on N.

For further use, we introduce $\mathcal{G}_+(n)$, the set of games $v \in \mathcal{G}(n)$ such that $v \geq 0$ (i.e., $v(S) \geqslant 0$ for all $S \in 2^N$) and $v(N) = 1$, and for every $\alpha \in \mathbb{R}$ we introduce $\mathcal{G}_\alpha(n)$, the set of games $v \in \mathcal{G}(n)$ such that $v(N) = \alpha$. In addition, we will often use the following families of games, which are bases of the $(2^n - 1)$-dimensional vector space $\mathcal{G}(n)$:

1. The *Dirac games* δ_S, $\emptyset \neq S \subseteq N$, defined by
$$\delta_S(T) = \begin{cases} 1, & \text{if } T = S \\ 0, & \text{otherwise.} \end{cases}$$
2. The *unanimity games* u_S, $\emptyset \neq S \subseteq N$, defined by
$$u_S(T) = \begin{cases} 1, & \text{if } T \supseteq S \\ 0, & \text{otherwise.} \end{cases}$$

Assuming that all players agree to form the grand coalition N, we look for a way to share the benefit $v(N)$ among all players, that is, for an *allocation* or *payment vector* $x \in \mathbb{R}^N$, where coordinate x_i indicates the payoff given to player i. For any coalition S, we denote by $x(S) := \sum_{i \in S} x_i$ the total payoff given to the players in S. An allocation is *efficient* if $x(N) = v(N)$. A systematic way of assigning a set of allocations to a game is called a *solution concept*.

In this paper we focus on one of the best known solution concepts, which is the *core* (Gillies [6]). The core is the set of efficient allocations satisfying *coalitional rationality*, which means that $x(S) \geqslant v(S)$ for all coalitions S. Under this condition, no coalition S has an incentive to leave the grand coalition N to form a subgame on S. The core of a game (N, v) is denoted by $C(N, v)$ (or $C(v)$ for short if N is fixed):

$$C(v) := \{x \in \mathbb{R}^n : x(S) \geqslant v(S),\ \forall S \in 2^N, x(N) = v(N)\}.$$

The core is a closed convex polytope that may be empty, as illustrated in the following example.

Example 1. The unanimity game u_S has a nonempty core for any $\emptyset \neq S \subseteq N$. Indeed, $x = 1/|S|\mathbf{1}^S$ is a core allocation, where $\mathbf{1}^S$ is the characteristic vector of S, that is, $\mathbf{1}^S_i = 1$ if $i \in S$ and zero otherwise. Moreover, it can be easily checked that

$$C(u_S) = \{x \in \mathbb{R}^n : x(S) = 1, x_j = 0, j \notin S\}.$$

However, any Dirac game δ_S has an empty core, except when $S = N$, as it is easy to check.

A condition for nonemptiness of the core has been given by Bondareva [1] and Shapley [19], which we detail below.

A *balanced collection* $\mathcal{B}$ on N is a family of nonempty subsets of N such that there exist positive (balancing) weights $\lambda^{\mathcal{B}}_S$, $S \in \mathcal{B}$, satisfying

$$\sum_{\substack{S \in \mathcal{B} \\ S \ni i}} \lambda^{\mathcal{B}}_S = 1,\ \forall i \in N.$$

This notion is an extension of the notion of partition, as any partition is a balanced collection with balancing weights all equal to one. A balanced collection is *minimal* if it contains no balanced proper subcollection. We denote by $\mathfrak{B}^*(n)$ the set of all minimal balanced collections on a set N of cardinality n, excluding the collection $\{N\}$. It can be shown that minimal balanced collections (abbreviated hereafter by m.b.c.) have a unique set of balancing weights and that their cardinality is at most n. Moreover, if $\mathcal{B}$ is a m.b.c. with balancing weights $(\lambda^{\mathcal{B}}_S)_{S \in \mathcal{B}}$, then $\overline{\mathcal{B}} := \{N \setminus S : S \in \mathcal{B}\}$ is also a m.b.c. with balancing weights

$$\lambda^{\overline{\mathcal{B}}}_S = \frac{\lambda^{\mathcal{B}}_{N \setminus S}}{\sum_{T \in \mathcal{B}} \lambda^{\mathcal{B}}_T - 1}.$$

A game v is *balanced* if

$$\sum_{S \in \mathcal{B}} \lambda^{\mathcal{B}}_S v(S) \leqslant v(N), \quad \forall \mathcal{B} \in \mathfrak{B}^*(n). \tag{1}$$

The following result holds (Bondareva [1], Shapley [19]).

Theorem 1. *Consider a game (N, v). Then, $C(v) \neq \emptyset$ if and only if v is balanced.*

We will denote by $\mathcal{BG}(N)$ the set of games in $\mathcal{G}(N)$ being balanced, that is, the set of games having a nonempty core. For the same reasons as before, we simplify the notation to $\mathcal{BG}(n)$. Hence, applying Theorem 1 and (1), we get

$$\mathcal{BG}(n) = \left\{ v \in \mathcal{G}(n) : \sum_{S \in \mathcal{B}} \lambda^{\mathcal{B}}_S v(S) - v(N) \leqslant 0, \quad \forall \mathcal{B} \in \mathfrak{B}^*(n) \right\}. \tag{2}$$

For application purposes and further studies, the following subsets of $\mathcal{BG}(n)$ are of interest:

1. The set of balanced games $v \in \mathcal{G}_\alpha(n)$ for some $\alpha \in \mathbb{R}$. We denote by $\mathcal{BG}_\alpha(n)$ the set of such games, that is,

$$\mathcal{BG}_\alpha(n) = \left\{ v \in \mathcal{G}_\alpha(n) : \sum_{S \in \mathcal{B}} \lambda^{\mathcal{B}}_S v(S) \leqslant \alpha, \quad \forall \mathcal{B} \in \mathfrak{B}^*(n) \right\}. \tag{3}$$

The study of this set is motivated as follows: When searching for the closest balanced game v' to a given nonbalanced game v, it is natural to impose that $v'(N) = v(N)$, as this is the total benefit which has to be distributed among players.

2. The set of balanced games $v \in \mathcal{G}_+(n)$. We denote by $\mathcal{BG}_+(n)$ the set of such games, that is,

$$\mathcal{BG}_+(n) = \left\{ v \in \mathcal{G}_+(n) : \sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} v(S) \leqslant 1, \quad \forall \mathcal{B} \in \mathfrak{B}^*(n) \right\}. \tag{4}$$

Studying this set is motivated by the fact that many applications concern games with nonnegative values. Therefore, it would make no sense to find a closest balanced game taking negative values.

Observe that if $v \geqslant 0$ and $v(N) = \alpha \neq 0$, then v is balanced if and only if (iff) $(1/\alpha)v \in \mathcal{BG}_+(n)$. This shows that studying the balancedness of nonnegative games is equivalent to study the balancedness of nonnegative games with $v(N) = 1$, hence the set $\mathcal{BG}_+(n)$, except if $v(N) = 0$, but the latter case is trivial as only $v = 0$ is balanced.

Note that $\mathcal{BG}_+(n), \mathcal{BG}(n)$ and $\mathcal{BG}_\alpha(n)$ are convex polyhedra. The next sections are devoted to the study of the structure of these polyhedra.

3. Polyhedron $\mathcal{BG}(n)$

Because for any game $v \in \mathcal{G}(n)$, $v(\emptyset)$ is fixed, we consider v as being a vector in $\mathbb{R}^{2^N \setminus \{\emptyset\}}$. We start by showing a technical result, which will be useful in the sequel.

Lemma 1. *Let v be such that*

$$\sum_{S \in \Pi} v(S) = v(N)$$

for all partitions Π of N of the following form: either $\Pi = N^\perp := \{\{1\}, \ldots, \{n\}\}$ or $\Pi = \{S, (N \setminus S)^\perp\}$ for any $S \subset N, |S| > 1$, with $T^\perp = \{\{i\}, i \in T\}$. Then

$$\sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} v(S) = v(N), \quad \forall \mathcal{B} \in \mathfrak{B}^*(n).$$

Proof. Assume v satisfies the assumption. Take any m.b.c. $\mathcal{B}$ with balancing weights $(\lambda_S^{\mathcal{B}})_{S \in \mathcal{B}}$ and denote $\overline{\mathcal{B}} = \{N \setminus S, S \in \mathcal{B}\}$. Then, $\overline{\mathcal{B}}$ is a m.b.c. with weights $\lambda_S^{\overline{\mathcal{B}}} = \frac{\lambda_{N \setminus S}^{\mathcal{B}}}{\sum_{T \in \mathcal{B}} \lambda_T^{\mathcal{B}} - 1}$ (see Section 2). It follows that

$$\begin{aligned}
\sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} v(S) &= \sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} \left(v(N) - \sum_{i \in N \setminus S} v(\{i\}) \right) = v(N) \sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} - \sum_{i \in N} \left[v(\{i\}) \sum_{\substack{N \setminus S \in \overline{\mathcal{B}} \\ N \setminus S \ni i}} \lambda_S^{\mathcal{B}} \right] \\
&= v(N) \sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} - \sum_{i \in N} v(\{i\}) \sum_{\substack{N \setminus S \in \overline{\mathcal{B}} \\ N \setminus S \ni i}} \left[\lambda_{N \setminus S}^{\overline{\mathcal{B}}} \left(\sum_{T \in \mathcal{B}} \lambda_T^{\mathcal{B}} - 1 \right) \right] \\
&= v(N) \sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} - \left(\sum_{T \in \mathcal{B}} \lambda_T^{\mathcal{B}} - 1 \right) \underbrace{\sum_{i \in N} v(\{i\})}_{v(N)} = v(N). \quad \square
\end{aligned}$$

Remark that the reciprocal holds as all partitions are minimal balanced collections with balanced weights equal to one.

Let us first establish the structure of $\mathcal{BG}(n)$.

3.1. Structure of $\mathcal{BG}(n)$

Theorem 2. *Let $n \geqslant 2$. Then, $\mathcal{BG}(n)$ is a $(2^n - 1)$-dimensional polyhedral cone, which is not pointed. Its lineality space has dimension n, with basis $(w_i)_{i \in N}$, with $w_i = \sum_{S \ni i} \delta_S$, where δ_S is the Dirac game centered on S (see Section 2).*

Proof. We make the proof in two steps.

1. First, observe that the zero game is balanced, and if v is balanced, then αv with $\alpha \geqslant 0$ is also balanced. Then $\mathcal{BG}(n)$ is a cone. Moreover, it is a polyhedral cone because it is defined by a finite number of linear inequalities. Next, observe that no equality can be implied by System (2), as for any $S \neq \emptyset$, all coefficients of $v(S)$ are of the same sign. Because, in addition there is no equality, the cone is full dimensional.

2. Denoting by $Av \leqslant 0$ the system of inequalities in (2), the lineality space is the set of solutions of $Av = 0$. From Lemma 1, we infer that this is equivalent to solve the system

$$\sum_{S\in\Pi} v(S) - v(N) = 0$$

for all partitions Π of N of the following form: either $\Pi = N^{\perp} := \{\{1\},\dots,\{n\}\}$ or $\Pi = \{S,(N\setminus S)^{\perp}\}$ for $S \subset N, |S| > 1$.

Therefore, we obtain that for any $S \subset N, |S| > 1$, $v(S) = v(N) - \sum_{i\in N\setminus S} v(\{i\})$, and $v(N) = \sum_{i\in N} v(\{i\})$. It follows that its set of solutions is, expressing all coordinates in terms of those of $v(\{i\})$, $i = 1,\dots,n$,

$$\left\{\left(v_1,\dots,v_n, v_1+v_2,\dots,\sum_{i\in S} v_i,\dots,\sum_{i\in N} v_i\right) : v_1,\dots,v_n \in \mathbb{R}\right\}.$$

A basis for this subspace is $\{w_1,\dots,w_n\}$ with

$$w_i = \sum_{S\ni i} \delta_S. \quad \square$$

The reader may compare this result with lemma 3.1 in Kroupa and Studený [10], where it is proved that modular games form the lineality space of the cone of balanced games, totally balanced games, and exact games.

As $\mathcal{BG}(n)$ is not pointed, it can be decomposed as the direct sum of its lineality space of dimension n (that we will denote by $\mathsf{Lin}(\mathcal{BG}(n))$) and a supplementary subspace (not necessarily orthogonal) of dimension $2^n - n - 1$, which is a pointed cone and whose extremal rays can be found in the usual way. However, because there is no unique way to choose a supplementary space, there is no unique representation of $\mathcal{BG}(n)$ by extremal rays. It seems that in the case of $\mathcal{BG}(n)$ the orthogonal supplement does not yield simple results, and we will use instead the supplement where the coordinates corresponding to singletons are zero. We denote this set as $\mathcal{BG}^0(n)$, and hence we have

$$\mathcal{BG}(n) = \mathsf{Lin}(\mathcal{BG}(n)) \oplus \mathcal{BG}^0(n).$$

We study in the next section the extremal rays of $\mathcal{BG}(n)$.

3.2. Extremal Rays of $\mathcal{BG}(n)$

Theorem 3. *Let $n \geqslant 2$. The extremal rays of $\mathcal{BG}(n)$ are*

- *The $2n$ extremal rays corresponding to $\mathsf{Lin}(\mathcal{BG}(n))$: $w_1,\dots,w_n,-w_1,\dots,-w_n$;*
- *The $2^n - n - 2$ extremal rays of the form $r_S = -\delta_S$, $S \subset N, |S| > 1$;*
- *The n extremal rays of the form*

$$r_i = \sum_{S\ni i,\, |S|>1} \delta_S, \quad i \in N.$$

This yields in total $2^n + 2n - 2$ extremal rays.

Proof. We study each part separately.

1. The $2n$ extremal rays corresponding to $\mathsf{Lin}(\mathcal{BG}(n))$ space come from the vectors obtained in Theorem 2.

2. Let us consider $S \subset N, |S| > 1$ and show that $r_S = -\delta_S$ is an extremal ray. Let us denote by $r_S(T)$ the coordinate of r_S corresponding to subset T and similarly for all rays. First, r_S is a ray of $\mathcal{BG}^0(n)$ because it satisfies System (2) of inequalities and $r_S(\{i\}) = 0$ for all $i \in N$.

Suppose that it is not extremal. Then, there exist two rays $r, r' \in \mathcal{BG}^0(n)$ not proportional to r_S such that $r_S = r + r'$. Suppose that $r(T) > 0$ (w.l.o.g. $r(T) = 1$) for some $T \neq S$, $1 < |T| < n$. Then, $r'(T) = -1$. Using the partition $\{T,(N\setminus T)^{\perp}\}$, because $r(\{i\}) = 0$ for all $i \in N$, the corresponding inequality in (2) yields $r(N) \geqslant 1$ because the weights for a m.b.c. being a partition are one, and because $r'(N) = -r(N)$, we get $r'(N) \leqslant -1$. Taking now the partition $N^{\perp}$, we obtain

$$0 - r'(N) \leqslant 0,$$

which is impossible. Letting $r(T) < 0$ yields the same contradiction by inverting the roles of r and r'. We conclude that $r(T) = 0$, $\forall T \neq S$. Finally, observe that $r(S) > 0$ is not possible as the inequality for partition $\{S,(N\setminus S)^{\perp}\}$ would not be satisfied. This finishes the proof.

3. Let us take $i \in N$ and show that r_i is an extremal ray. First, as

$$\sum_{S\in\mathcal{B},\, S\ni i} \lambda_S^{\mathcal{B}} = 1,\ \forall \mathcal{B} \in \mathfrak{B}^*(n),$$

it follows that r_i satisfies all inequalities in (2); moreover, $r_i(\{j\}) = 0$ for all $j \in N$ and hence, r_i is a ray of $\mathcal{BG}^0(n)$. To show that it is extremal, we need to show that the set of solutions of the subsystem of (2) formed by tight inequalities has dimension 1 in $\mathcal{BG}^0(n)$. Consider a m.b.c. $\mathcal{B}$ and its corresponding inequality:

$$\sum_{S\in\mathcal{B}} \lambda_S^{\mathcal{B}} r_i(S) - r_i(N) \leqslant 0.$$

We obtain

$$\sum_{S\in\mathcal{B}} \lambda_S^{\mathcal{B}} r_i(S) - r_i(N) = \sum_{\substack{S\in\mathcal{B}\\ S\ni i\\ |S|>1}} \lambda_S^{\mathcal{B}} - 1.$$

Hence, the inequality is tight iff $\{i\} \notin \mathcal{B}$. Let us call (2)′ the subsystem of tight inequalities in v, with $v \in \mathcal{BG}^0(n)$. Consider the partition $\{S, (N \setminus S)^{\perp}\}$ with $1 < |S| < n$. In terms of (2)′, the corresponding equality is $v(S) = v(N)$. Now, consider any m.b.c. $\mathcal{B}$ such that $\{i\} \notin \mathcal{B}$. The corresponding equality reads

$$\sum_{\substack{S\in\mathcal{B}\\ S\ni i}} \lambda_S^{\mathcal{B}} v(S) - v(N) = 0.$$

Substituting $v(S)$ by $v(N)$ yields $v(N) - v(N) = 0$. Therefore, the subspace of solutions is given by $\{(\alpha, \dots, \alpha), \alpha \in \mathbb{R}\}$, of dimension 1.

4. It remains to prove that there is no other extremal ray. Consider a ray w in $\mathcal{BG}^0(n)$, hence satisfying $w(\{i\}) = 0$ for all $i \in N$, and (2). Suppose w is not a conic combination of the extremal rays r_S and r_i, $S \subset N, |S| > 1$, $i \in N$; that is, the following system has no solution in α_S, α_i:

$$\begin{aligned} \sum_{S\subset N,\, |S|>1} \alpha_S r_S + \sum_{i\in N} \alpha_i r_i &= w \\ \alpha_S &\geqslant 0, \quad S \subset N, |S| > 1 \\ \alpha_i &\geqslant 0, \quad i \in N. \end{aligned} \tag{5}$$

Using definitions of r_S and r_i and omitting coordinates for singletons in r_S, r_i, w as they are all zero, we obtain that the previous system can be written as

$$\begin{aligned} -\alpha_S + \sum_{i\in S} \alpha_i &= w(S), \quad S \subset N, |S| > 1 \\ \sum_{i\in N} \alpha_i &= w(N) \\ \alpha_S &\geqslant 0, \quad S \subset N, |S| > 1 \\ \alpha_i &\geqslant 0, \quad i \in N. \end{aligned} \tag{6}$$

We may denote (with some abuse) the whole system (6) by $A\alpha \geqslant b$ in matrix notation. If this system has no solution, then, by Farkas' lemma, this is equivalent to say that there exists a vector $[y\ z\ t]$ with coordinates $y_S \in \mathbb{R}$, $S \subseteq N, |S| > 1$, $z_T \geqslant 0$, $T \subset N, |T| > 1$, and $t_i \geqslant 0$, $i \in N$, such that

$$[y\ z\ t]^{\top} A = 0, \quad [y\ z\ t]^{\top} b > 0.$$

Observe that the only vectors $[y\ z\ t]$ such that $[y\ z\ t]^{\top} A = 0$ have the form:

$$y_S = 1, \quad z_S = 1\,(S \subset N, |S| > 1), \quad y_N = -2^{n-1} + 2 - t, \quad t_1 = \cdots = t_n = t,$$

up to a multiplicative factor $K > 0$.

Then, we obtain

$$[y\ z\ t]^\top b = \sum_{S\subset N,\,|S|>1} w(S) - (2^{n-1}-2+t)w(N).$$

Observe that $\{S \subset N, |S| > 1\}$ is a balanced collection (not minimal if $n > 3$) with balancing weights $\frac{1}{2^{n-1}-2}$. It follows that

$$\sum_{S\subset N,\,|S|>1} w(S) \leqslant (2^{n-1}-2)w(N).$$

Therefore, $[y\ z\ t]^\top b \leqslant (2^{n-1}-2)w(N) - (2^{n-1}-2+t)w(N) = -tw(N) \leqslant 0$. The last inequality follows from $t \geqslant 0$ and (6). Hence, this system has no solution, and consequently, System (6) always has a solution. Therefore, w is not extremal and the result follows. □

Compare also this result with lemma 5.4 in Kroupa and Studený [10], giving condition for a ray to be extreme in the dual cone, and corollary 5.1. mentioning the extremal rays $-\delta_S$.

The following result is immediate.

Lemma 2. *The cores of w_i, $-w_i$, r_i, and r_S for all $i \in N$, $S \subset N$, $|S| > 1$ are singletons (respectively, $\{\mathbf{1}^{\{i\}}\}$, $\{-\mathbf{1}^{\{i\}}\}$, $\{\mathbf{1}^{\{i\}}\}$, $\{0\}$).*

3.3. Facets

Theorem 4. *Each inequality in (2) defines a facet, that is, minimal balanced collections in $\mathfrak{B}^*(n)$ are in bijection with the facets of $\mathcal{BG}(n)$.*

Proof. Let us consider $\mathcal{B} \in \mathfrak{B}^*(n)$ and the inequality

$$\sum_{S\in\mathcal{B}} \lambda_S^{\mathcal{B}} v(S) - v(N) = 0. \tag{7}$$

It suffices to show that this face contains $2^n - 2$ independent extremal rays. We already know by definition of the lineality space that any face contains the n rays $w_1, \ldots, w_n$ of the lineality space.

Among the extremal rays r_S, $S \subset N$, $|S| > 1$, it is easy to check that only those such that $S \notin \mathcal{B}$ satisfy (7). There are $2^n - n - 2 - |\mathcal{B}| + k_{\mathcal{B}}$ such rays, where $k_{\mathcal{B}}$ is the number of singletons in $\mathcal{B}$.

Next, define $B = N \setminus \cup_{\{i\}\in\mathcal{B}}\{i\}$. Then $|B| = n - k_{\mathcal{B}}$. Now, observe that any r_i with $i \in \mathcal{B}$ is satisfying (7). Indeed,

$$\sum_{S\in\mathcal{B}} \lambda_S^{\mathcal{B}} r_i(S) - r_i(N) = \sum_{\substack{S\in\mathcal{B}\\ S\ni i\\ |S|>1}} \lambda_S^{\mathcal{B}} - 1 = 0.$$

This makes another set of $n - k_{\mathcal{B}}$ rays. Therefore, we have in total $2^n - 2 + n - |\mathcal{B}| \geqslant 2^n - 2$, as $\mathcal{B}$ is minimal (see Section 3). It remains to prove independence. Observe that every $\delta_{\{i\}}$, $i \in N$, is used by w_i, every δ_S, $|S| > 1$, $S \notin \mathcal{B}$, is used by r_S, and every δ_S for $S \in \mathcal{B}$ is used in some of the r_i's with $i \in B$, unless $S \subseteq N \setminus B$. However, then consider $\mathcal{B}' = \mathcal{B} \setminus \{S\}$ and weights

$$\lambda_T^{\mathcal{B}'} = \begin{cases} \lambda_i^{\mathcal{B}} + \lambda_S^{\mathcal{B}}, & T = \{i\}, i \in S \\ \lambda_T^{\mathcal{B}'}, & \text{otherwise,} \end{cases}$$

and hence, $\mathcal{B}$ is not minimal, a contradiction. If $|\mathcal{B}| = n$, then we have exactly $2^n - 2$ extremal rays, which are independent by the above argument. If $|\mathcal{B}| < n$, then $n - |\mathcal{B}|$ rays r_i have to be removed to make the family independent. □

4. Polyhedron $\mathcal{BG}_\alpha(n)$

Let us now study the set $\mathcal{BG}_\alpha(n)$, defined by (see (3)):

$$\mathcal{BG}_\alpha(n) = \left\{ v \in \mathcal{G}(n) : \sum_{S\in\mathcal{B}} \lambda_S^{\mathcal{B}} v(S) \leqslant \alpha,\ \ \forall \mathcal{B} \in \mathfrak{B}^*(n) \right\}.$$

We follow the same notation as in Section 3, except that now v is a vector in $\mathbb{R}^{2^N\setminus\{\emptyset,N\}}$. As the proof techniques are similar to the case of $\mathcal{BG}(n)$, the proofs are relegated to the appendix or omitted. We begin by expliciting the structure of $\mathcal{BG}_\alpha(n)$.

Theorem 5. *Let $n \geqslant 2$, $\alpha \in \mathbb{R}$. Then, $\mathcal{BG}_\alpha(n)$ is a (2^n-2)-dimensional affine cone,*[1] *which is not pointed. Its lineality space has dimension $n-1$, with base $(w_i)_{i\in N\setminus\{n\}}$, where*

$$w_i = \sum_{\substack{S\ni i\\ S\not\ni n}} \delta_S - \sum_{\substack{S\not\ni i\\ S\ni n}} \delta_S.$$

The affine space is given by

$$\sum_{i\in N} v(\{i\}) = \alpha$$
$$v(S) + \sum_{i\in N\setminus S} v(\{i\}) = \alpha, \quad S \subset N, |S| > 1.$$

In Theorem 5, element n plays a particular role, but this choice is arbitrary. Indeed, the affine space in $\mathcal{BG}_\alpha(n)$ contains in particular the games $\alpha u_{\{i\}}$, for all $i \in N$.

To find the extreme rays of $\mathcal{BG}_\alpha(n)$, we proceed as for $\mathcal{BG}(n)$. Using the notation from the proof of Theorem 5, we write

$$\mathcal{BG}_\alpha(n) = \alpha u_{\{n\}} + \mathcal{C}_\alpha(n),$$

where $\mathcal{C}_\alpha(n)$ is a cone defined by System (12). Now,

$$\mathcal{C}_\alpha(n) = \mathsf{Lin}(\mathcal{C}_\alpha(n)) \oplus \mathcal{C}^0_\alpha(n),$$

where $\mathsf{Lin}(\mathcal{C}_\alpha(n))$ is the lineality space of $\mathcal{C}_\alpha(n)$ and $\mathcal{C}^0_\alpha(n)$ is its supplement where the coordinates of the singletons $\{1\},\dots,\{n-1\}$ are zero.

Theorem 6. *Let $n \geqslant 2$ and $\alpha \in \mathbb{R}$. The extremal rays of $\mathcal{BG}_\alpha(n)$ are*

- *The $2n-2$ extremal rays corresponding to $\mathsf{Lin}(\mathcal{C}_\alpha(n))$: $w_1,\dots,w_{n-1},-w_1,\dots,-w_{n-1}$;*
- *The 2^n-n-2 extremal rays of the form $r_S = -\delta_S$, $S \subset N$, $|S| > 1$;*
- *The n extremal rays of the form*

$$r_i = \sum_{\substack{S\ni i\\ S\not\ni n\\ |S|>1}} \delta_S - \sum_{\substack{S\not\ni i\\ S\ni n}} \delta_S, \quad i \in N\setminus\{n\},$$

and $r_n = -\delta_{\{n\}}$.

This yields in total $2^n + 2n - 4$ extremal rays.

Observe that by definition of the lineality space, the extremal rays w_i, $-w_i$ belong to every facet of $\mathcal{C}_\alpha(n)$. We also have the following result (proof is omitted).

Lemma 3. *The core of w_i, $-w_i$ for all $i \in N\setminus\{n\}$, r_i for all $i \in N$, r_S for all $S \subset N$, $|S| > 1$ or $S = \{n\}$, are reduced to singletons, which are $\{\mathbf{1}^{\{i\}} - \mathbf{1}^{\{n\}}\}$, $-\{\mathbf{1}^{\{i\}} + \mathbf{1}^{\{n\}}\}$, $\{\mathbf{1}^{\{i\}} - \mathbf{1}^{\{n\}}\}$, and $\{0\}$, respectively.*

Theorem 7. *Each inequality in (3) defines a facet; that is, minimal balanced collections in $\mathfrak{B}^*(n)$ are in bijection with the facets of $\mathcal{BG}_\alpha(n)$.*

The proof is similar to the one of Theorem 4 and is omitted.

5. Polytope $\mathcal{BG}_+(n)$

Finally, let us study $\mathcal{BG}_+(n)$. Recall that $\mathcal{BG}_+(n)$ is defined by (see (4)):

$$\mathcal{BG}_+(n) = \left\{ v \in \mathcal{G}_+(n) : \sum_{S\in\mathcal{B}} \lambda^{\mathcal{B}}_S v(S) \leqslant 1, \ \forall \mathcal{B} \in \mathfrak{B}^*(n) \right\}.$$

5.1. Dimension, Boundedness

Proposition 1. *The set $\mathcal{BG}_+(n)$ is a (2^n-2)-dimensional polytope.*

Proof. First, $\mathcal{BG}_+(n)$ is defined by a set of linear equations. Hence, it is a polyhedron of dimension at most 2^n-2 as $v(N) = 1$ is fixed. Moreover, the 2^n-1 unanimity games u_S, $S \neq \emptyset$ belong to $\mathcal{BG}_+(n)$ (see Example 1), and they are affinely independent because the 2^n-2 games $u_S - u_N$, $S \neq \emptyset, N$, are linearly independent, so that the polyhedron is 2^n-2-dimensional.

Second, the polyhedron is bounded. To see this, if $v \in \mathcal{BG}_+(n)$, it follows that $v(S) \leqslant 1$, $\forall S \neq \emptyset, N$. Indeed, if $v(S) > 1$, as v is balanced, there exists $x \in \mathbb{R}^N$ such that $x(T) \geqslant v(T)$ for all $T \subseteq N$, and $x(N) = 1$. However, as $v(\{i\}) \geqslant 0$, we have $x_i \geqslant 0$, which implies $x(S) \leqslant 1$, a contradiction. □

5.2. Vertices

Theorem 8. *Consider $v \in \mathcal{G}_+(n)$. Then, v is a vertex of $\mathcal{BG}_+(n)$ if and only if v is balanced and 0-1 valued.*

Proof. ⇐) Suppose v is balanced and 0-1 valued. Assume v is not extremal. Then, there exist $v', v'' \in \mathcal{BG}_+(n)$ such that $v = (v' + v'')/2$. Take (if it exists) $S \neq N$ such that $v(S) = 1$. We have already shown in the proof of Proposition 1 that $v'(S) > 1$ is impossible. Similarly, $v'(S) < 1$ is impossible as it imposes $v''(S) > 1$. It follows that $v'(S) = v''(S) = 1$ for all $S \subseteq N$ such that $v(S) = 1$.

Take now $\emptyset \neq S \subset N$ such that $v(S) = 0$. Taking $v'(S) > 0$ forces $v''(S) < 0$, which is impossible. Therefore, $v'(S) = v''(S) = 0$ for all $S \subset N$ s.t. $v(S) = 0$. We conclude that $v' = v'' = v$; that is, v is an extreme point.

⇒) Take v balanced and extremal and suppose by contradiction that there exists $S \subset N$ s.t. $0 < v(S) < 1$. We distinguish two cases. Suppose first that there exists some core element $x \in C(v)$ such that $x(S) > v(S)$. Then, consider the two games v', v'' that differ from v only inasmuch as $v'(S) = v(S) - \epsilon$ and $v''(S) = v(S) + \epsilon$, with $0 < \epsilon < x(S) - v(S)$. Then v', v'' are balanced because $x \in C(v')$ and $x \in C(v'')$. Therefore, v is not extremal as $v = (v' + v'')/2$.

Suppose now that no such core element exists, that is, $x(S) = v(S)$ for all $x \in C(v)$. As $x(S) = v(S) < 1$, there exists $j \in N \setminus S$ such that $x_j > 0$. Similarly, as $0 < v(S) = x(S)$, we conclude that there exists $i \in S$ such that $x_i > 0$. We define the two games v', v'' by

$$\begin{aligned} &v'(T) = v(T) + \epsilon, \quad v''(T) = v(T) - \epsilon, \quad \forall T \text{ such that } i \in T, j \notin T \\ &v'(T) = v(T) - \epsilon, \quad v''(T) = v(T) + \epsilon, \quad \forall T \text{ such that } j \in T, i \notin T \\ &v'(T) = v(T), \quad v''(T) = v(T), \quad \text{otherwise,} \end{aligned}$$

with $\epsilon > 0$ small enough so that $v'(T), v''(T) \in [0,1]$, and $0 < x_i - \epsilon < x_i + \epsilon < 1$, $0 < x_j - \epsilon < x_j + \epsilon < 1$. Clearly, $v = (v' + v'')/2$. Observe that v', v'' are balanced, as $x', x'' \in \mathbb{R}^N$ defined by

$$x'_i = x_i + \epsilon, x'_j = x_j - \epsilon, \quad x''_i = x_i - \epsilon, x''_j = x_j + \epsilon, \quad x'_k = x''_k = x_k, \ \forall k \neq i, j,$$

are core elements of v' and v'', respectively. Hence, v is not extremal. □

Hence, a vertex v of $\mathcal{BG}_+(n)$ is characterized in terms of the subsets $S \in 2^N \setminus \{\emptyset, N\}$ such that $v(S) = 1$. Let us denote the family of such subsets by $\mathcal{D}$. In next result, we treat the reciprocal problem; that is, we give the necessary and sufficient conditions for a family of subsets $\mathcal{D}$ in $2^N \setminus \{\emptyset, N\}$ to determine a vertex of $\mathcal{BG}_+(n)$.

Theorem 9. *Let $\mathcal{D}$ be a family of subsets $\mathcal{D}$ in $2^N \setminus \{\emptyset, N\}$. Then, $\mathcal{D}$ defines a vertex of $\mathcal{BG}_+(n)$ iff either $\mathcal{D} = \emptyset$ or $\cap \mathcal{D} \neq \emptyset$.*

Proof. The case $\mathcal{D} = \emptyset$ defines $v(S) = 0$, $\forall S \subset N$, which is clearly balanced. For the rest of the proof, we consider $\cap \mathcal{D} \neq \emptyset$.

Choose some family $\mathcal{D}$ s.t. $\cap \mathcal{D} \ni i$ for some $i \in N$, and construct the corresponding v. By definition, v is 0-1-valued. Hence, by Theorem 8, we just have to check balancedness. Take $x \in \mathbb{R}^N$ s.t. $x_i = 1$, and $x_j = 0$ for $j \neq i$. Then $x \in C(v)$.

Conversely, suppose $\mathcal{D} \neq \emptyset$ and $\cap \mathcal{D} = \emptyset$. We show that the game v corresponding to $\mathcal{D}$ is not balanced. Consider a maximal family $B_1, \ldots, B_r \in \mathcal{D}$ s.t. $B_1 \cap \cdots \cap B_r = T \neq \emptyset$. Suppose $C(v) \neq \emptyset$ and take a core element $x \in C(v)$. As $x(B_i) \geqslant v(B_i) = 1$, $x \geqslant 0$ and $x(N) = 1$, it follows that $x(B_i) = 1$ for $i = 1, \ldots, r$. Therefore,

$$x(B_i \setminus T) = 1 - x(T), \quad i = 1, \ldots, r.$$

Summing up we find

$$(r-1)\sum_{i \notin T} x_i \geqslant \sum_{i=1}^{r} x(B_i \setminus T) = r(1 - x(T)),$$

where the first inequality comes from the fact that each $i \in N \setminus T$ belongs at most to $r-1$ sets in the family $B_1 \setminus T, \ldots, B_r \setminus T$. If $r > 1$, we obtain

$$\sum_{i \notin T} x_i \geqslant \frac{r}{r-1}(1 - x(T)) > 1 - x(T),$$

which implies $x(N) > 1$, a contradiction. Therefore, it must be that $r = 1$; that is, there exists $S, S' \in \mathcal{D}$ such that $S \cap S' = \emptyset$. As $v(S) = v(S') = 1$, a core element x should satisfy $x(S) = x(S') = 1$, which implies $x(N) \geqslant 2$, a contradiction. As a conclusion, no core element x exists. □

The vertex corresponding to the empty collection is u_N, the unanimity game centered on N (equivalently, the Dirac game δ_N).

The next result gives explicitely the core of each vertex.

Proposition 2. *Let v be a vertex of $\mathcal{BG}_+(n)$, with associate collection $\mathcal{D}$. If $\mathcal{D}$ is not the empty collection, then*

$$C(v) = \operatorname{conv}\{\mathbf{1}^{\{i\}} : i \in \bigcap \mathcal{D}\}, \tag{8}$$

which implies that the dimension of the core of v is $|\cap \mathcal{D}| - 1$. If $\mathcal{D}$ is the empty collection, then $v = u_N$, whose core is the simplex $\Delta(n) := \{x \in \mathbb{R}^N_+ : \sum_{i \in N} x_i = 1\}$.

Proof. x is a core element iff $x(\cap \mathcal{D}) = 1$, from which the result follows. □

Proposition 2 generalizes some known results for simple games. Recall that a *simple game* v is a 0-1–valued game which is monotonic; that is, $S \subseteq T$ implies $v(S) \leqslant v(T)$. The collection $\mathcal{D}$ is called the collection of *winning coalitions*, and $\cap \mathcal{D}$ is the set of *veto players*. It is well known that the core of a simple game is empty if and only if there is no veto player, and when nonempty, it is expressed by (8) (Peters [16]). Our result is more general as not all vertices of $\mathcal{BG}_+(n)$ are simple games. We can also deduce the following (a simple game is *proper* if $S \in \mathcal{D}$ implies $N \setminus S \notin \mathcal{D}$).

Corollary 1. *If a voting game is balanced, then it is proper.*

Proof. Suppose there exists a balanced voting game v that is not proper. Then, there exists S s.t. $S, N \setminus S \in \mathcal{D}$. However, then, $\cap \mathcal{D} = \emptyset$, contradicting Theorem 9. □

5.3. Vertex Enumeration and Generation

We deduce from Theorem 9 that the enumeration of vertices amounts to the enumeration of the collections $\mathcal{D}$ whose intersection is nonempty. In the next result, we obtain a recursive formula to compute the number of vertices of $\mathcal{BG}_+(n)$, which we denote by b_n. Also, for further use, we introduce a number of notations. We denote by $\mathcal{A}_N$ the set of all collections of sets in $2^N \setminus \{\emptyset, N\}$, including the empty collection. The cardinality of $\mathcal{A}_N$ is $t_n = 2^{2^n - 2}$. We introduce also $\mathcal{F}_N$ and $\mathcal{S}_N$ the set of all *nonempty* collections in $\mathcal{A}_N$ with a nonempty intersection, and empty intersection, respectively. Their cardinalities are denoted by f_n and s_n, respectively. We have by definition $t_n = f_n + s_n + 1$.

Theorem 10. *The number of vertices b_n of $\mathcal{BG}_+(n)$ is given by $b_n = f_n + 1$ where f_n can be obtained recursively as follows:*

$$f_n = \sum_{k=1}^{n-1} \binom{n}{k} (2^{2^k - 1} - f_k - 1), \ \forall n > 1 \text{ and } f_1 = 0.$$

Proof. Vertices of $\mathcal{BG}_+(n)$ are in bijection with collections $\mathcal{D} \in \mathcal{A}_N$ such that $\cap \mathcal{D} \neq \emptyset$, plus the empty collection (associated to u_N).

Now, observe that f_n can be split by considering the different choices for $\cap \mathcal{D}$. Suppose that the intersection is the set S, that is, $\cap \mathcal{D} := S$, with $|S| = k$. There are two possibilities: either $S \in \mathcal{D}$ or $S \notin \mathcal{D}$. If $S \notin \mathcal{D}$, then $\mathcal{D}$ is associated to a nonempty collection $\mathcal{B}$ given by $\mathcal{B} = \{B \setminus S : B \in \mathcal{D}\}$. Observe that $\mathcal{B} \in \mathcal{S}_{N \setminus S}$. If $S \in \mathcal{D}$, then $\mathcal{D}$ is associated to a collection $\mathcal{B} \in \mathcal{A}_{N \setminus S}$ given by $\mathcal{B} = \{B \setminus S : B \in \mathcal{D}\}$ if $\mathcal{D} \neq \{S\}$, and $\mathcal{B} = \emptyset$ otherwise. This way, summing over all S, we obtain

$$f_n = \sum_{k=1}^{n-1} \binom{n}{k} (s_{n-k} + t_{n-k}) = \sum_{k=1}^{n-1} \binom{n}{k} (2t_{n-k} - f_{n-k} - 1) = \sum_{k=1}^{n-1} \binom{n}{k} (2^{2^{n-k} - 1} - f_{n-k} - 1).$$

Making the change $k' = n - k$ and using $\binom{n}{k} = \binom{n}{n-k}$, the result holds. □

The first values of b_n can be seen in Table 1. As we can see, the number of vertices grows exponentially.

Remark 1. The integer sequence of the number of vertices given in Table 1 happens to be already known: It appears as sequence A051381 in the Online Encyclopedia of Integer Sequences (OEIS) (Sloane [20]) and is referred to as the "number of Boolean functions of n variables from Post class $F(5, inf)$." The founding paper is by

Table 1. Number of vertices of $\mathcal{BG}_+(n)$.

	n									
	1	2	3	4	5	6	7	8	9	10
b_n	1	3	19	471	162,631	12,884,412,819	6.456*e*19	1.361*e*39	5.210*e*77	6.703*e*154

Jojović and Kilibarda [9]. It gives an explicit (nonrecursive) formula for this sequence, shown in a very general context and with a long proof. We have kept our recursive formula with its short proof, as we need it hereafter for random vertex generation.

Table 2 enumerates the 19 vertices for $n = 3$. Vertices for $n = 4$ are available as supplementary material. For simplicity, braces and commas are omitted for writing sets, for example, 12 instead of {1, 2}, and so on. Also, we used the following notation: For any collection $\mathcal{C} \subseteq 2^N \setminus \{\emptyset, N\}$, we define the game

$$d_{\mathcal{C}} := \sum_{S \in \mathcal{C}} \delta_S + \delta_N,$$

for example, $d_{1,12} = \delta_1 + \delta_{12} + \delta_{123}$.

Based on the result of Theorem 10 and its proof, we develop an algorithm that generates vertices of $\mathcal{BG}_+(n)$ uniformly at random. For this, consider the recursive expression (see proof of Theorem 10):

$$f_n = \sum_{k=1}^{n-1} \binom{n}{k} (s_{n-k} + t_{n-k}). \tag{9}$$

This formula comes from the following fact. Suppose $\cap\, \mathcal{D} = S$, with $|S| = k$. If $S \in \mathcal{D}$, removing S from $\mathcal{D}$ and from each set in $\mathcal{D}$ yields a collection in $\mathcal{A}_{N\setminus S}$, while if $S \notin \mathcal{D}$, removing S from each set in $\mathcal{D}$ yields a collection in $\mathcal{S}_{N\setminus S}$. Generating a collection $\mathcal{D}$ randomly in $\mathcal{A}_N$ is very simple. It suffices to sort all $2^n - 2$ sets (for example, lexicographically), and generate a vector of $2^n - 2$ zeros and ones, the ones marking the sets that are in $\mathcal{D}$.

The algorithm goes as follows. First, we consider the possibility of drawing the vertex u_N. As $b_n = f_n + 1$, we choose u_N with probability

$$p_0(n) := \frac{1}{1 + f_n}.$$

Table 3 shows the first values for $p_0(n)$. As it can be seen, this value is almost zero for $n > 4$.

Table 2. List of vertices of $\mathcal{BG}_+(n)$ for $n = 3$.

	Vertices					
	1	2	3	12	13	23
u_{123}	0	0	0	0	0	0
u_1	1	0	0	1	1	0
u_2	0	1	0	1	0	1
u_3	0	0	1	0	1	1
u_{12}	0	0	0	1	0	0
u_{13}	0	0	0	0	1	0
u_{23}	0	0	0	0	0	1
$u_{12} \vee u_{13}$	0	0	0	1	1	0
$u_{12} \vee u_{23}$	0	0	0	1	0	1
$u_{13} \vee u_{23}$	0	0	0	0	1	1
d_1	1	0	0	0	0	0
d_2	0	1	0	0	0	0
d_3	0	0	1	0	0	0
$d_{1,12}$	1	0	0	1	0	0
$d_{1,13}$	1	0	0	0	1	0
$d_{2,12}$	0	1	0	1	0	0
$d_{2,23}$	0	1	0	0	0	1
$d_{3,13}$	0	0	1	0	1	0
$d_{3,23}$	0	0	1	0	0	1

Note. "$\vee$" indicates the maximum.

Table 3. First values of $p_0(n)$.

	n								
	1	2	3	4	5	6	7	8	9
$p_0(n)$	1	0.333	0.052	0.002	$6.14e-06$	$7.76e-11$	$1.54e-20$	$7.34e-40$	$1.91e-78$

If u_N is not chosen, then we must generate a random element $\mathcal{D}$ of $\mathcal{F}_N$. Using (9), we start by choosing the number of elements in $\cap\,\mathcal{D}$. The probability of $|\cap\,\mathcal{D}| = k$ is given by

$$p_1^k(n) = \frac{\binom{n}{k}(s_{n-k} + t_{n-k})}{f_n}.$$

Table 4 shows the first values of $p_1^k(n)$.

As we can see, by far the most likely is that the intersection of the sets of $\mathcal{D}$ has cardinal 1. For $n \geq 5$, we may consider the probability distribution $p_1^k(n)$ to be approximately the Dirac distribution at $k = 1$.

The next step in the algorithm is to choose a set S of cardinal k at random among the $\binom{n}{k}$ possibilities and then to decide if either $S \in \mathcal{D}$ or $S \notin \mathcal{D}$. For a given selected k, the probability of $S \in \mathcal{D}$ is given by

$$p_2(n-k) = \frac{t_{n-k}}{s_{n-k} + t_{n-k}}.$$

Table 5 gives the first values of $p_2(n)$.

As we can see, for $n \geq 5$, the probabilities of whether the set S is in the selected collection $\mathcal{D}$ or not are approximately the same. If $S \in \mathcal{D}$, it suffices to generate uniformly an element $\mathcal{A}$ in $\mathcal{A}_{N\setminus S}$ with the procedure described above, and the final collection will be

$$\mathcal{D} = \{S\} \bigcup_{A \in \mathcal{A}} \{S \cup A\}.$$

If the set $S \notin \mathcal{D}$, it suffices to generate a random element $\mathcal{S}$ of $\mathcal{S}_{N\setminus S}$. To achieve this task, we use the fact that the quotient s_n/t_n is almost one for $n \geq 5$, as it can be seen in Table 6.

Hence, for $n \geq 3$, an element $\mathcal{S}$ in $\mathcal{S}_N$ can be generated uniformly at random via a rejection sampler on $\mathcal{A}_N$. That is, we generate an element $\mathcal{A}$ of $\mathcal{A}_N$. If $\mathcal{A} \in \mathcal{S}_N$, then we return $\mathcal{A}$. Otherwise $\mathcal{A} \notin \mathcal{S}_N$ and we repeat the procedure until we obtain an element in $\mathcal{S}_N$. For $n \leq 2$, we have $s_1 = 0$ and $s_2 = 1$, so these cases are trivial. Once $\mathcal{A} \in \mathcal{S}_N$ is generated, the chosen collection is

$$\mathcal{D} = \bigcup_{A \in \mathcal{A}} \{S \cup A\}.$$

Here, we provide a summary of this algorithm.

Algorithm 1 (Vertex Sampler for $\mathcal{BG}_+(n)$)

Step 1: With probability $p_0(n)$ **return** $\mathcal{D} = \emptyset$ (vertex u_N). With probability $1 - p_0(n)$ go to Step 2.

Step 2: Choose some $k \in \{1, \ldots, n-1\}$ with probability $p_1^k(n)$. Also, generate a set $S \subset N$ with k elements. Go to Step 3.

Step 3: With probability $p_2(n-k)$ go to Step 4 and with probability $1 - p_2(n-k)$ to Step 5.

Step 4: Generate at random some $\mathcal{A} \in \mathcal{A}_{N\setminus S}$. Then **return** $\mathcal{D}$ such that

$$\mathcal{D} = \{S\} \bigcup_{A \in \mathcal{A}} \{S \cup A\}.$$

Step 5: Generate at random (by rejection) some $\mathcal{A} \in \mathcal{S}_{N\setminus S}$. Then **return** $\mathcal{D}$ such that

$$\mathcal{D} = \bigcup_{A \in \mathcal{A}} \{S \cup A\}.$$

Lemma 4. *The previous algorithm generates a vertex of $\mathcal{BG}_+(n)$ uniformly at random.*

Table 4. First values of $p_1^k(n)$.

	k					
	1	2	3	4	5	6
$p_1^k(2)$	1					
$p_1^k(3)$	0.83	0.17				
$p_1^k(4)$	0.93	0.06	0.01			
$p_1^k(5)$	0.993	0.006	0.001	≈ 0		
$p_1^k(6)$	0.999	≈ 0	≈ 0	≈ 0	≈ 0	
$p_1^k(7)$	≈ 1	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0

Proof. It suffices to show that any family $\mathcal{D}$ defining a vertex of $\mathcal{BG}_+(n)$ has probability $1/b_n$. Suppose $S = \cap\,\mathcal{D}$ and $S \in \mathcal{D}$. If $|S| = k$, the probability of selecting $\mathcal{D}$ in the algorithm is given by

$$P(\mathcal{D}) = \frac{f_n}{1+f_n} \cdot \frac{\binom{n}{k}(s_{n-k}+t_{n-k})}{f_n} \cdot \frac{1}{\binom{n}{k}} \cdot \frac{t_{n-k}}{s_{n-k}+t_{n-k}} \cdot \frac{1}{t_{n-k}} = \frac{1}{1+f_n} = \frac{1}{b_n}.$$

Similarly if $S \notin \mathcal{D}$, we get

$$P(\mathcal{D}) = \frac{f_n}{1+f_n} \cdot \frac{\binom{n}{k}(s_{n-k}+t_{n-k})}{f_n} \cdot \frac{1}{\binom{n}{k}} \cdot \frac{s_{n-k}}{s_{n-k}+t_{n-k}} \cdot \frac{1}{s_{n-k}} = \frac{1}{1+f_n} = \frac{1}{b_n}.$$

As the probability of selecting the empty family is $1/b_n$, the result follows. □

Let us finally deal with the computational complexity of generating a vertex according to the previous algorithm. First, let us establish a result about the growth of the number of vertices in terms of n.

Proposition 3. *The asymptotic growth rate of the number of vertices, b_n, in $\mathcal{BG}_+(n)$ is*

$$b_n = O(n2^{2^{n-1}}). \tag{10}$$

Proof. As we know $b_n = f_n + 1$ where f_n can be written as

$$f_n = \sum_{k=1}^{n-1} \binom{n}{k} (s_{n-k} + t_{n-k}).$$

Because $s_k \le t_k$, we get

$$\sum_{k=1}^{n-1} \binom{n}{k} t_{n-k} \le f_n \le 2 \sum_{k=1}^{n-1} \binom{n}{k} t_{n-k}.$$

Therefore,

$$b_n = O\left(\sum_{k=1}^{n-1} \binom{n}{k} t_{n-k} \right).$$

Considering that $t_k = 2^{2^k-2}$, the dominant term in the expression $\sum_{k=1}^{n-1} \binom{n}{k} t_{n-k}$ is the one associated with $k = n-1$ because the binomial coefficients are polynomial. Thus,

$$b_n = O(n2^{2^{n-1}-2}) = O(n2^{2^{n-1}}). \quad \square$$

Table 5. First values of $p_2(n)$.

	n								
	1	2	3	4	5	6	7	8	9
$p_2(n)$	1	0.80	0.58	0.51	0.50	0.50	0.50	0.50	0.50

Table 6. First values of s_n/t_n.

	n								
	1	2	3	4	5	6	7	8	9
s_n/t_n	0	0.25	0.70	0.97	0.99	≈ 1	≈ 1	≈ 1	≈ 1

The following result refers to the computational complexity of the worst-case computation time required to generate a vertex by the previous algorithm.

Proposition 4. *The computational complexity of Algorithm 1 is* $O(2^n)$.

Proof. As usual, we will assume that generating a random number has a complexity of $O(1)$. We also assume that the probabilities $p_0(n), p_1^k(n), p_2(n)$ are known a priori. With these considerations, we know that steps 1 and 3 have a complexity of $O(1)$ because only a random number needs to be generated. For step 2, we can choose k with complexity $O(n)$, and we can generate the set S with complexity $O(n)$ as well. For step 4, we can generate an element of $\mathcal{A}_{N\setminus S}$ by listing all possible subsets and generating a zero or one for each case, which means a complexity of $O(2^n)$. Finally, for step 5, as the rejection rate tends to zero very quickly, the rejection rate will only affect small n, in which case it would multiply the computational complexity by a constant, $O(C2^n)$. Taking all this into account, the total computational complexity would be $O(2^n)$. □

We can see that in relation to the asymptotic number of vertices (10), the complexity is very reduced.

5.4. Adjacency of Vertices and Related Properties

Recall that two vertices are adjacent if they both belong to the same edge (one-dimensional face of the polytope). The aim of this section is twofold: first, we will see that $\mathcal{BG}_+(n)$ is a combinatorial polytope and, as a consequence, that there exists a Hamiltonian path connecting each pair of vertices. Second, we will characterize adjacency of vertices corresponding to collections $\mathcal{D}$ such that their intersection $\cap\,\mathcal{D}$ is a singleton, which is by far the most common case, according to Table 4. In addition, these vertices have a core reduced to a singleton. This characterization is given in Theorems 13–15.

To prove that $\mathcal{BG}_+(n)$ is Hamilton connected, we first show that $\mathcal{BG}_+(n)$ is a combinatorial polytope.

Definition 1 (Nadder and Pulleyblank [14]). A polytope $\mathcal{P}$ is said to be *combinatorial* if the two following conditions hold:

- All vertices of $\mathcal{P}$ are 0-1 valued.
- Given two vertices v_1, v_2 of $\mathcal{P}$, if they are not adjacent, then there exist two other different vertices v_3, v_4 such that

$$v_1 + v_2 = v_3 + v_4.$$

For a combinatorial polytope, the following can be shown.

Theorem 11 (Naddef and Pulleyblank [14]). *Let G be the adjacency graph of a combinatorial polytope. Then G is either a hypercube or is Hamilton connected.*

Now, the following holds.

Lemma 5. *Let v_1, v_2 be two distinct vertices of $\mathcal{BG}_+(n)$, with associate collections $\mathcal{D}_1, \mathcal{D}_2$. Then v_1, v_2 are not adjacent if and only if there exist vertices v_3, v_4 distinct from v_1, v_2, such that*

$$v_1 + v_2 = v_3 + v_4,$$

In addition, the associated collections $\mathcal{D}_3, \mathcal{D}_4$ satisfy

$$\mathcal{D}_3 \cup \mathcal{D}_4 = \mathcal{D}_1 \cup \mathcal{D}_2, \quad \mathcal{D}_3 \cap \mathcal{D}_4 = \mathcal{D}_1 \cap \mathcal{D}_2.$$

Proof. Let us consider two vertices v_1, v_2 with collections $\mathcal{D}_1, \mathcal{D}_2$. They are not adjacent if and only if there exist $\lambda, \lambda' \in [0,1]$ and vertices v_3, v_4 distinct from v_1, v_2 such that

$$\lambda v_1 + (1-\lambda)v_2 = \lambda' v_3 + (1-\lambda')v_4 =: v.$$

Let us denote by $\mathcal{D}_3, \mathcal{D}_4$ the collections associated to v_3, v_4. We have the following, using the decomposition into v_1, v_2:

$$v(S) = \begin{cases} \lambda, & S \in \mathcal{D}_1 \setminus \mathcal{D}_2 \\ 1, & S \in \mathcal{D}_1 \cap \mathcal{D}_2 \\ 1-\lambda, & S \in \mathcal{D}_2 \setminus \mathcal{D}_1 \\ 0, & S \notin \mathcal{D}_1 \cup \mathcal{D}_2. \end{cases}$$

Doing similarly with v_3, v_4, we deduce that $\mathcal{D}_1 \cup \mathcal{D}_2 = \mathcal{D}_3 \cup \mathcal{D}_4$, $\mathcal{D}_1 \cap \mathcal{D}_2 = \mathcal{D}_3 \cap \mathcal{D}_4$. As $\mathcal{D}_3, \mathcal{D}_4$ are distinct from $\mathcal{D}_1, \mathcal{D}_2$, this imposes that

1. If $\mathcal{D}_1 \setminus \mathcal{D}_2 \neq \emptyset$ and $\mathcal{D}_2 \setminus \mathcal{D}_1 \neq \emptyset$, then either (a) $\mathcal{D}_3 = \mathcal{D}_1 \cup \mathcal{D}_2$ and $\mathcal{D}_4 = \mathcal{D}_1 \cap \mathcal{D}_2$ (or the converse), or (b) at least one of these collections must intersect $\mathcal{D}_3$ and $\mathcal{D}_4$.
2. If $\mathcal{D}_1 \subset \mathcal{D}_2$ then $\mathcal{D}_2 \setminus \mathcal{D}_1$ must intersect $\mathcal{D}_3$ and $\mathcal{D}_4$;
3. Similar when $\mathcal{D}_2 \subset \mathcal{D}_1$.

In any case, this implies $\lambda = \lambda' = 1 - \lambda' = 1 - \lambda$, that is, $\lambda = \lambda' = 1/2$. □

Hence, as a consequence of Theorem 8 and Lemma 5, the following holds.

Theorem 12. *The polytope $\mathcal{BG}_+(n)$ is combinatorial. Moreover, the adjacency graph of $\mathcal{BG}_+(n)$ is Hamilton connected.*

Proof. $\mathcal{BG}_+(n)$ is combinatorial as a straight consequence of Theorem 8 and Lemma 5. Now, from Theorem 11, the adjacency graph of this polytope is either Hamilton connected or a hypercube. For $n = 1$, $\mathcal{BG}_+(1)$ is reduced to a singleton; therefore, the result holds trivially. For $n = 2$, the number of vertices is three, and therefore $\mathcal{BG}_+(2)$ is not a hypercube. For $n \geqslant 3$, observe that for any distinct $i, j \in N$, $\hat{\delta}_{\{i\}} := \delta_{\{i\}} + \delta_N$ and $\hat{\delta}_{\{j\}}$ are vertices of $\mathcal{BG}_+(n)$; however, the game $\delta_{\{i\}} + \delta_{\{j\}} + \delta_N$ is not a vertex of $\mathcal{BG}_+(n)$. □

Let us now look more closely to adjacency. For further use, we illustrate on Figure 1 the condition of nonadjacency when $\mathcal{D}_1 \setminus \mathcal{D}_2 \neq \emptyset$ and $\mathcal{D}_2 \setminus \mathcal{D}_1 \neq \emptyset$.

We now focus on the vertices whose associated collections have intersection reduced to a singleton.

Theorem 13. *Consider two vertices v_1, v_2 of $\mathcal{BG}_+(n)$, associated to $\mathcal{D}_1, \mathcal{D}_2$ respectively, and $\cap \mathcal{D}_1 = \{i\} = \cap \mathcal{D}_2$. Then v_1 and v_2 are adjacent iff either $\mathcal{D}_1 \subseteq \mathcal{D}_2$ or the converse, and $|\mathcal{D}_1 \Delta \mathcal{D}_2| = 1$.*

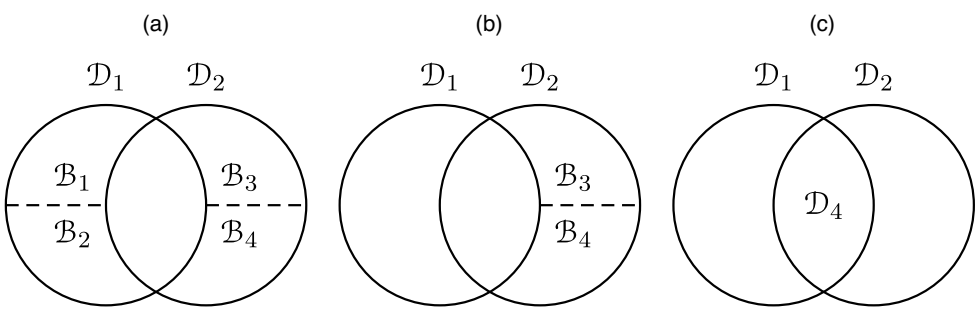
Proof. Consider two vertices v_1, v_2 as above and $v = \frac{1}{2}(v_1 + v_2)$. By definition, $v(S) = 1$ iff $S \in \mathcal{D}_1 \cap \mathcal{D}_2$, $v(S) = 1/2$ iff $S \in \mathcal{D}_1 \Delta \mathcal{D}_2$, and $v(S) = 0$ otherwise.

⇒) Assume v_1, v_2 are adjacent and suppose $\mathcal{D}_1 \setminus \mathcal{D}_2 \ni S_1$ and $\mathcal{D}_2 \setminus \mathcal{D}_1 \ni S_2$. Consider v_3 generated by $\mathcal{D}_1 \cup \{S_2\}$ and v_4 generated by $\mathcal{D}_2 \setminus \{S_2\}$. Then v_3, v_4 differ from v_1, v_2 and $v = (v_3 + v_4)/2$, contradicting that v_1, v_2 are adjacent. Consequently, $\mathcal{D}_1 \subseteq \mathcal{D}_2$ (or the converse).

Assuming the former, let us prove that $|\mathcal{D}_2 \setminus \mathcal{D}_1| = 1$. Suppose on the contrary that there exist distinct $S_1, S_2 \subseteq N$ s.t. $S_1, S_2 \in \mathcal{D}_2 \setminus \mathcal{D}_1$. Hence, we can consider the games v_3, v_4 generated by $\mathcal{D}_1 \cup \{S_1\}$ and $\mathcal{D}_2 \setminus \{S_1\}$, respectively. Thus, defined, v_3, v_4 differ from v_1, v_2 and $v = (v_3 + v_4)/2$, a contradiction.

⇐) Suppose by contradiction that v_1, v_2 are not adjacent. Then by Lemma 5 there exist vertices $v_3, v_4 \in \mathcal{BG}_+(n)$ different from v_1, v_2 such that $v_1 + v_2 = v_3 + v_4$, with associated collections $\mathcal{D}_3, \mathcal{D}_4$ satisfying $\mathcal{D}_1 \cap \mathcal{D}_2 \subseteq \mathcal{D}_3 \subseteq \mathcal{D}_1 \cup \mathcal{D}_2$ and similarly for $\mathcal{D}_4$.

Figure 1. Nonadjacency of v_1, v_2, with associated collections $\mathcal{D}_1, \mathcal{D}_2$. Case (a): $\mathcal{D}_3 = \mathcal{B}_1 \cup (\mathcal{D}_1 \cap \mathcal{D}_2) \cup \mathcal{B}_3$, $\mathcal{D}_4 = \mathcal{B}_2 \cup (\mathcal{D}_1 \cap \mathcal{D}_2) \cup \mathcal{B}_4$; Case (b): $\mathcal{D}_3 = \mathcal{D}_1 \cup \mathcal{B}_3$, $\mathcal{D}_4 = (\mathcal{D}_1 \cap \mathcal{D}_2) \cup \mathcal{B}_4$ (similar when $\mathcal{D}_1, \mathcal{D}_2$ exchanged); Case (c): $\mathcal{D}_3 = \mathcal{D}_1 \cup \mathcal{D}_2$, $\mathcal{D}_4 = \mathcal{D}_1 \cap \mathcal{D}_2$.



If $\mathcal{D}_1 \not\subseteq \mathcal{D}_2$ and $\mathcal{D}_2 \not\subseteq \mathcal{D}_1$ we are done considering $\mathcal{D}_3 = \mathcal{D}_1 \cap \mathcal{D}_2, \mathcal{D}_4 = \mathcal{D}_1 \cup \mathcal{D}_2$. Otherwise, assume $\mathcal{D}_1 \subseteq \mathcal{D}_2$. The above constraints resume to $\mathcal{D}_1 \subseteq \mathcal{D}_3 \subseteq \mathcal{D}_2$ and the same for $\mathcal{D}_4$. As v_3, v_4 differ from v_1, v_2, strict inclusion must hold throughout, which implies $|\mathcal{D}_1 \Delta \mathcal{D}_2| > 1$. The case $\mathcal{D}_2 \subseteq \mathcal{D}_1$ is similar. Hence, the result holds. □

Lemma 6. *Let v_1, v_2 be two vertices of $\mathcal{BG}_+(n)$ with associated collections $\mathcal{D}_1, \mathcal{D}_2$ such that $\cap \mathcal{D}_1 = \{i\}$, $\cap \mathcal{D}_2 = \{j\}$. If $i \neq j$, then $\mathcal{D}_1 \setminus \mathcal{D}_2 \neq \emptyset$, $\mathcal{D}_2 \setminus \mathcal{D}_1 \neq \emptyset$.*

Proof. Suppose $i \neq j$. This implies that in $\mathcal{D}_1$ there must exist S such that $S \not\ni j$ (otherwise we would have $\cap \mathcal{D}_1 \supseteq \{i,j\}$). Hence, $S \in \mathcal{D}_1 \setminus \mathcal{D}_2$. Similarly, there must exist $T \in \mathcal{D}_2$ such that $T \not\ni i$, which implies $T \in \mathcal{D}_2 \setminus \mathcal{D}_1$. □

Theorem 14. *Consider two vertices v_1, v_2 of $\mathcal{BG}_+(n)$, associated to families $\mathcal{D}_1, \mathcal{D}_2$ respectively, such that $\cap \mathcal{D}_1 = \{i\} \neq \cap \mathcal{D}_2 = \{j\}$ and suppose $\mathcal{D}_1 \cap \mathcal{D}_2 = \emptyset$. Then v_1 and v_2 are adjacent iff there do not exist collections $\mathcal{D}_3, \mathcal{D}_4$ distinct from $\mathcal{D}_1, \mathcal{D}_2$ satisfying*

1. $\mathcal{D}_3 \cap \mathcal{D}_4 = \emptyset$,
2. $\mathcal{D}_3 \cup \mathcal{D}_4 = \mathcal{D}_1 \cup \mathcal{D}_2$,
3. $\cap \mathcal{D}_3 \neq \emptyset$ *and* $\cap \mathcal{D}_4 \neq \emptyset$.

Proof. $\Rightarrow$) Suppose there exist $\mathcal{D}_3, \mathcal{D}_4$ satisfying the above conditions. Then, because $\cap \mathcal{D}_3$ and $\cap \mathcal{D}_4$ are nonempty by condition 3, they determine vertices v_3 and v_4, respectively. By conditions 1 and 2, it follows that

$$\frac{1}{2}v_1 + \frac{1}{2}v_2 = \frac{1}{2}v_3 + \frac{1}{2}v_4,$$

showing that v_1 and v_2 are not adjacent.

$\Leftarrow$) Suppose v_1 and v_2 are not adjacent. By Lemma 5, it follows that there exist vertices v_3 and v_4 such that $v_1 + v_2 = v_3 + v_4$, determined by collections $\mathcal{D}_3, \mathcal{D}_4$, respectively, satisfying conditions 1 and 2. Now, condition 3 is implied by the fact that v_3 and v_4 are vertices. □

Lemma 7. *Consider two vertices v_1, v_2 of $\mathcal{BG}_+(n)$, associated to $\mathcal{D}_1, \mathcal{D}_2$, respectively, such that $\cap \mathcal{D}_1 = \{i\} \neq \cap \mathcal{D}_2 = \{j\}$ and $\mathcal{D}_1 \cap \mathcal{D}_2 = \emptyset$. If v_1, v_2 are adjacent, then*

$$\forall S \in \mathcal{D}_1, j \notin S, \quad \forall T \in \mathcal{D}_2, i \notin T.$$

Proof. It suffices to note that if there exists $S \in \mathcal{D}_1$ such that $j \in S$, then we can consider $\mathcal{D}_3 := \mathcal{D}_1 \setminus \{S\}$ and $\mathcal{D}_4 := \mathcal{D}_2 \cup \{S\}$, which fulfill the three conditions of Theorem 14. □

Theorem 15. *Consider two vertices v_1, v_2 of $\mathcal{BG}_+(n)$, associated to $\mathcal{D}_1, \mathcal{D}_2$ respectively, and $\cap \mathcal{D}_1 = \{i\} \neq \cap \mathcal{D}_2 = \{j\}$. Suppose $\mathcal{D}_1 \cap \mathcal{D}_2 \neq \emptyset$ and denote $\cap (\mathcal{D}_1 \cap \mathcal{D}_2) = T \cup \{i,j\}$, with $T \subseteq N \setminus \{i,j\}$. Then, v_1 and v_2 are adjacent iff the following two conditions are satisfied:*

1. *For all $S \in \mathcal{D}_1 \setminus \mathcal{D}_2$, $j \notin S$, and for all $S \in \mathcal{D}_2 \setminus \mathcal{D}_1$, $i \notin S$*
2. *For every disjoint $K_1, K_2 \subseteq T$ and disjoint $K_3, K_4 \subseteq T$ such that $K_1 \cap K_3 \neq \emptyset$ and $K_2 \cap K_4 \neq \emptyset$, there exists either $S \in \mathcal{D}_1 \setminus \mathcal{D}_2$ s.t. $K_1 \not\subseteq S$ and $K_2 \not\subseteq S$, or $S' \in \mathcal{D}_2 \setminus \mathcal{D}_1$ s.t. $K_3 \not\subseteq S'$ and $K_4 \not\subseteq S'$.*

Proof. $\Rightarrow$) Let us consider two adjacent vertices v_1, v_2 as above, and let $v = (v_1 + v_2)/2$. By Lemma 6, we know that $\mathcal{D}_1 \setminus \mathcal{D}_2 \neq \emptyset$ and $\mathcal{D}_2 \setminus \mathcal{D}_1 \neq \emptyset$. Suppose there exists $S_1 \in \mathcal{D}_1 \setminus \mathcal{D}_2$ s.t. $j \in S_1$. Consider v_3, v_4 generated by $\mathcal{D}_1 \setminus \{S_1\}$ and $\mathcal{D}_2 \cup \{S_1\}$, respectively. Then $v = (v_3 + v_4)/2$, a contradiction. The argument is the same with the existence of $S_2 \in \mathcal{D}_2 \setminus \mathcal{D}_1$ s.t. $i \in S_2$. This proves the first condition.

We prove the second condition. Our strategy is to show that the nonexistence of a partition of $\mathcal{D}_1 \cup \mathcal{D}_2$ like in Figure 1 (which is equivalent to adjacency) implies the second condition. We first observe that the first condition implies that a partition of $\mathcal{D}_1 \cup \mathcal{D}_2$ like in cases (b) and (c) of Figure 1 can never occur. Indeed, in these cases, by the first condition, it follows that $\cap \mathcal{D}_3$ contains neither i nor j, and because $\cap \mathcal{D}_3 \neq \emptyset$, it must contain some other element, say k. But then, $k \in \cap \mathcal{D}_1$, a contradiction. Consequently, we only have to consider case (a). Let us call (a)-partition a possible partition like in Case (a) and show that it is not possible to build such a partition. We distinguish different cases in terms of $\mathcal{D}_1 \cap \mathcal{D}_2$.

1. Suppose $\cap (\mathcal{D}_1 \cap \mathcal{D}_2) = \{i,j\}$. The first condition implies that $j \notin \cap \mathcal{B}_1$ (same for $\cap \mathcal{B}_2$), and $i \notin \cap \mathcal{B}_3$ (same for $\cap \mathcal{B}_4$). It follows that $\cap \mathcal{D}_3 = \emptyset$. Hence no (a)-partition can exist.

2. Suppose $\cap (\mathcal{D}_1 \cap \mathcal{D}_2) = \{i,j,k\}$. In this case, by the first condition, it follows that $k \in \cap \mathcal{B}_1$ and $k \in \cap \mathcal{B}_3$ to ensure $\cap \mathcal{D}_3 \neq \emptyset$, and the same holds for $\cap \mathcal{B}_2$ and $\cap \mathcal{B}_4$. But then $\cap \mathcal{D}_1 \ni k$, a contradiction. Hence no (a)-partition can exist.

3. Suppose $\cap (\mathcal{D}_1 \cap \mathcal{D}_2) = T \cup \{i,j\}$ with $|T| \geqslant 2$. There exists an (a)-partition iff one can have $(\cap \mathcal{B}_1 \cap (\cap \mathcal{B}_3)) \cap T \neq \emptyset$, $(\cap \mathcal{B}_2 \cap (\cap \mathcal{B}_4)) \cap T \neq \emptyset$ (to ensure nonemptiness of $\cap \mathcal{D}_3$, $\cap \mathcal{D}_4$), and $\cap \mathcal{B}_1 \cap (\cap \mathcal{B}_2) \cap T = \emptyset$, $\cap \mathcal{B}_3 \cap (\cap \mathcal{B}_4) \cap$

$T = \emptyset$ (to ensure $\cap \mathcal{D}_1 = \{i\}$ and $\cap \mathcal{D}_2 = \{j\}$). By letting $K_i := \cap \mathcal{B}_i \cap T$ for $i = 1, \ldots, 4$, this is equivalent to: $\exists K_1, K_2, K_3, K_4 \subseteq T$, $K_1 \cap K_2 = \emptyset$, $K_3 \cap K_4 = \emptyset$, $K_1 \cap K_3 \neq \emptyset$, $K_2 \cap K_4 \neq \emptyset$ such that for every $S \in \mathcal{D}_1 \setminus \mathcal{D}_2$, either $K_1 \subseteq S$ or $K_2 \subseteq S$, and for $S' \in \mathcal{D}_2 \setminus \mathcal{D}_1$, either $K_3 \subseteq S'$ or $K_4 \subseteq S'$. Therefore, there is no (a)-partition iff: $\forall K_1, K_2, K_3, K_4 \subseteq T$ such that $K_1 \cap K_2 = \emptyset$, $K_3 \cap K_4 = \emptyset$, $K_1 \cap K_3 \neq \emptyset$, $K_2 \cap K_4 \neq \emptyset$, either $\exists S \in \mathcal{D}_1 \setminus \mathcal{D}_2$ s.t. $K_1 \nsubseteq S, K_2 \nsubseteq S$, or $\exists S' \in \mathcal{D}_2 \setminus \mathcal{D}_1$ s.t. $K_3 \nsubseteq S', K_4 \nsubseteq S$.

$\Leftarrow$) Suppose v_1, v_2 are not adjacent. Then by Lemma 5, there exist $\mathcal{D}_3, \mathcal{D}_4$ such that either an (a)-partition, a (b)-partition, or a (c)-partition of $\mathcal{D}_1 \cup \mathcal{D}_2$ exists (Figure 1).

Suppose there exists an (a)-partition. If the first condition is not satisfied, we are done. Hence, assume that the first condition holds. Then, proceeding as in cases 1, 2, 3 in the $\Rightarrow$) part, we deduce that if $|T| < 2$, no (a)-partition can exist, and if $|T| \geqslant 2$, the existence of a (a)-partition implies the negation of the second condition, as desired.

Suppose there exists a (b)-partition. Then $\mathcal{D}_3 := \mathcal{D}_1 \cup \mathcal{B}_3$ satisfies $\cap \mathcal{D}_3 \neq \emptyset$, say it contains k. Then, $k \in \cap \mathcal{D}_1$, which implies $k = i$. This in turn implies $i \in \cap \mathcal{B}_3$, which violates the first condition.

Suppose there exists a (c)-partition. Observe this cannot occur. Indeed, $\cap \mathcal{D}_3 = \cap (\mathcal{D}_1 \cup \mathcal{D}_2) \neq \emptyset$, say $k \in \cap (\mathcal{D}_1 \cup \mathcal{D}_2)$. In particular, $k \in \cap \mathcal{D}_1$, which implies $k = i$. But then $i \in \cap \mathcal{D}_2$, a contradiction. □

Observe that if $T = \emptyset$ in the above theorem, then only the first condition remains.

In Figure 2, the adjacency graph of vertices of $\mathcal{BG}_+(n)$ can be seen. This graph has the vertices as nodes, and there is an edge between two nodes if they are adjacent. Finally, to illustrate Theorem 12, in Figure 3, a Hamilton path joining u_{123} and $d_{2,23}$ is given.

5.5. Facets

The following result gives the facets of $\mathcal{BG}_+(n)$.

Theorem 16. *The following holds:*

1. *Each equality $v(S) = 0$, $S \neq \emptyset, N$, defines a facet.*
2. *Each equality $\sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} v(S) = 1$, $\mathcal{B} \in \mathfrak{B}^*(n)$, defines a facet.*
3. *The number of facets is $2^n + b(n) - 3$, with $b(n)$ the number of minimal balanced collections on N.*

Proof. Recall from (4) that

$$\mathcal{BG}_+(n) := \left\{ v \in \mathcal{G}(n) : \sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} v(S) \leqslant 1, \ \forall \mathcal{B} \in \mathfrak{B}^*(n), v(S) \geqslant 0, \ \forall \emptyset \neq S \subset N, v(N) = 1 \right\}.$$

All the facets of a polytope can be obtained by converting into equality one of the inequalities that define it. However, some inequalities might define a lower-dimensional face.

1. We claim that every equality $v(S) = 0, S \neq N, \emptyset$, leads by intersection with $\mathcal{BG}_+(n)$ to a face of dimension $2^n - 3$ and hence a facet. To see this, consider the games $\hat{\delta}_T = \delta_T + \delta_N, T \neq S, N, \emptyset$, that is,

$$\hat{\delta}_T(S') = \begin{cases} 1, & \text{if } S' = T \text{ or } S' = N \\ 0, & \text{otherwise.} \end{cases}$$

Clearly, the games $\hat{\delta}_T$ for every $T \neq \emptyset, S, N$ are vertices of $\mathcal{BG}_+(n)$ and satisfy $v(S) = 0$. In addition, the game u_N also belongs to the face $v(S) = 0$. These $2^n - 2$ games are affinely independent because the $2^n - 3$ games $\hat{\delta}_T - u_N = \delta_T$ are linearly independent, which proves the claim.

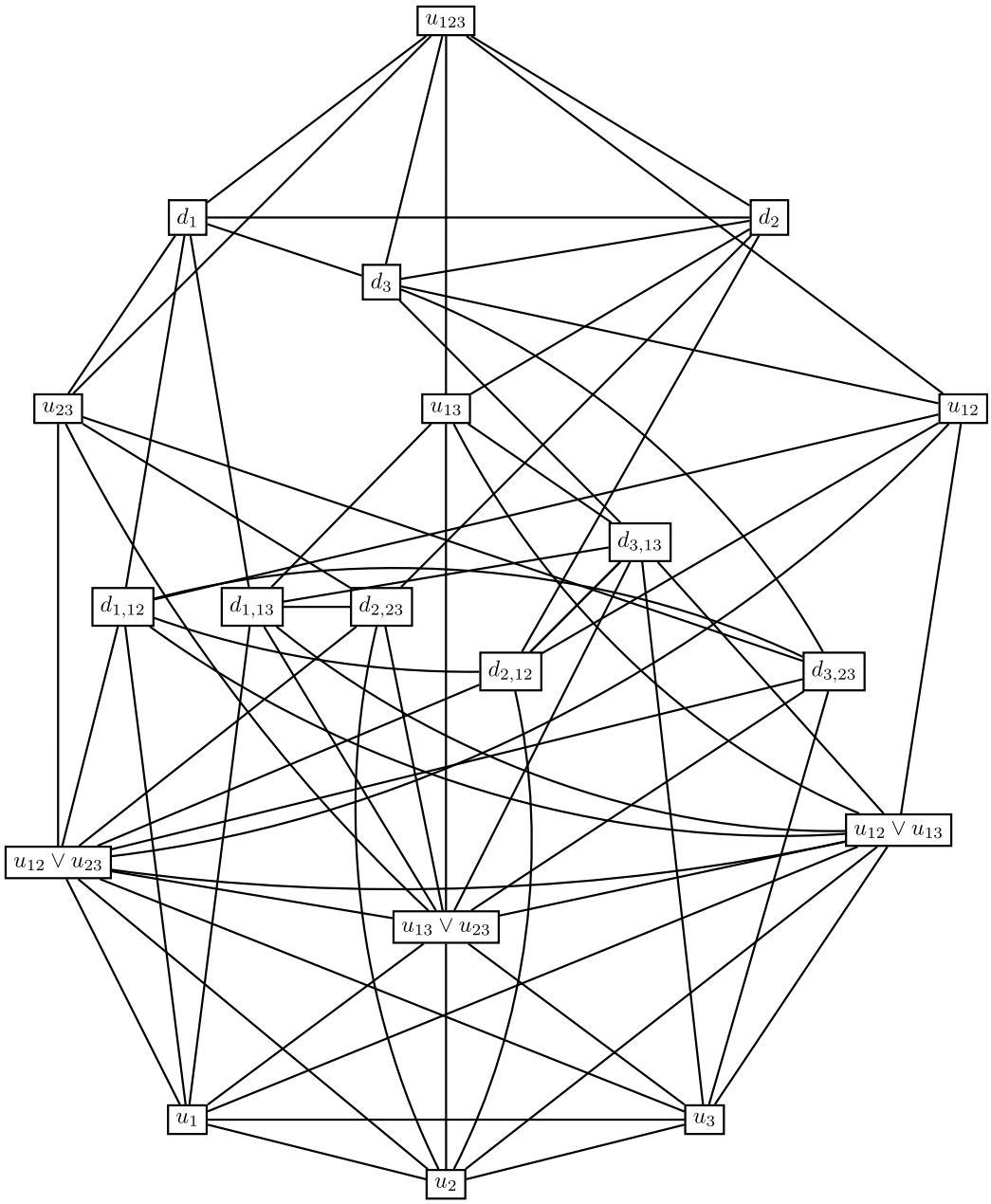
2. Consider any m.b.c. $\mathcal{B} \in \mathfrak{B}^*(n)$ and the corresponding inequality. For any $i \in N$ define the game v^i by

$$v^i(S) = \begin{cases} 1, & \text{if } S \ni i \text{ and } S \in \mathcal{B}, \text{ or } S = N \\ 0, & \text{otherwise.} \end{cases}$$

Observe that each v^i is a vertex of $\mathcal{BG}_+(n)$, and v^i satisfies $\sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} v^i(S) = 1$. Moreover, the number of distinct v^i is $|\mathcal{B}| =: p$. Let us call $\mathcal{D}^1, \ldots, \mathcal{D}^n$ the corresponding collections defining $v^1, \ldots, v^n$. Define successively the collections

- $\mathcal{D}^1 \cup \{S\}$, with $S \ni 1$, $S \notin \mathcal{B}$
- $\mathcal{D}^2 \cup \{S\}$, with $S \ni 2$, $S \not\ni 1$, $S \notin \mathcal{B}$
- $\mathcal{D}^3 \cup \{S\}$, with $S \ni 3$, $S \not\ni 2$, $S \not\ni 1$, $S \notin \mathcal{B}$
- And so on …
- $\mathcal{D}^n \cup \{\{n\}\}$ (if $\{n\}$ not already present in $\mathcal{D}^n$)

Figure 2. Adjacency graph of $\mathcal{BG}_+(n)$ for $n = 3$. Vertices are defined in Table 2.



Observe that $\mathcal{D}^1, \ldots, \mathcal{D}^n$ plus all the above ones yields $2^n - 2$ distinct collections, where each $S \in 2^N \setminus \{\emptyset, N\}$ is present at least once. Moreover, each collection defines a game that is a vertex belonging to the face defined by $\sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} v(S) = 1$. It follows that these $2^n - 2$ games are affinely independent (as, for example, the family of the $2^n - 3$ games $v - v^1$ with v any game as defined above forms a linearly independent family), and therefore the inequality $\sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} v(S) \leqslant 1$ defines a facet.

3. This is an immediate consequence of 1 and 2. □

6. When Is the Core Reduced to a Singleton?

We address in this section the question of which balanced game in any of the three sets of balanced games under consideration in this paper has a core reduced to a singleton. Let us call this for simplicity a *point core*.

A first observation is the following: Because $\mathcal{BG}_\alpha(n)$ and $\mathcal{BG}_+(n)$ are subsets of $\mathcal{BG}(n)$, it suffices to find all games in $\mathcal{BG}(n)$ that have a point core and then to check if they belong to $\mathcal{G}_+(n)$ or satisfy $v(N) = \alpha$. Therefore, we put our effort into finding all balanced games in $\mathcal{BG}(n)$ having a point core.

Figure 3. (Color online) Adjacency graph of $\mathcal{BG}_+(n)$ for $n = 3$. The Hamiltonian path joining u_{123} and $d_{2,23}$ is indicated in thick line.



6.1. Case of $\mathcal{BG}(n)$

For a given balanced game v in $\mathcal{BG}(n)$, let us denote by $\mathcal{E}(v)$ the collection

$$\mathcal{E}(v) = \{S \subseteq N : x(S) = v(S), \ \forall x \in C(v)\}.$$

The set $\mathcal{E}(v)$ is called the set of *effective* coalitions for v. Obviously, $N \in \mathcal{E}(v)$, and $\mathcal{E}(v) = \{N\}$ if and only if the core is full-dimensional, that is, $(n-1)$-dimensional. The following result of Laplace Mermoud et al. [11] is central in our investigation.

Lemma 8. *$\mathcal{E}(v)$ is the union of all minimal balanced collection $\mathcal{B}$ such that*

$$\sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} v(S) = v(N).$$

An immediate consequence of this lemma is the following general result.

Proposition 5. *If v belongs to the interior of $\mathcal{BG}(n)$, then v has not a point core.*

Indeed, for such a game v, no inequality in (2) is tight, which implies by Lemma 8 that $\mathcal{E}(v)$ is reduced to N, which means that the core is full-dimensional.

We begin with a simple result.

Lemma 9. *Any game in* $\mathsf{Lin}(\mathcal{BG}(n))$ *has a point core.*

Proof. Take $v \in \mathsf{Lin}(\mathcal{BG}(n))$. Then, this game can be written as

$$v = \sum_{i \in N} \alpha_i w_i,\ \alpha_1, \ldots, \alpha_n \in \mathbb{R}.$$

As each $w_i = u_{\{i\}}$ is an additive game, so is v, and therefore v has a point core. □

We turn to the examination of facets, which by Theorem 4 corresponds to minimal balanced collections in $\mathfrak{B}^*(n)$.

Theorem 17. *Consider a minimal balanced collection $\mathcal{B} \in \mathfrak{B}^*(n)$ and its corresponding facet $\mathcal{F}$ in $\mathcal{BG}(n)$. The following holds:*

1. *If $|\mathcal{B}| = n$, every game in $\mathcal{F}$ has a point core.*
2. *Otherwise, no game in the relative interior of $\mathcal{F}$ has a point core.*

Proof. Consider a minimal balanced collection $\mathcal{B}$ and its corresponding facet $\mathcal{F}$ in $\mathcal{BG}(n)$.

1. Take v in the relative interior of $\mathcal{F}$. Any core element $x \in C(v)$ satisfies the system

$$x(S) = v(S), \quad S \in \mathcal{E}(v). \tag{11}$$

By Lemma 8, we have that $\mathcal{E}(v) = \mathcal{B} \cup \{N\}$. Observe that the inequality $x(N) = v(N)$ is redundant with the others as $\mathcal{B}$ is balanced. Therefore, in (11), we can remove $S = N$. Consider now the system

$$\sum_{S \in \mathcal{B}} \lambda_S \mathbf{1}^S = \mathbf{1}^N.$$

As $\mathcal{B}$ is minimal, the above system has a unique solution. Hence, the rank of the matrix M of this linear system is $|\mathcal{B}|$. Conversely, the matrix of System (11) (without N) is the transpose of M, and therefore has rank $|\mathcal{B}|$. It follows that the solution of (11) is unique iff $|\mathcal{B}| = n$.

2. It remains to prove that games on the frontier of a facet $\mathcal{F}$ defined by a m.b.c. $\mathcal{B}$ s.t. $|\mathcal{B}| = n$ have also a point core. Take v in the frontier of $\mathcal{F}$. Then, v also belongs to other facets, say $\mathcal{F}_1, \ldots, \mathcal{F}_r$, associated with m.b.c. $\mathcal{B}_1, \ldots, \mathcal{B}_r$, respectively. It follows from Lemma 8 that any core element $x \in C(v)$ satisfies the system

$$x(S) = v(S), \quad S \in \mathcal{B} \cup \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_r.$$

As the system $x(S) = v(S)$ for $S \in \mathcal{B}$ has a unique solution, it follows that either the above system has no solution or a unique one. However, the system has solution by the fact that v is balanced. □

Remark 2. As the lineality space is the intersection of all facets, Lemma 9 is obtained as a corollary of the above theorem.

Part 2 of the previous proof allows for a characterization of games with a point core in $\mathcal{BG}(n)$.

Theorem 18. *Consider a face $\mathcal{F} = \mathcal{F}_1 \cap \cdots \cap \mathcal{F}_p$, with $\mathcal{F}_1, \ldots, \mathcal{F}_p$ the facets associated to m.b.c. $\mathcal{B}_1, \ldots, \mathcal{B}_p$, respectively. Then, any game in $\mathcal{F}$ has a point core iff the rank of the matrix $\{\mathbf{1}^S, S \in \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_p\}$ is n.*

Proof. Using the same argument as in the proof of Theorem 17, any core element x of a game v in the relative interior of $\mathcal{F}$ satisfies the system

$$x(S) = v(S), \quad S \in \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_p$$

and the conclusion follows immediately. Now, if v is in the frontier, proceed as in the proof of Theorem 17. □

The above results plus the result on the interior of $\mathcal{BG}(n)$ completely characterizes the set of games with a point core.

Table 7. Structure of the facets and extremal rays.

m.b.c.	$-\delta_{12}$	$-\delta_{13}$	$-\delta_{23}$	r_1	r_2	r_3
$\mathcal{B}_1 = \{1,2,3\}$	×	×	×			
$\mathcal{B}_2 = \{1,23\}$	×	×			×	×
$\mathcal{B}_3 = \{2,13\}$	×		×	×		×
$\mathcal{B}_4 = \{3,12\}$		×	×	×	×	
$\mathcal{B}_5 = \{12,13,23\}$				×	×	×

Example 2. Consider $n = 4$, and the facets $\mathcal{F}_1, \mathcal{F}_2$ associated with the m.b.c. $\mathcal{B}_1 = \{12,34\}$ and $\mathcal{B}_2 = \{1,234\}$. Then, games in the relative interior of $\mathcal{F}_1 \cap \mathcal{F}_2$ have no point core. Although $|\mathcal{B}_1| + |\mathcal{B}_2| = 4 = n$, the rank of the matrix $\{1^S, S \in \mathcal{B}_1 \cup \mathcal{B}_2\}$ is three; therefore, games in the face $\mathcal{F}_1 \cap \mathcal{F}_2$ have no point core but a core of dimension 1.

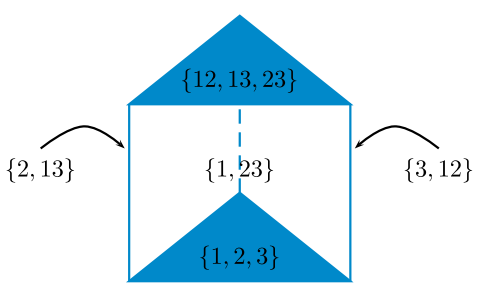
Consider now the face $\mathcal{F}_3$ associated with $\mathcal{B}_3 = \{13,124,23\}$. Games in the relative interior of this facet have no point core. However, games in $\mathcal{F}' = \mathcal{F}_1 \cap \mathcal{F}_3$ have a point core because the rank of $\{\mathbf{1}^S, S \in \mathcal{B}_1 \cup \mathcal{B}_3\}$ is four.

Example 3. Let us give a complete analysis with $n = 3$. The lineality space has basis $\{u_{\{1\}}, u_{\{2\}}, u_{\{3\}}\}$. The extremal rays are $-\delta_{12}$, $-\delta_{13}$, $-\delta_{23}$, and

$$r_1 = \delta_{12} + \delta_{13} + \delta_{123}$$
$$r_2 = \delta_{12} + \delta_{23} + \delta_{123}$$
$$r_3 = \delta_{13} + \delta_{23} + \delta_{123}.$$

Table 7 gives the facets and which extremal rays (except those of the lineality space that all belong to every facet) belong to them.

This shows the neighborhood relations between facets (two facets are neighbors if they have a common extremal ray that is not in the lineality space), and consequently all faces. Interestingly, the figure below of the five facets gives a faithful representation of the neighborhood relation, although it is not a correct geometrical representation. The part in blue indicates where are the games with a point core.



6.2. Case of $\mathcal{BG}_\alpha(n)$

Because the facet of $\mathcal{BG}_\alpha(n)$ corresponding to some m.b.c. $\mathcal{B} \in \mathfrak{B}^*(n)$ is the intersection of the facet of $\mathcal{BG}(n)$ corresponding to $\mathcal{B}$ and the hyperplane $v(N) = \alpha$, every result established for $\mathcal{BG}(n)$ applies without any change to $\mathcal{BG}_\alpha(n)$.

6.3. Case of $\mathcal{BG}_+(n)$

We know by Theorem 16 that every m.b.c. $\mathcal{B} \in \mathfrak{B}^*(n)$ determines a facet of $\mathcal{BG}_+(n)$, which is therefore a subset of the corresponding facet in $\mathcal{BG}(n)$. Therefore, once again, Theorem 17 applies without any change.

On the other hand, we know from Theorem 16 that each equality $v(S) = 0$, $S \neq \emptyset, N$, defines a facet. Taking a game v in the relative interior of this facet, as no inequality $\sum_{S \in \mathcal{B}} \lambda_S^{\mathcal{B}} v(S) \leqslant 1$ is tight, v is in the interior of $\mathcal{BG}(n)$, and by Proposition 5, v has not a point core. Summarizing, we have found the following.

Theorem 19. *The following holds for $\mathcal{BG}_+(n)$:*

1. *Every game in a facet defined by $\mathcal{B} \in \mathfrak{B}^*(n)$ s.t. $|\mathcal{B}| = n$ has a point core.*
2. *No game in the relative interior of any other facet has a point core.*
3. *Consider a face $\mathcal{F} = \mathcal{F}_1 \cap \cdots \cap \mathcal{F}_p \cap \mathcal{F}_{p+1} \cap \cdots \cap \mathcal{F}_r$, with $\mathcal{F}_1, \ldots, \mathcal{F}_p$ being facets associated to m.b.c. $\mathcal{B}_1, \ldots, \mathcal{B}_p$, and $\mathcal{F}_{p+1}, \ldots, \mathcal{F}_r$ being facets associated to sets $S_{p+1}, \ldots, S_r$. Then any game in $\mathcal{F}$ has a point core iff the rank of the matrix $\{\mathbf{1}^S, S \in \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_p\}$ is n.*

In 3 of the above theorem, the facets associated to sets $S_{p+1}, \ldots, S_r$ play no role.

Another approach is to address the question under the point of view of vertices, instead of faces. Proposition 2 gives immediately the answer to our question as far as vertices are concerned.

Corollary 2. *A vertex $v \in \mathcal{BG}_+(n)$ associated to collection $\mathcal{D}$ has a point core iff $|\cap \mathcal{D}| = 1$.*

Another point is to investigate whether games in a face whose all vertices have a point core have also a point core. Surprisingly, this is false in general already for edges, that is, faces of dimension 1. The next theorem clarifies the situation for edges.

Theorem 20. *Consider two adjacent vertices v_1, v_2 of $\mathcal{BG}_+(n)$, with associated collections $\mathcal{D}_1, \mathcal{D}_2$, respectively, and $\cap \mathcal{D}_1 = \{i\}$, $\cap \mathcal{D}_2 = \{j\}$. Consider $v = \lambda v_1 + (1-\lambda)v_2$ with $\lambda \in]0,1[$, that is, a game in the edge between v_1, v_2. Then:*

1. *If $i = j$, then $C(v)$ is a singleton, that is, v has a point core.*
2. *If $i \neq j$ and $n \leqslant 4$, then v has a point core.*

Proof. We prove the result in two steps.

1. Suppose $i = j$. Then by Theorem 13, we may suppose $\mathcal{D}_1 \subseteq \mathcal{D}_2$. Take $x \in C(v)$. Then for any $S \in \mathcal{D}_1$, $v(S) = 1$, therefore $x(S) \geqslant 1$, which implies $x \in C(v_1)$. As v is balanced and $C(v_1)$ is a singleton, so is $C(v)$.
2. Suppose w.l.o.g. $i = 1, j = 2$, and $n \leqslant 4$.
   2.1. Suppose $\mathcal{D}_1 \cap \mathcal{D}_2 = \emptyset$ and consider $x \in C(v)$. Suppose $\{1\} \in \mathcal{D}_1$. Then $x_1 \geqslant \lambda$. Similarly, if $\{2\} \in \mathcal{D}_2$, $x_2 \geqslant 1-\lambda$. As $x(N) = 1$ and $x \geqslant 0$, both facts force $x_1 = \lambda$ and $x_2 = 1-\lambda$, and therefore $x_k = 0$ for all $k \neq 1,2$, proving that v has a point core. Otherwise, if $\{2\} \notin \mathcal{D}_2$, then $\mathcal{D}_2$ must contain at least two sets, whose intersection yields $\{2\}$. This is possible with n = 4 and then $\mathcal{D}_2$ must contain $\{2,3\}, \{2,4\}$ by Lemma 7. Then, $x_2 + x_3 \geqslant 1-\lambda$ and $x_2 + x_4 \geqslant 1-\lambda$, which together with $x_1 \geqslant \lambda$ forces equality everywhere, from which we deduce $x_3 = x_4$, then $x_1 + x_2 + x_3 = 1$, and so $x_4 = x_3 = 0$ and $x_2 = 1-\lambda$. Finally, if $\{1\} \notin \mathcal{D}_1$ and $\{2\} \notin \mathcal{D}_2$, we get $\mathcal{D}_1$ contains $\{1,3\}, \{1,4\}$ and $\mathcal{D}_2$ contains $\{2,3\}, \{2,4\}$, which contradicts adjacency (see case (a) of Figure 1).
   2.2. Suppose $\mathcal{D}_1 \cap \mathcal{D}_2 \neq \emptyset$ and consider $x \in C(v)$. Any set in $\mathcal{D}_1 \cap \mathcal{D}_2$ must contain {1, 2}. Suppose first $\{1,2\} \in \mathcal{D}_1 \cap \mathcal{D}_2$. Then $x_1 + x_2 \geqslant 1$, and because $x(N) = 1$ and $x \geqslant 0$, we obtain $x_1 + x_2 = 1$ and $x_k = 0$ for $k \neq 1,2$. Because there must exist $S \in \mathcal{D}_1 \setminus \mathcal{D}_2$ s.t. $2 \notin S$, this implies that $x(S) = x_1 \geqslant \lambda$. Similarly, there exists $S \in \mathcal{D}_2 \setminus \mathcal{D}_1$ such that $1 \notin S$, and $x(S) = x_2 \geqslant 1-\lambda$. Combining this with $x_1 + x_2 = 1$, we obtain equality throughout, which proves that v has a point core.

Suppose now with $2 < n \leqslant 4$ that $\{1,2,3\} \in \mathcal{D}_1 \cap \mathcal{D}_2$. Proceeding as above we get $x_1 + x_2 + x_3 = 1$ and $x_k = 0$ for $k \neq 1,2,3$. Now, there must exist $S \in \mathcal{D}_1 \setminus \mathcal{D}_2$ such that $3 \notin S$: either $S = \{1\}$ or, if $n = 4$, $S = \{1,4\}$. This yields $x_1 \geqslant \lambda$. Similarly with $\mathcal{D}_2 \setminus \mathcal{D}_1$, we get $x_2 \geqslant 1-\lambda$, and we can conclude as above. The case $\{1,2,4\} \in \mathcal{D}_1 \cap \mathcal{D}_2$ is similar.

Finally, with $n = 4$, $\{1,2,3,4\} \in \mathcal{D}_1 \cap \mathcal{D}_2$ is possible. Then, $\mathcal{D}_1 \setminus \mathcal{D}_2$ must contain $\{1\}$ or both $\{1,3\}, \{1,4\}$, and similarly for $\mathcal{D}_2 \setminus \mathcal{D}_1$. It suffices to proceed as for the case $\mathcal{D}_1 \cap \mathcal{D}_2 = \emptyset$. □

The following counterexample shows that the result is no more true for $n > 4$.

Example 4. Let us take $n = 5$ and two collections $\mathcal{D}_1, \mathcal{D}_2$ defined by

$$\mathcal{D}_1 = \{\{1,2,3,4,5\}, \{1,3\}, \{1,4\}, \{1,5\}\}, \quad \mathcal{D}_2 = \{\{1,2,3,4,5\}, \{2,3,4\}, \{2,4,5\}, \{2,3,5\}\}.$$

We have $\cap \mathcal{D}_1 = \{1\}$ and $\cap \mathcal{D}_2 = \{2\}$. Therefore, $\mathcal{D}_1, \mathcal{D}_2$ determine vertices, which we denote by v_1, v_2, respectively. It can be checked (via Theorem 15 or simply Figure 1) that these two vertices are adjacent.

Let us consider $v = (v_1 + v_2)/2$. One can check that x, y are two distinct core elements of v, and hence it is not reduced to a singleton:

$$x = \left(\frac{1}{4} \quad 0 \quad \frac{1}{4} \quad \frac{1}{4} \quad \frac{1}{4}\right)$$

$$y = \left(\frac{1}{2} \quad \frac{1}{2} \quad 0 \quad 0 \quad 0\right).$$

7. Applications

The fact that the polytope $\mathcal{BG}_+(n)$ is combinatorial has profound implications on optimization issues associated with the polytope. In Matsui and Tamura [12], the importance of a polytope being combinatorial when performing optimization on it is explained. In this work, the property of being combinatorial is referred to as Property C. Some results of this work require a stronger property called Property B (Property B implies Property C). Next, we can see that the polytope $\mathcal{BG}_+(n)$ also satisfies Property B, and therefore all the results of Matsui and Tamura [12] are applicable to our case.

Proposition 6. *Let v_1, v_2 and v_3 three vertices of $\mathcal{BG}_+(n)$ such that $v_1 \leqslant v_2 \leqslant v_3$ (coordinate-wise), then $v_1 + v_3 - v_2$ is a vertex of $\mathcal{BG}_+(n)$ (Property B).*

Proof. Let $\mathcal{D}_1, \mathcal{D}_2$ and $\mathcal{D}_3$ be the collections associated with each vertex. Because $\mathcal{D}_1 \subseteq \mathcal{D}_2 \subseteq \mathcal{D}_3$, it follows that the sets attaining value 1 in $v_1 + v_3 - v_2$ are exactly those in $\mathcal{D}_3 \setminus (\mathcal{D}_2 \cup \mathcal{D}_1)$. As this set is contained in $\mathcal{D}_3$, it follows that

$$\bigcap (\mathcal{D}_3 \setminus (\mathcal{D}_2 \cup \mathcal{D}_1)) \supseteq \bigcap \mathcal{D}_3 \neq \emptyset,$$

and therefore $v_1 + v_3 - v_2 \in \mathcal{BG}_+(n)$. □

For combinatorial polytopes, more efficient vertex enumeration algorithms can be defined. In Merino and Mütze [13], one can appreciate how the presence of Hamiltonian paths between vertices can be leveraged to make enumeration algorithms more efficient.

Many interesting problems in cooperative game theory can be expressed as an optimization problem on $\mathcal{BG}_+(n)$, especially approximation problems. As explained above, the fact that $\mathcal{BG}_+(n)$ is combinatorial and even satisfies the stronger Property B permits to benefit from more efficient vertex enumeration algorithms and therefore to obtain better performance in linear optimization.

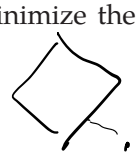
Many operators on games are linear, for example, the Harsanyi dividends (a.k.a. Möbius transform), the Shapley value, and its generalization the interaction transform $T : v \mapsto I^v$ defined by

$$I^v(S) = \sum_{T \subseteq N \setminus S} \frac{(n - |S| - |T|)!|T|!}{(n - |S| + 1)!} \sum_{K \subseteq S} (-1)^{|S \setminus K|} v(K \cup T)$$

(see Grabisch et al. [8] and Grabisch [7] for many other linear operators on games). Note that $I^v(\{i\})$ is the Shapley value for player i. It follows that the maximization/minimization of $I^v(S)$ over $v \in \mathcal{BG}_+(n)$ for some S, or any linear combination of such terms, is a LP problem that can be solved efficiently, taking advantage of the fact that $\mathcal{BG}_+(n)$ is combinatorial, and even satisfies property B.

Another class of optimization problems is the approximation problem. As explained in the Introduction, the main motivation behind this work is to be able to solve the projection problem on $\mathcal{BG}(n)$, or on $\mathcal{BG}_\alpha(n), \mathcal{BG}_+(n)$, which is a quadratic optimization problem. One can also consider to minimize the L_1 norm instead of the L_2 norm, which leads to a LP problem (after standard linearization):

$$\min_{v \in \mathcal{BG}_+(n)} \sum_{S \subseteq N} |v(S) - w(S)|,$$



with $w \in \mathcal{G}_+(n)$. However, the linearization implies to introduce new variables and constraints; therefore, it is not obvious if one can still benefit from the properties of $\mathcal{BG}_+(n)$. Some further investigation is needed here.

8. Concluding Remarks

Our study has permitted to have a complete description of the polyhedral structure of the set of balanced games $\mathcal{BG}(n)$, as well as of its subsets $\mathcal{BG}_\alpha(n)$ and $\mathcal{BG}_+(n)$, in terms of extremal rays, vertices, facets, and adjacency relations between vertices. In addition, we provided an algorithm for the uniform random generation of the vertices of $\mathcal{BG}_+(n)$. Unexpectedly, the polytope $\mathcal{BG}_+(n)$ of nonnegative normalized balanced games seems to be related to some well-known combinatorial structures, as its number of vertices is a known sequence in OEIS, related to the number of Boolean functions in some Post classes. Moreover, this polytope is combinatorial, which means that the adjacency graph of its vertices is Hamiltonian. In the last part of the paper, we have given a characterization of faces of these polyhedra that contain games with a core reduced to a singleton.

However, some issues would need a deeper analysis, especially in adjacency relations. Although we provided a characterization of adjacent vertices for $\mathcal{BG}_+(n)$, we did not perform this analysis for the extremal rays of $\mathcal{BG}(n)$, $\mathcal{BG}_\alpha(n)$ or for the facets of these polyhedra. This would help to solve the projection problem we intend to address in a future work: finding the closest balanced game for a given nonbalanced game.

Another topic of future research would be to study the set of monotone balanced games. Monotone games are games satisfying the following property: if $S \subseteq T$, then $v(S) \leqslant v(T)$. As $v(\emptyset) = 0$, these games are nonnnegative and therefore form a subset of $\mathcal{BG}_+(n)$. This class of games has a great importance: They are known under the name of *capacities* (Choquet [2]) in decision theory and nonadditive integral theory (Schmeidler [18], and Grabisch [7]). Unfortunately, the analysis of the set of balanced capacities reveals to be extremely difficult: with $n = 4$, the number of vertices is already equal to 9,002, and most of them are not 0-1 valued. Finding an analytical characterization of them seems to be challenging.

Appendix A. Proofs of Section 4

A.1. Proof of Theorem 5

1. Let us prove that $\mathcal{BG}_\alpha(n)$ is an affine cone, that is, the translation of a cone. Observe that a particular game in $\mathcal{BG}_\alpha(n)$ is $\alpha u_{\{n\}}$, that is, the unanimity game centered on $\{n\}$ and multiplied by α.

Define $\mathcal{C}_\alpha(n) := \mathcal{BG}_\alpha(n) - \alpha u_{\{n\}}$ and let us show that

$$\mathcal{C}_\alpha(n) = \left\{ v' : \sum_{S\in\mathcal{B}} \lambda_S^{\mathcal{B}} v'(S) \leqslant 0 \right\}.$$

Pick $v \in \mathcal{BG}_\alpha(n)$ and consider $v' := v - \alpha u_{\{n\}}$. Then, for any $\mathcal{B} \in \mathfrak{B}^*(n)$,

$$\sum_{S\in\mathcal{B}} \lambda_S^{\mathcal{B}} v'(S) = \sum_{S\in\mathcal{B}} \lambda_S^{\mathcal{B}} v(S) - \sum_{S\in\mathcal{B},\, S\ni n} \lambda_S^{\mathcal{B}} \alpha \leqslant \alpha - \alpha = 0.$$

Hence, the claim is proved. Conversely, suppose that a game v'' satisfies

$$\sum_{S\in\mathcal{B}} \lambda_S^{\mathcal{B}} v''(S) \leqslant 0, \ \forall \mathcal{B} \in \mathfrak{B}^*(n). \tag{A.1}$$

Then, $v'' + \alpha u_{\{n\}}$ satisfies

$$\sum_{S\in\mathcal{B}} \lambda_S^{\mathcal{B}} [v''(S) + \alpha u_{\{n\}}(S)] = \sum_{S\in\mathcal{B}} \lambda_S^{\mathcal{B}} v''(S) + \alpha \leqslant \alpha,$$

for any $\mathcal{B} \in \mathfrak{B}^*(n)$, that is, $v'' + \alpha u_{\{n\}} \in \mathcal{BG}_\alpha(n)$ and thus $v'' \in \mathcal{C}_\alpha(n)$.

Finally, take $v \in \mathcal{C}_\alpha(n)$ and $\beta \geqslant 0$. We have $\beta v \in \mathcal{C}_\alpha(n)$ because it satisfies (A.1). Therefore, $\mathcal{C}_\alpha(n)$ is a cone.

2. Proceeding as for the proof of Theorem 2, the affine space contained in $\mathcal{BG}_\alpha(n)$ is obtained by replacing $v(N)$ by α in the equations giving the lineality space of $\mathcal{BG}(n)$. This yields the system

$$\sum_{i\in N} v(\{i\}) = \alpha$$
$$v(S) + \sum_{i\in N\setminus S} v(\{i\}) = \alpha, \quad S \subset N, |S| > 1.$$

To obtain the corresponding vector space, we just replace α by 0, thus obtaining

$$\sum_{i\in N} v(\{i\}) = 0$$
$$v(S) + \sum_{i\in N\setminus S} v(\{i\}) = 0, \quad S \subset N, |S| > 1.$$

The set of solutions for this system is, expressed in terms of $v(\{1\}), \ldots, v(\{n-1\})$,

$$\left\{ \left(v_1, \ldots, v_{n-1}, -\sum_{i=1}^{n-1} v_i, v_1 + v_2, \ldots, \underbrace{\sum_{i\in S} v_i}_{S\not\ni n}, \ldots, \underbrace{-\sum_{i\notin S} v_i}_{S\ni n}, \ldots \right) : v_1, \ldots, v_{n-1} \in \mathbb{R} \right\}.$$

This yields the basis $(w_i)_{i\in N\setminus\{n\}}$ given above.

A.2. Proof of Theorem 6

We follow the same steps as for Theorem 3.

Observe that because $r_n = -\delta_{\{n\}}$, it could be considered as a ray of type r_S with $S = \{n\}$. We follow this way in steps 2 and 3 of the proof.

1. The $2n-2$ extremal rays corresponding to the lineality space come from its basis given in Theorem 5.

2. Consider $S \subset N, |S| > 1$ or $S = \{n\}$, and let us show that r_S is extremal. Clearly (A.1) is satisfied and $r_S(\{i\}) = 0$ for all $i \in N \setminus \{n\}$, hence r_S is a ray of $\mathcal{C}_\alpha^0(n)$. Suppose it is not extremal. Then there exist two rays $r, r' \in \mathcal{C}_\alpha^0(n)$ such that $r_S = r + r'$. Suppose that $r(T) > 0$, say, $r(T) = 1$ for some $T \neq S$, $1 < |T| < n$ or $T = \{n\}$. Then, $r'(T) = -1$.

Using the partition $\{T, (N \setminus T)^\perp\}$, the corresponding inequality in (A.1) for r becomes either $1 \leqslant 0$ if $n \in T$, or $1 + r(\{n\}) \leqslant 0$ if $n \notin T$. The first case being a contradiction, let us study the second case. For this case, it follows that $r(\{n\}) \leqslant -1$, which in turn implies $r'(\{n\}) \geqslant 1$. Considering then the partition $\{N^\perp\}$, the corresponding inequality in (A.1) for r' becomes $0 + r'(\{n\}) \leqslant 0$, a contradiction.

The case $r(T) < 0$ can be treated similarly. We conclude that r must have zero coordinates, possibly excepted for $T = S$. Observe that as before, $r(S) > 0$ is not possible because the inequality for $\{S, (N \setminus S)^{\perp}\}$ would not be satisfied (proceed as above with T). Hence, r_S is extremal.

3. Let us take $i \in N \setminus \{n\}$ and show that r_i is an extremal ray. First, we prove it is a ray in $\mathcal{C}^0_\alpha(n)$. By definition, $r_i(\{j\}) = 0$ for all $j \neq n$. It remains to prove that r_i is a solution of (A.1). Taking any m.b.c. $\mathcal{B}$, the inequality becomes

$$\sum_{\substack{S\in\mathcal{B}\\ S\ni i\\ S\not\ni n\\ |S|>1}} \lambda_S^{\mathcal{B}} - \sum_{\substack{S\in\mathcal{B}\\ S\not\ni i\\ S\ni n}} \lambda_S^{\mathcal{B}} = \sum_{\substack{S\in\mathcal{B}\\ S\ni i\\ S\not\ni n\\ |S|>1}} \lambda_S^{\mathcal{B}} - \left(1 - \sum_{\substack{S\in\mathcal{B}\\ S\ni i\\ S\ni n}} \lambda_S^{\mathcal{B}}\right) = \sum_{\substack{S\in\mathcal{B}\\ S\ni i\\ |S|>1}} \lambda_S^{\mathcal{B}} - 1 \leqslant 0.$$

This is obviously satisfied as $\sum_{\substack{S\in\mathcal{B}\\ S\ni i}} \lambda_S^{\mathcal{B}} = 1$ and $\lambda_S^{\mathcal{B}} \geqslant 0$. Observe that the inequality is tight iff $\mathcal{B} \not\ni \{i\}$.

To show that r_i is extremal, we need to show that the set of solutions of the system of tight inequalities has dimension 1. We have already observed that the inequality is tight iff $\{i\} \notin \mathcal{B}$. Let us consider this system in v, that is,

$$\sum_{S\in\mathcal{B}} \lambda_S^{\mathcal{B}} v(S) = 0, \ \forall \mathcal{B} \in \mathfrak{B}^*(n) \text{ s.t. } \{i\} \notin \mathcal{B}, \tag{A.2}$$

with $v \in \mathcal{G}_\alpha(n)$ s.t. $v(i) = 0, i \neq n$. Take the partition $\mathcal{B} = \{S, S^{\perp}\}$ with $i \in S$. The corresponding equality in (A.2) reads $v(S) = 0$ if $S \ni n$, and $v(S) + v(\{n\}) = 0$ otherwise, that is, $v(S) = -v(\{n\})$. It follows that the set of solutions has the form $\{(\underbrace{\beta}_{\{n\}}, \underbrace{0 \cdots 0}_{S\ni n,\, |S|>1}, \underbrace{-\beta \cdots -\beta}_{S\not\ni n,\, |S|>1}), \beta \in \mathbb{R}\}$, and hence has dimension 1.

4. It remains to prove that there is no other extremal ray. Consider a ray $w \in \mathcal{C}^0_\alpha(n)$, hence satisfying $w(\{i\}) = 0$ for all $i \in N \setminus \{n\}$ and (A.1). If w is not a conic combination of r_S and r_i, $S \subset N, |S| > 1$, $i \in N$, the system (5) has no solution in α_S, α_i. Using definitions of r_S and r_i and omitting coordinates for singletons (except n) and N in r_S, r_i, w, we obtain

$$\begin{aligned}
-\alpha_S - \sum_{i\notin S} \alpha_i &= w(S), \quad S \subset N, |S| > 1, S \ni n\\
-\alpha_S + \sum_{i\in S} \alpha_i &= w(S), \quad S \subset N, |S| > 1, S \not\ni n\\
\alpha_S &\geqslant 0, \quad S \subset N, |S| > 1\\
\alpha_i &\geqslant 0, \quad i \in N.
\end{aligned}$$

We may denote the whole system by $A\alpha \geqslant b$ in matrix notation (with some abuse). Then, by Farkas' lemma, this system has no solution iff there exists a vector $[y\ z\ t]$ with coordinates $y_S \in \mathbb{R}, S \subseteq N, |S| > 1, z_T \geqslant 0, T \subset N, |T| > 1$, and $t_i \geqslant 0, i \in N$, such that $[y\ z\ t]^{\top} A = 0$ and $[y\ z\ t]^{\top} b > 0$. Observe that the only vector $[y\ z\ t]$ solution of $[y\ z\ t]^{\top} A = 0$ is

$$y_S = 1\,(S \subset N, |S| > 1), \quad z_S = 1\,(S \subset N, |S| > 1), \quad t_1 = \cdots = t_n = 0.$$

For this solution, we obtain

$$[y\ z\ t]^{\top} b = \sum_{S\subset N,\, |S|>1} w(S).$$

Considering again the balanced collection $\{S \subset N, |S| > 1\}$ with balancing weights $\frac{1}{2^{n-1}-2}$, it follows that

$$\sum_{S\subset N,\, |S|>1} w(S) \leqslant 0.$$

Therefore, $[y\ z\ t]^{\top} b \leqslant 0$ and the second system has no solution. Hence, the first system has always a solution.

Endnote

[1] That is, a cone plus a point.

References

[1] Bondareva O (1963) Some applications of linear programming to the theory of cooperative games. *Problemy Kibernetiki* 10:119–139.
[2] Choquet G (1953) Theory of capacities. *Ann. Institute. Fourier (Grenoble)* 5:131–295.
[3] Edmonds J (1970) Submodular functions, matroids, and certain polyhedra. Guy R, Hanani H, Sauer N, Schönheim J, eds. *Proc. Calgary Internat. Conf. Combinatorial Structures Their Appl.* (Gordon and Breach, New York), 66–87.
[4] Foldes S, Hammer P (2005) Submodularity, supermodularity and higher order monotonicities of pseudo-Boolean functions. *Math. Oper. Res.* 30:453–461.
[5] Fujishige S (2005) *Submodular Functions and Optimization*, Annals of Discrete Mathematics, vol. 58, 2nd ed. (Elsevier, Amsterdam).
[6] Gillies D (1953) Some theorems on n-person games. PhD thesis, Princeton University, Princeton, NJ.

[7] Grabisch M (2016) *Set Functions, Games, and Capacities in Decision Making*, Theory and Decision Library C, vol. 46 (Springer, Berlin).
[8] Grabisch M, Marichal J-L, Roubens M (2000) Equivalent representations of set functions. *Math. Oper. Res.* 25(2):157–178.
[9] Jovović V, Kilibarda G (1999) On the number of Boolean functions in the Post classes $\mathcal{F}_8^{\mu_*}$. *Discrete Math. Appl.* 9:593–605.
[10] Kroupa T, Studený M (2019) Facets of the cone of totally balanced games. *Math. Methods Oper. Res.* 90:271–300.
[11] Laplace Mermoud D, Grabisch M, Sudhölter P (2023) Minimal balanced collections and their application to core stability and other topics of game theory. *Discrete Appl. Math.* 341:60–81.
[12] Matsui T, Tamura S (1995) Adjacency on combinatorial polyhedra. *Discrete Appl. Math.* 56:311–321.
[13] Merino A, Mütze T (2023) Traversing combinatorial 0/1-polytopes via optimization. *Proc. 64th IEEE Annual Sympos. Foundations Comput. Sci.* (IEEE, Piscataway, NJ), 1282–1291.
[14] Naddef D, Pulleyblank WR (1981) Hamiltonicity and combinatorial polyhedra. *J. Combinatorial Theory Ser. B* 31:297–312.
[15] Peleg B, Sudhölter P (2003) *Introduction to the Theory of Cooperative Games* (Kluwer Academic Publisher, New York).
[16] Peters H (2008) *Game Theory: A Multilevel Approach* (Springer, Berlin).
[17] Rutkowski KE (2017) Closed-form expressions for projectors onto polyhedral sets in Hilbert spaces. *SIAM J. Optim.* 27:1758–1771.
[18] Schmeidler D (1989) Subjective probability and expected utility without additivity. *Econometrica* 57(3):571–587.
[19] Shapley LS (1967) On balanced sets and cores. *Naval Res. Logist. Quart.* 14:453–460.
[20] Sloane NJA The online encyclopedia of integer sequences (OEIS). Accessed July 23, 2024, https://oeis.org.

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